

ACADEMIC ELECTIVE



Learning Activity Sheets for Contemporary Literature 1

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LEARNING ACTIVITY SHEET

Name:		Grade & Section:	
Learning Area	Finite Mathematics 1		

Unit 1: GEOMETRY OF DESIGN

Lesson 1.1. SYMMETRIES IN NATURE AND ARTS

Objectives:	At the end of the lesson, you will able to: 1. identify and describe symmetries in nature and art; 2. identify different types of symmetry such as reflection and rotational symmetry; 3. determine the center of rotation, angle of rotation, and order of symmetry using cut-outs or GeoGebra; and 4. describe the rotational symmetry of objects found in nature and Filipino design.
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Key Ideas and Examples:	Symmetry in Nature In nature, symmetry is often a sign of balance, health, or function. Many plants and animals have bilateral symmetry, meaning one side is a mirror image of the other. It is an exact correspondence of form and constituent configuration on opposite sides of a dividing line or plane or about a center or an axis. Other natural objects show radial symmetry a type of symmetry where the parts of an object are arranged around a central point—like the spokes of a bicycle wheel or slices of a pizza. Nature uses symmetry for many reasons: • To attract mates (like the patterns on peacocks) • For balance and movement (as in animals) • To grow in a stable and efficient way (like leaves and shells) Symmetry in Art Artists and architects have used symmetry for thousands of years to create beauty, harmony, and order. Whether it's the intricate patterns in Islamic tiles, the balance in classical architecture, or the radial designs in mandalas, symmetry helps art feel pleasing and structured.
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Symmetry

Fabric.



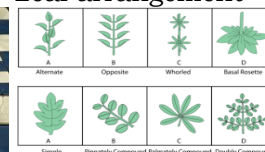
Brick wall.



Floor Tile



Leaf arrangement



Reflection (Mirror) Symmetry – occurs when one half of an object or shape is a mirror image of the other half. If you can draw a line down the middle (called the line of symmetry) and both sides match exactly, then the shape has reflection symmetry. It's like folding a picture in half and having both sides line up perfectly!

Rotational Symmetry – designs that repeat around a central point

Key Parts of Rotational Symmetry:

Center of rotation – the spot the shape rotates around.

Angle of rotation – how far you turn the shape each time it looks the same.

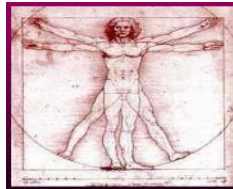
Order of symmetry – how many times the shape matches itself in one full turn.

The importance of symmetries in nature and art in terms of:

1. Reflection Symmetry (Mirror Symmetry)

Nature: Many living organisms exhibit mirror symmetry. For example, butterflies and human bodies are bilaterally symmetrical—one side is a mirror image of the other. This symmetry is often linked to biological function and evolutionary advantages like streamlined movement or balanced sensory input.

Art: In visual art and architecture, mirror symmetry is used to create harmony and a sense of order. Traditional Islamic art, for instance, heavily relies on reflection symmetry to produce visually pleasing patterns.



The 'Proportions of Man' is a famous work of art by Leonardo da Vinci that shows the symmetry of the human form.



Symmetry exists in architecture all around the world. The best known example of this is the Taj Mahal.

2. Rotational Symmetry

Nature: Flowers, starfish, and snowflakes often show rotational symmetry. For instance, a five-petaled flower has fivefold rotational symmetry. This type of symmetry helps in maximizing exposure to sunlight or efficient reproduction.

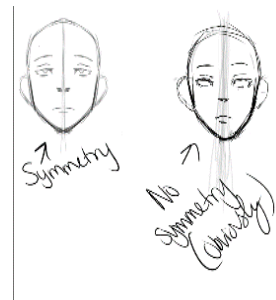
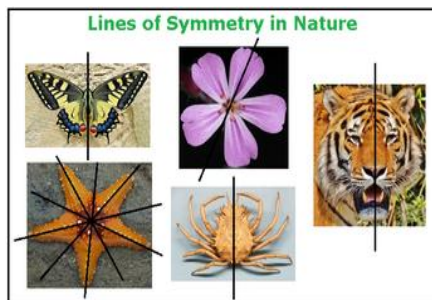
Arts: Mandalas and circular designs use rotational symmetry to evoke balance and spirituality. Artists use it to guide the viewer's eye around a central point, creating unity in the composition.



3. Line of Symmetry

Nature: A line of symmetry divides an object into two identical parts. Leaves, insects, and many animals possess a vertical line of symmetry. It often reflects functional optimization, like flight in birds or balance in humans.

Arts: Lines of symmetry are used to establish focal points or create balanced compositions. A portrait may use a vertical line of symmetry for a formal and composed look.



4. Center of Rotation

Nature: The center of a sunflower or a jellyfish is a clear example of where rotational symmetry originates. It helps distribute resources evenly (like seeds in a sunflower).

Art: In decorative art, like rose windows in Gothic cathedrals, the center of rotation is often the visual focus, radiating symmetry outward to engage the viewer.



5. Angle of Rotation

Nature: The angle of rotation defines how many times an object maps onto itself in a full 360° turn. For example, a snowflake typically has 60° rotational symmetry (sixfold).

Art: In pattern design or tessellations, the angle of rotation defines how elements repeat and interlock, essential for creating intricate and mesmerizing visuals.

Rotational Symmetry in Nature and Filipino Designs

Before performing the activity, the teacher should state the following: For a figure with rotational symmetry of order n , the smallest angle of rotation is computed as 360 degrees divided by n . For an idealized 5-arm starfish, $n = 5$, so the smallest angle is $360 / 5 = 72$ degrees. For an idealized 8-point parol, $n = 8$, so the smallest angle is $360 / 8 = 45$ degrees

Video Lesson:) <https://youtu.be/YuOlrrxFJL4?si=EmmK0rEb0UcL3hp1>

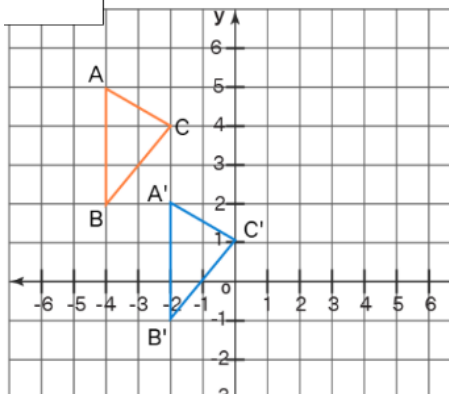
I. Activity 1:	Practicing Learned Skills																												
II. Materials:	concrete objects or images, worksheet or paper, pen																												
III. Instructions:	Instructions: 1. Complete the Symmetry Table: Using the objects provided (butterflies, flowers, starfish, parol designs, or circular textile patterns), fill in the table below. Observe each object carefully and identify its reflection and rotational symmetry properties: <table><tr><th>Object</th><th>Reflection Symmetry ✓-YES ✗-NO</th><th>Rotational Symmetry ✓-YES ✗-NO</th><th>Line of Symmetry</th><th>Center of Rotation</th><th>Angle of Rotation</th><th>Order of rotational symmetry</th></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> 2. Pair-Share Discussion: a. Work with a partner to compare your answers. b. Justify your reasoning using mathematical terms like line of symmetry, center of rotation, and angle of rotation. c. If you notice differences, discuss and re-examine the objects together.	Object	Reflection Symmetry ✓-YES ✗-NO	Rotational Symmetry ✓-YES ✗-NO	Line of Symmetry	Center of Rotation	Angle of Rotation	Order of rotational symmetry																					
Object	Reflection Symmetry ✓-YES ✗-NO	Rotational Symmetry ✓-YES ✗-NO	Line of Symmetry	Center of Rotation	Angle of Rotation	Order of rotational symmetry																							
V. Reflection:	Answer each question. 1. Why does the butterfly have only reflection symmetry? 2. Examine the starfish carefully. Identify all of its symmetries. Does it have reflection symmetry, rotational symmetry, or both? Justify your answer using the lines of symmetry and the angle of rotation. 3. How do multiple lines of symmetry affect a figure’s appearance?																												

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	images, worksheet or paper, pen
III. Instructions:	Observe the set of pictures shown by your teacher. Each image represents different real-life or artistic designs showing symmetry.

	2. For each picture, analyze carefully and answer the following:  - What type of symmetry is shown? (Reflection, Rotational, Both, or None) - Where do you see the symmetry in the image? - Justify your answer in one sentence. 3. Answer the guide the following questions: - Which part of the image shows the line of symmetry (if any)? Explain. - What point serves as the center of rotation (if applicable)? - If the image is rotated, what angle of rotation allows it to look the same again? - Which images show both reflection and rotational symmetry? Why? - Which image is the most symmetrical? Defend your answer.
V. Reflection:	Answer each question. - For each image you classified as having rotational symmetry, state the order of rotational symmetry and compute the smallest angle of rotation. Show your computation as $360 / \text{order} = \text{angle}$. - Which concept was easiest for you to understand? Which was more challenging? - How can identifying symmetry help you analyze patterns in real-life objects or designs?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials	drawing materials, worksheet or paper, pen
III. Instructions:	Option 1. Create Your Own Symmetrical Art - Choose Your Medium: Decide whether you want to create your design on paper, using colored pencils/markers, or digitally using GeoGebra or a drawing app. - Design Your Symmetry: - Create a design that shows either reflection symmetry or rotational symmetry. - Your design can be inspired by nature, Filipino cultural patterns, or geometric shapes. - Identify Symmetry Properties: - For reflection symmetry, draw and label the line(s) of symmetry. - For rotational symmetry, identify the center of rotation and calculate the angle of rotation. - Note the order of symmetry if applicable - Present Your Design: - Each learner presents their design to the class or in small groups. - Explain the type of symmetry used, how you determined the line of symmetry or center and angle of rotation, and how your design demonstrates balance, repetition, and harmony. - Peer Feedback: - Classmates may ask questions or give suggestions about the symmetry properties. - Reflect on how your design connects to real-life patterns in nature or Filipino art. Option 2. Gallery Walk Reflection - Walk around the classroom and observe the symmetry in your classmates' designs or artworks. - For each design you observe, try to identify the type of symmetry used (reflection symmetry, rotational symmetry, both, or neither). - Write a short comment or question on the feedback sheet provided. Your comment may include: - The type of symmetry you noticed - A positive observation about the design

	c. A question or suggestion about the pattern or symmetry used 4. After the gallery walk, return to your seat and discuss with a partner what you observed. 5. The class will then share interesting symmetry patterns discovered in the gallery.
IV. Reflection:	Answer each question. 1. What types of symmetry patterns were most common in your classmates' works? 2. How did the designs demonstrate reflection or rotational symmetry? 3. Which design impressed you the most and why? 4. How did symmetry help make the designs balanced, organized, or visually appealing?
*****END OF LESSON 1.1*****	

Lesson 1.2. TRANSLATION	
Objectives:	At the end of the lesson, the learners are will be able to: 1. define geometric transformation and identify its types; 2. define translation as a movement that slides a figure without changing its size and shape; 3. describe translation in terms of direction and distance on the coordinate plane; 4. perform translation of points and simple geometric figures using given rules; and 5. illustrate translation of figures on the coordinate plane accurately.
Key Ideas and Examples:	A transformation is a rule that maps each point of a figure, called the preimage, to a corresponding point of its image. A translation is a transformation that moves every point the same distance and direction. It does not turn, flip, stretch, or shrink the figure. Because every point moves by the same vector, a translation preserves length, angle measure, shape, size, and orientation. On the coordinate plane, a translation can be written as $(x, y) \rightarrow (x + a, y + b)$, where a tells how far the point moves horizontally and b tells how far it moves vertically. Translation means sliding a shape from one position to another without turning or flipping it. a. The shape keeps the same orientation. b. All points of the shape move the same distance in the same direction. The coordinate rule: $(x, y) \rightarrow (x + a, y + b)$ This image shows a translation, which is a geometric transformation that slides a figure from one position to another without rotating it, resizing it, or flipping it. Every point of the original figure (the pre-image, $\triangle ABC$) moves the same distance in the same direction to form the new figure (the image, $\triangle A'BC'$).  To find the exact translation rule, we can track how much a single vertex moves along the grid lines from start to finish: 1. Identify the Coordinates Original Point : A(-4, 5), Translated Point : A')-2, 2) 2. Calculate the Shift Horizontal Movement (-axis): To go from an x -coordinate of -4 to 2 , you move 2 units to the right. Vertical Movement (-axis): To go from a y-coordinate of 5 to 3 , you move 3 units down. 3. Summary of the Rule This specific transformation can be written in two common mathematical formats: Coordinate Notation: $(x, y) \rightarrow (x + 2, y - 3)$

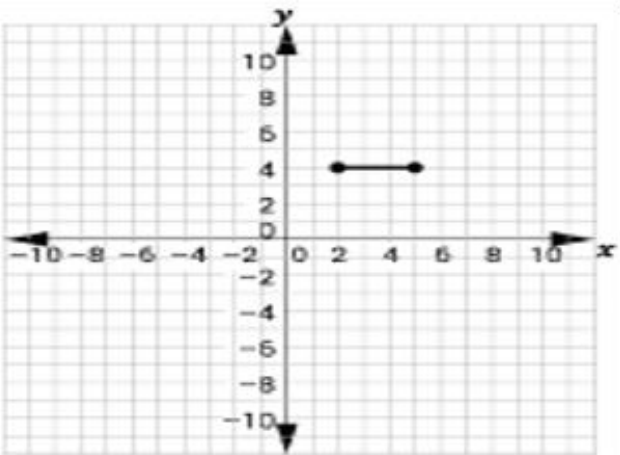
	Vector Notation: (2, -3)
Video Lesson	https://youtu.be/GqHWdTLL8Qw

I. Activity 1:	Practicing Learned Skills
II. Material:	pen and paper
III. Instructions:	1. Given $\triangle ABC$ with $A(2, 1)$, $B(5, 1)$, $C(2, 4)$, translate it by $(-3, 2)$. Find A' , B' , C' . 2. Compare AB and $A'B'$. What stayed the same? What changed?
V. Reflection:	Answer each question: 1. How did adding the translation vector $(-3, 2)$ to each vertex help determine the coordinates of the image $A'B'C'$? 2. Why do the lengths of AB and $A'B'$ remain the same after the translation, even though the coordinates of the points changed?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	Perform the given activities, write your answers on sheet of paper. 1. When the transformation is performed on point P , its image is point $P'(-3, 4)$. What are the coordinates of P ? A. $(-7, 11)$ B. $(-1, 3)$ C. $(1, -3)$ D. $(7, -11)$ 2. Consider the following points:) and . R undergoes the translation , such that $R'(5, 1)$ and $U'(16, -10)$. Part A: Complete the following algebraic description. Part B: Compute the lengths of RU and $R'U'$. How does $ R'U' $ compare to $ RU $? What does this tell you about translation? 3. Which transformation slides a figure without changing its size or shape? A. Rotation B. Reflection C. Translation D. Dilation 4. What is the image of point under the transformation ? A. B. C. D. 5. Which transformation moves every point of a figure the same distance in the same direction? A. Rotation B. Reflection C. Translation D. Dilation 6. Find the image of the following points after translation 7. A point $P'(5, 7)$ is the image of point P after translation What are the coordinates of point P ? 8. A segment has endpoints and . After a translation, the endpoints become and . a. What translation rule was applied? _____ b. What is the length of the segment before and after translation? _____ c. What can you conclude about translation? _____
V. Reflection:	Answer each question: 1. How can you determine the translation rule if only the coordinates of a point and its image are given? 2. Why is translation considered a rigid transformation? Use evidence from the activities to support your answer.

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	pen and paper, GeoGebra App
III. Instructions:	Complete the following activities to strengthen understanding of translations and their properties.

	Translation Puzzle Cards Match each pre-image with its corresponding image by identifying the translation rule applied. Explain how the coordinates changed from the pre-image to the image. GeoGebra Mini-Challenge Using GeoGebra, create a polygon and apply appropriate translations to transform it so that it matches the given target image. Record the translation rule(s) used and verify that the size and shape of the figure remain unchanged.
V. Reflection:	Answer each question: - How would the image change if a different translation rule were applied to the same figure? - Why is translation considered a rigid transformation? Use evidence from the activities to support your answer.
*****END OF LESSON 1.2*****	

Lesson 1.3. DILATION CONCEPT	
Objectives:	At the end of the lesson, you will be able to - define dilation as a transformation that changes the size of a figure; - identify the center of dilation in given illustrations; - differentiate enlargement and reduction using scale factor; - describe dilation using appropriate mathematical terms; and - perform dilations on the coordinate plane using a given center and scale factor.
Key Ideas and Examples	Dilation enlarges or reduces the original figure proportionally from a fixed center, called the center of dilation. The scale factor k determines the change The center of dilation is a fixed point in the plane about which all points are expanded or contracted. "How different is one line from the other in the above figure?" The scale factor refers to how much the figure grows or shrinks, and it is denoted by: - If, the image gets smaller and closer to the center of dilation. - If, the image gets larger and farther from the center of dilation. Vector Rule: $CP' = k \cdot CP$ General Coordinate Formula: $P'(x', y') = (a + k(x - a), b + k(y - b))$ [Center $C = (a, b)$] Origin Coordinate Formula: $P'(x', y') = (kx, ky)$ [Center $C = (0, 0)$] Example: Given the illustration 

1. Identifying the Original Pre-Image

The graph contains a horizontal pre-image line segment located in the first quadrant. We can isolate its position by identifying its exact endpoint coordinates:

- **Left Endpoint:** (2, 4)
- **Right Endpoint:** (5, 4)

Using these points, the original length of the segment is easily calculated by finding the horizontal distance between the x-coordinates: $5 - 2 = 3 \text{ units}$

2. Dilation Formula (Center at Origin)

When a shape or segment is dilated from the origin, every coordinate pair (x, y) is multiplied directly by the scale factor k to produce the new mapped image points:

$$\text{Coordinate Mapping Rule:} \\ (x, y) \rightarrow (kx, ky)$$

3. Scaling Scenarios Based on Factor k

Depending on the value assigned to k, the line segment will transform into one of the following scenarios:

Scenario Type	Example Scale Factor	Resulting Endpoint Transformation
Enlargement ($k > 1$)	$k = 2$	$(2, 4) \rightarrow (4, 8)$ $(5, 4) \rightarrow (10, 8)$ Segment grows longer (6 units) and moves outward.
Reduction ($0 < k < 1$)	$k = 0.5$	$(2, 4) \rightarrow (1, 2)$ $(5, 4) \rightarrow (2.5, 2)$ Segment shrinks (1.5 units) and moves inward.

Video Lesson

<https://youtu.be/GqHWdTLL8Qw>

I. Activity 1:

Practicing Learned Skills

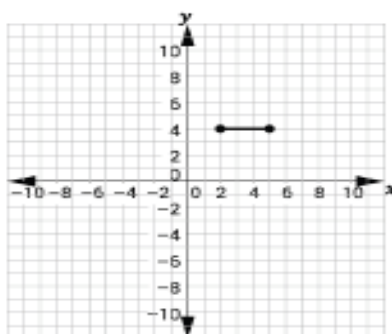
II. Materials:

pen and paper

III. Instructions:

Option 1 Collaborative Problem Solving

1. Pairs receive a worksheet with figures and Instructions: to dilate with a given scale factor from a specific center.
2. Pairs must calculate coordinates and draw the transformed figure accurately.



- a. How do you dilate the line segment on the above graph centered at a point on the same line?

	• Use (2, 4) as the center of dilation and complete the following: • If , then the dilated line segment will have • coordinates: _____ and _____. • If , then the dilated line segment will have • coordinates: _____ and _____. • When dilating a line that passes through the center of • dilation, the line is _____. b. How do you dilate the line segment on the above graph centered at the origin? • If , then the dilated line segment will have • coordinates: _____ and _____. • If , then the dilated line segment will have • coordinates: _____ and _____. • When dilating a line that does not pass through the center of dilation, the dilated line is _____ to the original. • $(x, y) \rightarrow (kx, ky)$ changes the size of the figure by a factor of k when the center of dilation is the origin. c. Given triangle with vertices: • Dilate the triangle using center and scale factor • Plot the original and the image • Label the new coordinates d. Rectangle Dilation • Given rectangle with vertices: • Dilate using center and scale factor • Plot and label the image Option 2. Real-Life Application Challenge 1. Choose or are assigned one scenario: • Enlarging a poster design • Scaling a house blueprint • Reducing a toy model 2. For your scenario: a. Draw a simple figure (e.g., triangle, rectangle, or design). b. Choose a center of dilation. c. Apply a scale factor (e.g., 2 or $\frac{1}{2}$). d. Show: • Original figure • Dilated figure • Coordinates (at least 3 points) e. Answer: • What is the scale factor? • Is it enlargement or reduction? • Why is dilation useful in your scenario? • Present your work to the class.
V. Reflection:	Answer each question: 1. How did the scale factor affect the size and position of the image compared to the original figure? 2. What similarities and differences did you observe between the original figure and its dilated image? 3. How can dilation be used in real-life situations such as designing posters, creating blueprints, or building scale models?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	Part I. Exit Ticket

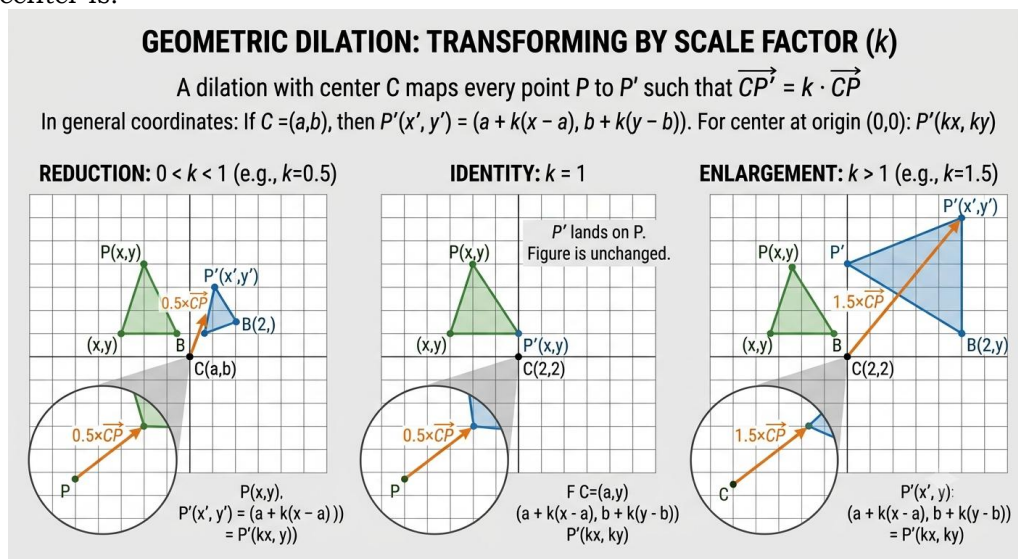
	Answer the following questions clearly and completely using appropriate mathematical terms. Show your computations where necessary and explain your reasoning. 1. Define dilation in your own words. Include the roles of the center of dilation and scale factor in your explanation. 2. Dilation on the Coordinate Plane Given: Pre-image: A(1, 1), B(3, 1), C(2, 4) Center of dilation: O(0, 0) Image: A'B'C': A'(2, 2), B'(6, 2), C'(4, 8) Answer the following: a. Identify the center of dilation. b. Determine the scale factor. Show how you obtained your answer. c. Describe how the image relates to the pre-image in terms of: d. size, shape, orientation, and distance from the center of dilation. e. State whether the transformation is an enlargement or a reduction and explain why. 3. Two Dilations of the Same Figure The center of dilation for both Image A and Image B is the origin, O(0, 0). Pre-image: DEF: D(4, 2), E(6, 2), F(5, 6) Image A ($D_1E_1F_1$): $D_1(8, 4)$, $E_1(12, 4)$, $F_1(10, 12)$ Image B: $D_2(2, 1)$, $E_2(3, 1)$, $F_2(2.5, 3)$ Answer the following: a. Identify which image represents an enlargement and which represents a reduction. b. Determine the scale factor for Image A and Image B. Show your solution. c. Compare the two images. Explain how the scale factor affects the size and position of each image relative to the pre-image. d. Explain your answer using the terms increase, decrease, proportional, and center of dilation.
V. Reflection:	Answer each question. 1. How did identifying the center of dilation help in determining the image and scale factor of the transformation? 2. What mathematical evidence did you use to decide whether a dilation was an enlargement or a reduction? 3. Why is it important to use precise mathematical terms such as scale factor, center of dilation, proportional, enlargement, and reduction when explaining a dilation?

I. Activity 3:	Extending and Reinforcing Activities																		
II. Materials:	pen and paper																		
III. Instructions:	Option 1. Online Simulation																		
	Instructions:																		
	1. Open GeoGebra App).																		
	2. Observe a triangle and the given center of dilation																		
	3. Change the scale factor using the slider.																		
	4. For each scale factor, complete the table below:																		
	<table><tr><td>Scale Factor</td><td>Image Size (Enlarged/ Reduced)</td><td>Orientation (Same/Opposite)</td><td>Location of Image</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></table>				Scale Factor	Image Size (Enlarged/ Reduced)	Orientation (Same/Opposite)	Location of Image											
Scale Factor	Image Size (Enlarged/ Reduced)	Orientation (Same/Opposite)	Location of Image																
	5. Answer the guide questions:																		
	a. What happens when the scale factor is greater than 1?																		
	b. What happens when the scale factor is between 0 and 1?																		
	c. What happens when the scale factor is negative?																		

	Option 2. Design Resizing Task Instructions: 1. Use resized versions of a printed or digital design (e.g., a logo, a Filipino textile pattern, or a poster element): one 50% reduction and one 200% enlargement, both keeping the original proportions. 2. Place the original, the 50% version, and the 200% version side by side on a single sheet. 3. Identify and label: a. Center of dilation (a fixed reference point used for the resizing) b. Scale factor ($k = 0.5$ for the reduction; $k = 2$ for the enlargement) c. Type: Enlargement or Reduction 4. Answer: a. What stayed the same across the original and the two resized versions? b. What changed, and how does the scale factor describe that change? Option 3. Group Quiz Game: Instructions: 1. Form groups of members. 2. Each group will be shown a figure and its image. 3. For each item, write your answer on a mini-board: a. Scale factor b. Center of dilation (given choices) c. Type (Enlargement/Reduction)
V. Reflection:	Answer each question. 1. What patterns did you observe when the scale factor was greater than 1, between 0 and 1, and negative? 2. How did the center of dilation influence the location of the image after the transformation? 3. Why is it important to preserve proportions when enlarging or reducing figures in real-life applications such as logos, blueprints, and digital designs?
*****END OF LESSON 1.3*****	

Lesson 1.4 DILATION GRAPH	
Objectives:	At the of the lesson, you will be able to: 1. determine the scale factor used in dilating a figure. 2. perform dilation of points and figures on the coordinate plane. 3. illustrate dilated images correctly using graphing techniques. 4. compare corresponding side lengths of original and dilated figures.
Key Ideas and Examples	A dilation is a transformation that changes the size of a figure but not its shape. It is defined by a center of dilation (C) and a scale factor (k). The key rule is that the position of the new point P' is determined by the vector relationship: $\vec{CP'} = k \cdot \vec{CP}$ This means P' lies on the line passing through C and P. The distance from C to P' is k times the distance from C to P. Our illustration provides three distinct scenarios using the same initial setup: Reduction ($0 < k < 1$): When the magnitude of k is between 0 and 1, the points move closer to the center. As shown in the left panel ($k = 0.5$), the entire figure shrinks by half. Unchanged ($k = 1$): When the scale factor is exactly 1, the mapping CP' means every point P' lands exactly on its original point P. The figure is invariant. Enlargement ($k > 1$): When k is greater than 1, the points move further away from the center. In the right panel ($k = 1.5$), the entire figure grows by 50%. The new points P' are 1.5 times farther from C than the original points P.

The diagram also includes the coordinate formulas you requested, showing how the math applies for an arbitrary center and the simplification when the center is.



Video Lesson:

<https://www.khanacademy.org/math/geometry/hs-geo-transformations/hs-geo-transformations-intro/v/introduction-to-transformations>

I. Activity 1:

Practicing Learned Skills

II. Materials:

pen and paper

III. Instructions:

Option 1. Partner Challenge

1. Work with a partner.
2. Plot each given figure on graph paper.
3. Use the given center of dilation and scale factor to draw the image.
4. Compare your answers with your partner and check for accuracy.

Figure 1: Triangle

Center of dilation:

Scale factor:

Figure 2: Triangle

Center of dilation:

Scale factor:

Figure 3: Quadrilateral

Center of dilation:

Scale factor:

Option 2. Reverse Dilation

1. Study each dilated figure on the coordinate plane.
2. Identify the center of dilation.
3. Determine the coordinates of the original figure. Hint: $P = (P' - \text{center})/k + \text{center}$. For a center at the origin, this simplifies to $P = P'/k$.
4. Explain your solution in your group.

Figure 1

Image points:

Center:

Scale factor: $k = 2$. Determine the coordinates of A, B, C.

Figure 2

Image points:

Center:

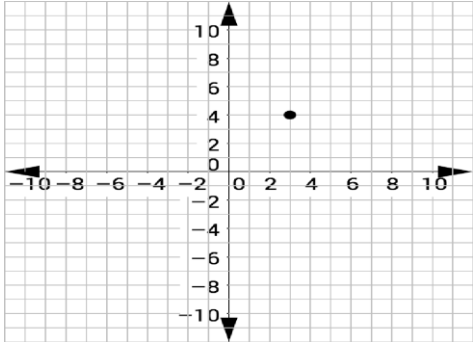
Scale factor: $k = 1/2$. Determine the coordinates of D, E, F.

	Figure 3 Image points: Center: Scale factor: $k = 3$. Determine the coordinates of P, Q, R.
V. Reflection:	Answer each questions. 1. How does the center of dilation help determine the location of an image after a dilation transformation? 2. What patterns did you observe about the size and position of figures when the scale factor was greater than 1 and when it was between 0 and 1? 3. How did the activities help deepen your understanding of dilation as a transformation, and where might this concept be useful in real-life situations?

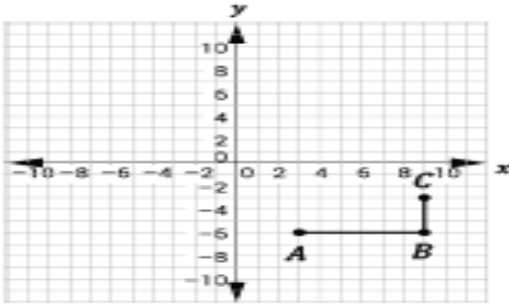
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	worksheet or paper, pencil or pen
III. Instructions:	Read each item carefully and complete all tasks. Show your solutions and draw your answers on graph paper when needed. 1. Determine scale factors Given triangle ABC and its image a. Determine the scale factor. b. State whether it is an enlargement or reduction. 2. Perform Dilation on the Coordinate Plane Triangle is given below: Center of dilation: Scale factor: a. Dilate the triangle. b. Write the coordinates of D', E', and F'. c. Draw the original and image on graph paper. 3. Compare Corresponding Side Lengths Triangle has the following side lengths: The triangle is dilated using a scale factor of . a. Find the lengths of , , and . b. Compare the original and image side lengths. c. What relationship do you observe? 4. Identify Corresponding Sides and Verify Dilation Given: Triangle $\rightarrow$ Triangle a. Identify corresponding sides. b. Compute the scale factor. c. Verify if the dilation is consistent
V. Reflection:	Answer the question: How can the concept of dilation be applied in real-life situations such as map scaling, architectural drawings, or digital image resizing?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	pen and paper, GeoGebra
III. Instructions:	1. Use GeoGebra to create dynamic dilations of polygons and shapes. 2. Optional challenge: Dilate generic Filipino-inspired geometric motifs digitally (e.g., simplified parol star, banig grid pattern, or Cordillera weave geometric units) and calculate scale factors. Avoid reproducing registered brand logos. 3. Discuss real-life examples where scale and proportionality are critical (architecture, clothing patterns, sports field layouts). 4. Optional enrichment for advanced learners (Negative Scale Factor). A dilation by negative scale factor k can be understood as a positive-k dilation followed by a 180-degree rotation about the center, which is equivalent to a point reflection through the center. This is treated formally in Finite Mathematics 2 / college geometry. Try: dilate triangle XYZ with X(2, 1), Y(4, 1), Z(3, 3) by $k = -1$ about O(0, 0) and describe the image
IV. Reflection:	Answer each question.

	1. How did using GeoGebra help you visualize and better understand the effect of different scale factors on the size and position of a figure during dilation? 2. What did you learn about the importance of scale and proportionality from the real-life examples discussed (e.g., architecture, clothing design, sports field layouts, or cultural geometric patterns)? 3. How did the enrichment activity deepen your understanding of dilation, particularly when using a negative scale factor? What similarities and differences did you observe between dilations with positive and negative scale factors?
*****END OF LESSON 1.4*****	

Lesson 1.5. ROTATION	
Objectives:	At the end of the lesson, the learners are will be able to 1. define rotation as a transformation that turns a figure about a fixed point. 2. identify the center of rotation in given situations. 3. determine the angle and direction of rotation (clockwise or counterclockwise). 4. illustrate rotation of geometric figures on the coordinate plane.
Key Ideas and Examples	A rotation is a rigid motion that turns every point of a figure through a given angle around a fixed point called the center of rotation. The image is congruent to the preimage. The distance of every point from the center stays the same. Size, shape, and angle measures are preserved. For rotations about the origin, the four rules above (90° CCW, 90° CW, 180°, 270° CCW) apply. A rotation changes the position of a figure by moving it around a fixed point through a specified angle. A clockwise rotation turns the figure in the same direction as the hands of a clock; a counterclockwise rotation turns it in the opposite direction. By mathematical convention, a positive angle measure denotes counterclockwise rotation, and a negative angle measure denotes clockwise rotation. • Center of rotation – the fixed point around which the figure turns • Angle of rotation – degree measure of turn Direction of rotation – clockwise or counterclockwise  Visualizing 90° Standard Rotations (Counterclockwise) If we take your point $P(3, 4)$ and rotate it around the origin $(0, 0)$ in increments of counterclockwise, look at how the coordinates shift: Rotation of 90° counterclockwise: * <i>Concept:</i> The point swings from the first quadrant into the second quadrant. The horizontal and vertical distances swap. <i>Rule:</i> $(x, y) \rightarrow (-y, x)$ <i>Your Point:</i> $P(3, 4) \rightarrow P'(-4, 3)$ Rotation of 180° (either direction): * <i>Concept:</i> The point travels in a straight line directly through the origin into the diagonally opposite quadrant (the third quadrant). The values stay in the same place, but their signs flip. <i>Rule:</i> $(x, y) \rightarrow (-x, -y)$ <i>Your Point:</i> $P(3, 4) \rightarrow P'(-3, -4)$

	Rotation of counterclockwise (or clockwise): * <i>Concept:</i> The point swings all the way down into the fourth quadrant. <i>Rule:</i> $(x, y) \rightarrow (y, -x)$ <i>Your Point:</i> $P(3, 4) \rightarrow P'(4, -3)$
Video Lesson:	https://www.khanacademy.org/math/geometry/hs-geo-transformations/hs-geo-transformations-intro/v/introduction-to-transformations

I. Activity 1:	Practicing Learned Skills
II. Materials:	graph paper, pencil, ruler
III. Instructions:	Part I. On graph paper, plot each figure and perform the indicated rotations about the given center. Record the new coordinates and sketch each image. Use the rotation rules from B.2. Figure 1: Triangle ABC with $A(2, 0)$, $B(2, 3)$, $C(5, 0)$; center $O(0, 0)$. Rotate 90° CCW, then 180°, then 90° CW. List the new coordinates after each rotation and sketch each image. Figure 2: Square DEFG with $D(1, 1)$, $E(3, 1)$, $F(3, 3)$, $G(1, 3)$; center $C(2, 2)$. Rotate 90° CCW, then 180°. Record the new coordinates after each rotation and sketch each image. Figure 3: Triangle PQR with $P(0, 0)$, $Q(4, 0)$, $R(2, 2)$; center $O(0, 0)$. Rotate 270° CW. Note: 270° CW is equivalent to 90° CCW Part II. Consider the following graph. The coordinate grid displays geometric figure ABC situated in the fourth quadrant. The vertices are mapped at connected perpendicular coordinates: $A(3, -6)$, $B(9, -6)$, and $C(9, -3)$.  a. Rotate figure ABC 270° counterclockwise about the origin. Graph the new figure on the coordinate plane, and complete each blank below with the appropriate coordinates. $A(\quad , \quad) \rightarrow A'(\quad , \quad)$ $B(\quad , \quad) \rightarrow B'(\quad , \quad)$ $C(\quad , \quad) \rightarrow C'(\quad , \quad)$ b. Rotate figure ABC 90° clockwise about the origin. Graph the new figure on the coordinate plane and complete each blank below with the appropriate coordinates. $A(\quad , \quad) \rightarrow A'(\quad , \quad)$ $B(\quad , \quad) \rightarrow B'(\quad , \quad)$ $C(\quad , \quad) \rightarrow C'(\quad , \quad)$
V. Reflection:	Answer each question. - How do you know the figure was rotated rather than reflected? - What distance from the center stayed the same? - How does the angle of rotation determine the new location? - Why does rotation preserve size and shape?

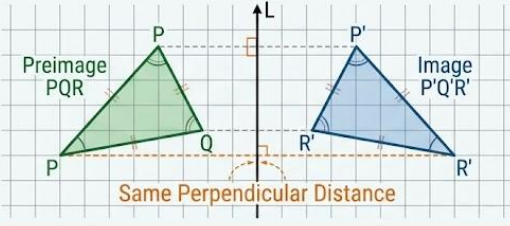
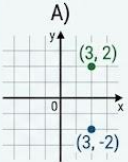
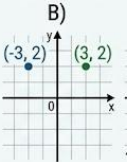
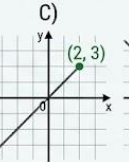
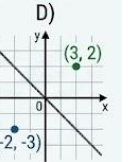
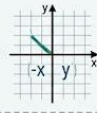
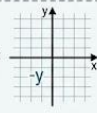
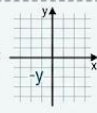
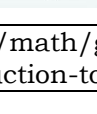
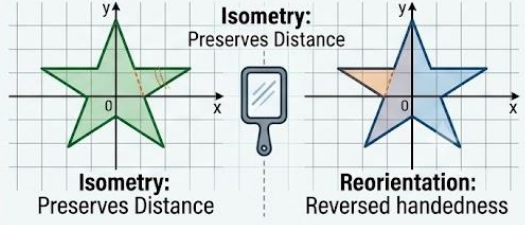
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	Part I. Solve each item. Show your solution and draw both the preimage and the image on graph paper, with the center $O(0, 0)$ clearly labeled. - Complete: A rotation is a transformation that turns a figure around a fixed point. - Rotate $P(3, 4)$ 90° CCW about the origin. Find P'.

	3. Rotate Q(3, 4) 90° CW about the origin. Find Q'. 4. Rotate R(3, 4) 180° about the origin. Find R'. 5. Triangle with vertices A(1, 2), B(4, 2), C(2, 5) is rotated 90° CW about the origin. Find A', B', C' and draw the preimage and image on the same coordinate plane with O(0, 0) labeled. Part II. Work in groups of 3–4. Solve the challenge as fast and accurately as possible. A rectangle has vertices, after rotation, the new vertices are: - Identify the center of rotation. - Determine the angle of rotation. - Identify the direction (clockwise or counterclockwise). - Explain how you got your answer.
V. Reflection:	Answer each question. - How did the coordinates of a point change when it was rotated by 90° clockwise, 90° counterclockwise, and 180° about the origin? What patterns did you observe? - How did drawing the preimage and image on the coordinate plane help you understand the relationship between the figure, the center of rotation, and the angle of rotation? - What strategies did your group use to identify the center, angle, and direction of rotation in the challenge activity, and how did these strategies help you solve the problem accurately?

I. Activity 3:	Extending and Reinforcing Learning												
II. Materials:	pen and paper, GeoGebra												
III. Instructions:	1. Open GeoGebra. 2. Plot the following figures on the coordinate plane: a. Figure 1: Triangle b. Figure 2: Square 3. Perform the following rotations about the origin: a. Rotate each figure 90° counterclockwise. b. Rotate the result again counterclockwise (total:) c. Rotate once more counterclockwise (total:) 4. Record your observations in the table: <table><tr><td>Rotation</td><td>What happens to the figure?</td><td>What stays the same?</td></tr><tr><td>90°</td><td></td><td></td></tr><tr><td>180°</td><td></td><td></td></tr><tr><td>270°</td><td></td><td></td></tr></table> 5. Answer the questions: a. What pattern do you notice after repeated rotations? b. After how many rotations do the figure return to its original position?	Rotation	What happens to the figure?	What stays the same?	90°			180°			270°		
Rotation	What happens to the figure?	What stays the same?											
90°													
180°													
270°													
V. Reflection:	Answer each question: 1. Why does a figure return to its original position after four consecutive 90° rotations about the same center? 2. How can recognizing rotation patterns help you predict the image of a figure without graphing every step? 3. Where might repeated rotations be observed or applied in real life, such as in art, engineering, architecture, logos, or technology? Explain your answer.												

*****END OF LESSON 1.5*****

Lesson 1.6. REFLECTION	
Objectives:	At the end of the lesson, you will be able to: - define reflection as a transformation producing a mirror image. - identify the line of symmetry in given figures. - perform reflection across the x-axis, y-axis, and other lines. - Illustrate reflected images accurately on the coordinate plane.

Key Ideas and Examples	A reflection is a rigid transformation that produces a mirror image of a figure across a line called the <i>line of reflection</i>. A reflection preserves size, shape, side lengths, and angle measures, so the image is congruent to the preimage. However, the <i>orientation is reversed</i>. Each point and its image are the same perpendicular distance from the line of reflection. The function $r_L(x, y)$ denotes the reflection of the point (x, y) across the line L. For example, $r_{x\text{-axis}}(3, 2)$ denotes the reflection of $(3, 2)$ across the x-axis. Computing the four reflections of $(3, 2)$: $r_{x\text{-axis}}(3, 2) = (3, -2)$; $r_{y\text{-axis}}(3, 2) = (-3, 2)$; $r_{y=x}(3, 2) = (2, 3)$; $r_{y=-x}(3, 2) = (-2, -3)$. GEOMETRIC REFLECTIONS: PRESERVING CONGRUENCE, REVERSING ORIENTATION DEFINITION AND PROPERTIES  REFLECTIONS OF A SINGLE POINT (3, 2) A)  A) Across x-axis $y = (3, -2)$ B)  B) Across y-axis $x = (-3, 2)$ C)  C) Across line $y=x = (2, 3)$ D)  D) Across line $y=-x$ $y=x = (-2, -3)$ COMPUTING REFLECTIONS (FORMULA SUMMARY) $r_{x\text{-axis}}(x, y) = (x, -y)$  $r_{y\text{-axis}}(x, y) = (-x, y)$  $r_{y=x}(x, y) = (y, x)$  $r_{y=-x}(x, y) = (-y, -x)$  VISUALIZING RIGID MOTION  Isometry: Preserves Distance Reorientation: Reversed handedness
Video Lesson:	https://www.khanacademy.org/math/geometry/hs-geo-transformations/hs-geo-transformations-intro/v/introduction-to-transformations

I. Activity 1:	Practicing Learned Skills
II. Materials:	pen and paper
III. Instructions:	Part I. Draw the following: Design 1: an L-shape with vertices $(1, 1), (1, 4), (3, 4), (3, 2), (4, 2), (4, 1)$; reflect across the y-axis. Design 2: a triangular pennant with vertices $(-2, 1), (-5, 3), (-2, 5)$; reflect across the line $y = x$. 1. Identify the line of symmetry or reflection axis. 2. Reflect the figure across the given line ($-axis, -axis$, or a vertical/horizontal line). 3. Draw both the original and reflected image on the coordinate plane. 4. Label all points clearly (and). 5. Compare your answer with another group. Part II. Solve the problems below. Show your solution on graph paper. 1. Reflect the point across the x-axis. 2. Reflect the point across the y-axis. 3. Reflect triangle across the y-axis. Draw both original and image.
V. Reflection:	Answer each question. 1. What properties of a figure remain unchanged after a reflection, and what property changes? 2. How did comparing your group's work with another group's work help strengthen your understanding of reflection? 3. If a figure is reflected twice across the same line, what do you think will happen to the figure? Explain your reasoning.

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	graph paper, pencil or pen, ruler
III. Instructions:	1. Read each item carefully.

	2. Answer using correct coordinates and clear graphing on coordinate plane. 3. Show your solution for each item. a. Complete the sentence: Reflection is a transformation that produces a _____ image of a figure. b. When a figure is folded along a line so that both halves match perfectly, what is that line called? In your own words, define the line of symmetry. c. Reflect the point across: • x -axis • y -axis a. Write the new coordinates. Triangle has the following coordinates: • Reflect the triangle across the y-axis. • Plot both original and reflected triangles on the coordinate plane. • Label all points clearly
V. Reflection:	Answer each question. 1. Why is a reflected image often described as a mirror image? How did the activity demonstrate this idea? 2. How does the line of symmetry or line of reflection help determine the location of corresponding points in a reflected figure? 3. How can the rules of reflection be applied in real-life situations such as mirror designs, architecture, art, or logo creation?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	worksheet or paper, pencil or pen, GeoGebra
III. Instructions:	1. Work in teams of 3 to 4 learners. 2. Each team receives the same number of reflection problems with equal difficulty or each item: 3. Predict the reflected image across the given line a. Draw your answer on the graph b. Fold the paper to check accuracy (or use tracing paper if available) c. Score 1 point for each correct reflection 4. The team with the highest score wins. 5. Game Rules: a. 1 round = 1 reflection problem b. No erasing during checking phase c. Must show complete working before folding
V. Reflection:	Answer each question. 1. Why is the line of reflection important when predicting the location of image points? Explain using examples from the game. 2. What mistakes were most common during the activity, and how can understanding reflection rules help avoid them? 3. How did working as a team help improve your understanding of reflections and symmetry on the coordinate plane?
*****END OF LESSON 1.6****	

Lesson 1.7. TILING AND TESSELLATION	
Objectives:	At the end of the lesson, you will be able to: 1. define tessellation. 2. identify polygons that can tessellate the plane (e.g., triangles, squares, hexagons) 3. generate tessellations using geometric transformation. 4. describe the vertex configuration of a tessellation and justify the tessellation using an interior-angle computation. 5. explain how translation, rotation, or reflection generates a tessellation with no gaps and no overlaps
Key Ideas and Examples:	Tessellation-a repetition of geometric figures on a two-dimensional plane that has no overlaps or gaps. What tessellates?

When we say that a polygon tessellates the plane, we mean that:

- Copies of the polygon can be repeated endlessly
- They fit together perfectly
- They leave no empty spaces and do not overlap.

Important rule:

If the angles add up to more than 360, the shapes will overlap. If the angles add up to less than 360, gaps will appear.

Translation means sliding a shape from one position to another without turning or flipping it.

- The shape keeps the same orientation.
- All points of the shape move the same distance in the same direction.

Rotation means turning a shape around a fixed point called the center of rotation.

- The shape turns by a certain angle (e.g., 60°, 90°, 120°)
- The shape remains congruent to the original.

Reflection means flipping a shape over a line, creating a mirror image. The reflected shape is reversed but remains the same size and shape.

Tessellating Polygons

A square has an interior angle of **90°**.

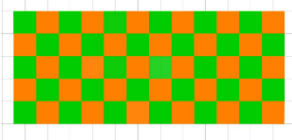
At a vertex:

$$90^\circ + 90^\circ + 90^\circ + 90^\circ = 360^\circ$$

✓ No gaps

✓ No overlaps

Transformation Used: Translation (sliding the square repeatedly)



An equilateral triangle has an interior angle of 60°.

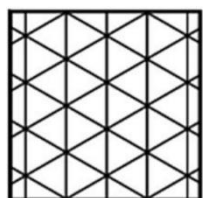
At a vertex:

$$60^\circ \times 6 = 360^\circ$$

✓ Six triangles meet perfectly

Transformations Used:

Rotation + Translation



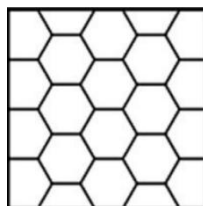
A regular hexagon has an interior angle of 120°.

At a vertex:

$$120^\circ + 120^\circ + 120^\circ = 360^\circ$$

✓ Common in nature (e.g., honeycomb pattern)

Transformation Used: Translation



Non- Tessellating Polygon

A regular pentagon has an interior angle of 108°.

Try fitting them:

- $108^\circ \times 3 = 324^\circ \rightarrow$ gap
- $108^\circ \times 4 = 432^\circ \rightarrow$ overlap

	Does NOT tessellate
Video Lessons:	Khan Academy. (n.d.). <i>Introduction to Transformations</i> [Video]. YouTube. https://www.youtube.com/watch?v=XiAoUDfrar0 Visual Impact Systems – Advertising and Marketing. (n.d.). <i>M.C. Escher – How to Create a Tessellation</i> [Video]. YouTube. https://www.youtube.com/watch?v=ZNVyrxdlrGQ

I. Activity No. 1	Practicing Learned Skills
II. Objectives:	At the end of the activity, you are expected to: 1. construct a tessellation using a chosen regular polygon; 2. apply at least two geometric transformations (translation, rotation, reflection); and 3. explain mathematically why the tessellation works using the vertex rule
III. Materials:	graph paper or a digital tool like GeoGebra
IV. Instructions:	Part I. Using graph paper or a digital tool like GeoGebra, choose one regular polygon (equilateral triangle, square, or regular hexagon) and perform the following: 1. Select a Base Shape: Draw your starting polygon. 2. Apply Transformations: Use at least two different types of geometric transformations (Translation, Rotation, or Reflection) to begin covering the plane. Label each transformation directly on your design. 3. Expand the Pattern: Continue the transformations until you have created a 5x5 grid or a cluster of at least 10 repeating shapes with no gaps and no overlaps. Part II. Underneath your design, complete the "Designer's Log" to prove your pattern works mathematically: • The Vertex Rule: Identify a single central point (vertex) where your shapes meet. List the interior angles of the shapes touching that point Sample Computation: $\rightarrow \angle 1 + \angle 2 + \angle 3 + \dots = 360^\circ$ • Transformation Map: Describe exactly how you moved the shape. ○ <i>Example:</i> "I translated the square 4 units to the right, then reflected the resulting row downward." Part 3: Peer Verification Swap your design with a partner. The partner must: • Check for any gaps or overlaps. • Verify if the interior angles at the vertex truly sum to 360°. Identify if the polygon used is a factor of 360° (e.g., 60° for triangles, 90° for squares, 120° for hexagons)
V. Reflection:	Answer each question, 1. Which part of the activity was easy for you to understand? Why? 2. Which part did you find difficult or confusing (e.g., applying transformations, explaining the vertex rule)? 3. What questions do you still have about tessellations or transformations?

I. Activity 2:	Assessing Learning Outcomes		
II. Materials:	pen and paper		
III. Instructions:	Part I. Identify whether the following polygons tessellate the plane. Write Yes or No and explain your answer.		
	Polygon	Tessellates?(Yes/No)	Reason
	Equilateral Triangle		
	Square		

	Regular Pentagon		
	Regular Hexagon		
	Regular Octagon		
	Part II. Analyze tessellation patterns and identify the geometric transformations used to generate the repeating design. 1. Study the tessellation patterns shown by the teacher or in the worksheet. For each pattern: 2. Identify the polygon used in the tessellation. 3. Determine the geometric transformation(s) used to generate the pattern. 4. Explain how the transformation helps create a repeating pattern without gaps or overlaps.		
IV. Reflection:	Answer each question. 1. Which questions were easy for you to answer? Why? 2. Which questions were challenging, and what made them difficult? 3. What concept do you still need help understanding?		

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	pen and paper
III. Instructions:	Remediation Activities Solve the following problems using the Vertex Rule, which states that the sum of the interior angles of polygons meeting at a single point (vertex) must equal exactly 360°. 1. A square has an interior angle of 90°. How many squares must meet at a vertex to form a tessellation? Show your computation. <i>Computation:</i> _____ 2. A regular pentagon has an interior angle of 108°. • If you place 3 pentagons together: $108^\circ \times 3 =$ _____ • If you place 4 pentagons together: $108^\circ \times 4 =$ _____ • Based on these totals, why does a regular pentagon fail to tessellate? 3. True or False: If the sum of angles at a vertex is 375°, the shapes will have gaps between them. • Answer: _____ • Correction (if False): _____ 4. Which of these sums allows for a perfect tessellation? a. 180° b. 270° c. 360° d. 400° Enrichment Activities 1. The "Escher" Method (Irregular Tessellations) Objective: Generate a non-polygonal tessellation by "nibbling" and "pasting." • The Activity: Start with a 3 x 3 inch square cardstock. • The Task: 1. Cut a unique shape out of the left side and translate (slide) it to the right side, taping it down. 2. Cut a shape from the top and translate it to the bottom. 3. Trace this new irregular "tile" repeatedly to cover a page. • The Challenge: Ask the LEARNER to identify why this irregular shape still satisfies the Vertex Rule. 2. Semi-Regular Tessellation Investigation Objective: Explore patterns that use more than one type of regular polygon. • The Activity: Provide a set of various regular polygons (triangles, squares, hexagons, dodecagons). • The Task: Find combinations of two or more different regular polygons that can surround a vertex and sum to 360°. ◦ <i>Example:</i> Two hexagons and two triangles ($120+120+60+60=360$). • Output: Create a "Transformation Map" showing how translation or rotation was used to propagate this multi-shape pattern across the plane.
V. Reflection:	Answer each question.

	1. How does the Vertex Rule help determine whether a shape or combination of shapes can form a tessellation? Use examples from the activities to support your answer. 2. What did you discover about the relationship between the interior angles of polygons and their ability to tessellate the plane without gaps or overlaps? 3. How were geometric transformations such as translation, rotation, or reflection used to create and extend tessellation patterns in the enrichment activities?
*****END OF LESSON 1.7*****	

Lesson 1.8. ESCHER-TYPE TESSELLATION	
Objectives:	At the end of the lesson, you will be able to: 1. define an Escher-type tessellation as a repeating artistic tessellation made by modifying a tessellating base tile into a figurative shape that interlocks without gaps or overlaps. 2. construct an Escher-type tessellation from a square, equilateral triangle, or regular hexagon by using a cut-and-attach process that preserves area and creates matching edges. 3. label the base tile, modified tile, cut edge, attached edge, and at least four interlocking copies of the modified tile. 4. write a short mathematical justification explaining why the modified tile still tessellates, using the ideas of area preservation and edge-matching.
Key Ideas and Examples:	What is an Escher-Type Tessellation? An Escher-type tessellation is an artistic, interlocking tiling where a regular polygon (such as a square, equilateral triangle, or hexagon) is modified using geometric transformations to create recognizable shapes like birds, fish, reptiles, or human figures. Named after the famous Dutch graphic artist M.C. Escher, these designs rely on a strict mathematical rule: whatever area you remove from one side of a tile, you must add back to another side. Because the total surface area remains perfectly invariant, the final stylized figures fit together flawlessly with zero gaps and zero overlaps. Three Fundamental Rules of Escher Tilings: Conservation of Area: The area of the final stylized tile must exactly equal the area of the base polygon template. Isometry (Rigid Motion): Transformations must preserve lengths and angle measures to ensure a seamless fit. Vertex Sums: The interior angles of the shapes meeting at any given intersection point must always sum up to precisely 360°. Geometric Transformation Methods Artists and mathematicians manipulate polygon boundaries using three main types of rigid geometric transformations to generate an endless variety of interlocking figures. 1. Translation Tessellations (The 'Nibble and Slide' Method) This technique typically starts with a square or a rectangle. You alter one side of the polygon by cutting out a unique boundary curve, then slide (translate) that exact modification directly across to the opposite parallel side where it is appended. Vector Mapping Rule: $T_{(w, 0)}(x, y) = (x + w, y)$ (where 'w' is the width of the baseline template square). 2. Rotation Tessellations (The 'Pivot' Method) Instead of sliding a cutout across to an opposite side, a rotation tessellation involves pivoting a modification around a shared vertex point (a corner) onto an adjacent side. This technique is heavily utilized with equilateral triangles (rotating 60°), squares (rotating 90°), and regular hexagons (rotating 120°). It often produces beautiful spiral arrangements of figures. 3. Reflection & Glide Reflection Tessellations A glide reflection combines a reflection over an axis line with a parallel translation (slide). Because a reflection flips the orientation of the shape, every alternating

	column or row of the tessellation will look like a mirror image ('left-handed' versus 'right-handed' versions of the same figure). This creates dynamic patterns where shapes appear to be marching or flying in alternating directions. Transformation Summary Comparison		
	Transformation Type	Base Polygon Options	Visual Behavior / Orientation
	Translation	Squares, Rectangles, Parallelograms	Tiles slide across; orientation stays completely identical.
	Rotation	Equilateral Triangles, Squares, Hexagons	Tiles pivot around corner vertices; creates rotational or circular symmetries.
	Glide Reflection	Squares, Rhombuses, Triangles	Tiles mirror and shift; flips orientation to produce alternating patterns.
Video Lesson	https://youtu.be/ZNVyrxdlrGQ		

I. Activity 1:	Practicing Learned Skills
II. Materials:	pen and paper
III. Instructions:	1. Take a square piece of scrap paper. 2. Cut a small “nibble” from one side of the square. 3. Transfer the cut-out piece by sliding it directly to the opposite side of the square. 4. Tape the piece in place. 5. Repeat Steps 2–4 for another pair of opposite sides. 6. Ensure that all movements are done by sliding only (translation)—do not flip or rotate the cut-out pieces. 7. Remember: flipping or rotating creates a different type of tessellation, which will be explored later in the enrichment activity. 8. Confirm that your modified shape has matching opposite sides. 9. Trace your shape repeatedly on a separate sheet of paper to form a repeating pattern. 10. Continue practicing until your tessellation forms a smooth, continuous, and gap-free design across the page.
V. Reflection:	Answer each question. 1. Why did you choose that specific design or shape for your tessellation? 2. How many sides did you modify in your design, and why did you choose that number? 3. What did you notice when your shape was repeated across the page?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	Read and follow each task carefully. 1. Sketch the Original Figure • Draw square ABCD with side length 6 cm. • Show and label the curved section cut from side AB. 2. Construct Tile T • Sketch the new shape after the cut piece has been attached to side DC. • Label the cut edge and the attached edge. 3. Create a Tessellation • Draw four interlocking copies of Tile T arranged in a 2 × 2 pattern. • Ensure that the tiles fit together without gaps or overlaps. 4. Explain Why the Tessellation Works • In 2–3 sentences, explain why Tile T can still tessellate the plane. • Use the terms area preservation and edge-matching in your explanation.

	5. Analyze an Alternative Situation - Suppose the cut piece was discarded instead of attached to the bottom edge. - Explain in one sentence what would happen to the tessellation and why.
V. Reflection:	Answer each question. - How did moving the cut piece from one side of the square to another affect the shape of the tile? Why was it still possible to create a tessellation without gaps or overlaps? - How did the concepts of area preservation and edge-matching help explain why Tile T could tessellate the plane successfully? - What did you learn about creating tessellations from a regular shape, and how might this process be used in art, design, architecture, or pattern-making?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	pen and paper, GeoGebra
III. Instructions:	- Draw a square or equilateral triangle on paper. - Cut a small shape from one side of the polygon. - Rotate the cut piece 180° about the midpoint of the same side from which it was cut. - Re-attach the rotated piece to the same side, on the opposite side of the midpoint. - The resulting shape now has matching curves on either side of the midpoint of that edge. Repeat steps 2 to 4 for the opposite edge if desired. - Trace the modified shape repeatedly across the page, rotating each tracing 180° about the corresponding midpoint to fill the plane. The shape tessellates by 180° rotation about the midpoint of each modified edge. Verify by tracing at least four copies; if gaps or overlaps appear, the construction is not valid. - Add details and color to turn your pattern into a creative design. - Write a short Artist Statement (3–5 sentences) explaining The transformation used (rotation)
V. Reflection:	Answer each question. - Why must the cut piece be rotated about the midpoint of the side before being reattached? How does this ensure that the edges will match during tessellation? - What evidence from your completed pattern shows that the tessellation is generated by rotational symmetry rather than another transformation? - How can rotational tessellations be used in real-world applications such as textile designs, floor patterns, wallpapers, logos, or visual art? Give specific examples.
***** END OF LESSON 1.8*****	

Lesson 1.9. FRIEZE PATTERNS	
Objectives:	By the end of the lesson, you should be able to: - define a Frieze pattern as a strip pattern that has translation symmetry along one direction. - identify the motif or generating unit and state the smallest translation vector that maps a given Frieze pattern onto itself. - identify additional symmetries in a Frieze pattern, such as vertical reflection, horizontal reflection, glide reflection, and half-turn rotation. - justify whether a given strip is or is not a Frieze pattern by checking for congruent motifs repeated at equal distances. - create a Frieze pattern strip with at least four motif repetitions and write a short mathematical justification of the transformations used.
Key Ideas and Examples:	A frieze pattern (or border pattern) is a mathematical design that repeats infinitely in a single linear direction. Commonly observed in architectural borders, ancient pottery, textile ribbons, and wallpaper trim, a frieze pattern is characterized by its unique set of symmetries. From a geometric standpoint, every frieze pattern maps onto itself via a minimum horizontal translation vector. This means that if you slide the entire pattern left or right by a specific distance, it will look completely identical to its original state.

The 4 Symmetries Governing Frieze Patterns:

1. Translation (T): The mandatory horizontal slide that shifts the pattern over by a fixed length.
2. Reflection (Horizontal or Vertical): Flipping the pattern across a horizontal centerline (H) or individual vertical lines (V).
3. Rotation (R): A half-turn (180° spin) around a series of specific midpoint vertex centers.
4. Glide Reflection (G): A combined transformation where the pattern slides horizontally and reflects over the horizontal axis simultaneously.

The Seven International Frieze Groups

In 1924, mathematician George Pólya demonstrated that despite an infinite variety of artistic details, there are exactly seven distinct mathematical ways to combine these symmetry operations. These seven categories are classified using the International Union of Crystallography (IUC) standard notation:

IUC Notation	Symmetry Elements	Structural Description / Behavior
p111	Translation only	The simplest group. Formed purely by shifting an asymmetric base shape repeatedly.
p1a1	Translation, Glide Reflection	Contains no individual reflections or rotations. Features staggered alternating figures.
p11m	Translation, Horiz. Reflection	The design is sliced horizontally into mirrored upper and lower halves.
p211	Translation, 180° Rotations	Features centers of rotation. Spinning the strip upside down perfectly reproduces the pattern.
pm11	Translation, Vert. Reflections	The border is divided into a series of vertical mirror lines, splitting individual segments left-to-right.
pma2	Trans., Vert. Refl., 180° Rot., Glide Refl.	Combines half-turns and vertical mirrors. Does not contain any horizontal line reflections.
pmm2	All Symmetries Active	The most complex group. Contains horizontal/vertical reflections, glide reflections, and 180° rotations.

Real-World and Historical Significance

Frieze patterns bridge the gap between abstract mathematics and creative arts. Historically, they are found on classical Greek temples (metopes and triglyphs), Roman friezes, and Native American basket weaving designs. In modern manufacturing, these repeating structures are critical in designing tire treads (to optimize traction and suppress rolling noise), automated conveyor belt textures, and repeating ironwork balustrades.

Video Lesson(	
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I. Activity 1:	Practicing Learned Skills
II. Materials:	paper, pen, drawing materials
III. Instructions:	Part I. The "Letter L" Analysis Challenge 1. Study the horizontal frieze strip below. A capital letter L is reflected over the horizontal midline of the strip and then slid 3 cm to the right. This exact "flip-and-slide" motion is repeated infinitely across the strip. 2. Analyze the pattern and answer the following questions in your groups: a. What single, composite transformation describes each individual "flip-and-slide" step? b. After exactly <i>one</i> flip-and-slide step, is the L upright or flipped? c. After <i>two</i> consecutive flip-and-slide steps, what happens to the orientation and position of the L? d. What is the length and direction of the smallest pure translation vector that maps this entire infinite strip onto itself? e. Explain mathematically why the answer to question 4 is <i>not</i> 3 cm. Part II. The Hands-On Design Challenge Now, you move from detective to designer. 1. Create your own original, asymmetric motif (like a comma, a footprint, or a jagged flag). 2. Using a ruler, construct a new frieze pattern by applying a 2 cm glide reflection. 3. On your final design, draw and state the exact length of the smallest pure translation vector required to map your pattern onto itself.
V. Reflection:	Answer each question. 1. How did the combination of reflection and translation create a glide reflection, and how did this transformation affect the orientation of the motif in the frieze pattern? 2. Why is the smallest pure translation vector different from the glide distance? Explain your answer using observations from the "Letter L" analysis challenge. 3. How did creating your own frieze pattern help you understand the relationship between symmetry, repetition, and geometric transformations?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	Read the each problem and perform the indicated tasks. Show reasoning, not only final answers. 1. A border repeats the same upright T every 4 cm along a horizontal strip. a. State he smallest translation vector. b. The letter T has a vertical line of symmetry. Where are the vertical reflection axes of the whole strip? 2. A letter L is reflected over the horizontal middle line of a strip and then slid 3 cm to the right. This flip-and-slide repeats. a. Name the transformation used in each step. b. State the smallest pure translation vector that maps the whole strip onto itself and explain why. 3. A flower border is arranged in a horizontal strip, but each flower is twice as large as the previous flower. Is this a Frieze pattern? Explain. 4. A triangle motif is rotated 180 degrees at varying intervals, with no consistent distance between motifs. Is this a Frieze pattern? Explain.
V. Reflection:	Answer each question. 1. How did identifying translations, reflections, and glide reflections help you determine whether a border pattern is a frieze pattern? 2. What conditions must a pattern satisfy to be considered a frieze pattern, and why do changes in size or inconsistent spacing prevent a pattern from being a frieze pattern?

	3. Among the patterns you analyzed, which transformation was easiest or most difficult to identify? Explain how you recognized the transformation and its effect on the pattern
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I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	pen and paper, GeoGebra
III. Instructions:	1. Look around your home, school, or neighborhood and find a Frieze pattern. 2. Take a clear photo of the pattern. 3. Upload the photo to the shared class folder. 4. Write a short caption that includes: a. The motif or generating unit b. The Frieze type (if identified) c. The translation vector or repeated distance and direction d. Additional symmetries present (reflection, rotation, glide reflection), if any e. One sentence explaining why the pattern is or is not a Frieze pattern 5. Submit your work on or before the start of the next class session. Non-Digital Option 1. If you are unable to take a photo: 2. Draw or sketch a Frieze pattern observed at home, in school, or in your community. a. Write a caption that includes: b. The motif or generating unit c. The estimated translation vector or repeated distance and direction d. Additional symmetries present, if any e. One sentence explaining why the pattern is or is not a Frieze pattern 3. Submit your sketch and caption during the next class session.
V. Reflection:	Answer each question. 1. How did identifying translations, reflections, and glide reflections help you determine whether a border pattern is a frieze pattern? 2. What conditions must a pattern satisfy to be considered a frieze pattern, and why do changes in size or inconsistent spacing prevent a pattern from being a frieze pattern
*****END OF LESSON 1.9*****	

Unit 1. SYMMETRIES IN NATURE AND ARTS

Lesson 2.1. THE GOLDEN RATIO

Objectives:	At the end of the lesson, you should be able to: 1. describe the golden ratio, golden rectangle, golden spiral, and golden angle using appropriate mathematical language and symbols and as observed in nature, architecture, and artworks. 2. explain the relationships among the golden ratio, golden rectangle, golden spiral, and golden angle in terms of proportion and pattern formation. 3. present real-life examples of the golden ratio in plants, shells, animals, architecture, and artworks based on their patterns and proportions. 4. construct models of the golden rectangle and golden spiral, and represent the golden angle using
Key Ideas:	The Golden Ratio in Nature and Art The Golden Ratio—often represented by the Greek letter Phi (Φ) is a mathematical constant approximately equal to 1.618. Two quantities are in the Golden Ratio if the ratio of their sum to the larger quantity is equal to the ratio of the larger quantity to the smaller one. Mathematically, it is expressed as: $\frac{a+b}{a} = \frac{a}{b} = \Phi$ By setting $\frac{a}{b} = \Phi$, we can substitute it back into the equation:

$$1 + \frac{a}{b} = \Phi \Rightarrow 1 + \frac{1}{\Phi} = \Phi$$

Multiplying the entire equation by Φ yields the characteristic quadratic equation:

$$\Phi^2 - \Phi - 1 = 0$$

Applying the quadratic formula and selecting the positive real root, we find the exact value of :

$$\Phi = \frac{1 + \sqrt{5}}{2} \approx 1.618..$$

When this ratio is applied geometrically, it creates the **Golden Rectangle**, **Golden Spiral**, and **Golden Angle**.

1. The Golden Rectangle

A **Golden Rectangle** is a rectangle whose side lengths are in the golden ratio, meaning the ratio of the width to the height is exactly $1:\Phi$.

Mathematical Property: If a Golden Rectangle has dimensions , partitioning a perfect square $a \times a$ of dimensions $a \times (a + b)$ from it leaves a remaining sub-rectangle with dimensions $b \times a$. The ratio of the sides of this new, smaller rectangle remains perfectly proportional:

$$\frac{a}{b} = \Phi$$

This creates a nested, infinitely recurring sequence of smaller Golden Rectangles.

2. The Golden Spiral

Connecting the opposite corners of the nested squares inside a Golden Rectangle with a smooth arc generates the famous **Golden Spiral**.

Polar Equation: In polar coordinates (r, θ) , the mathematical equation for a Golden Spiral is: $r = ae^{b\theta}$

Where r is the distance from the origin, θ is the angle of rotation in radians, a is an arbitrary scaling factor, and the constant growth factor b is defined as: $b = \frac{\ln(\Phi)}{\frac{\pi}{2}} =$

$$\frac{2 \ln(\Phi)}{\pi} \approx 0.3063489$$

Because of this specific growth factor, the spiral grows wider by a factor of exactly $\Phi \approx 1.618$ for every quarter-turn (90° or $\frac{\pi}{2}$ radians) it completes.

3. The Golden Angle (Ψ)

The **Golden Angle** is created by taking the total circumference of a circle (2π radians) and dividing it into a longer arc and a shorter arc in the Golden Ratio ($\frac{a}{b} = \Phi$). The Golden Angle, denoted by Ψ (Psi), is the central angle subtended by the smaller arc b .

Radian Derivation: $\Psi = 2\pi \left(1 - \frac{1}{\Phi}\right) = 2\pi (2 - \Phi)$ radians

Degree Conversion: $\Psi = 360^\circ \times (2 - \Phi) \approx 137.50^\circ$

Golden Ratio Manifests Across Nature, Art, and Design.

1. The Golden Ratio in Nature

In nature, the Golden Ratio is less about a "mystical blueprint" and more about **efficiency**. It is closely tied to the **Fibonacci Sequence** (1, 1, 2, 3, 5, 8, 13...), where dividing any number by its predecessor gets closer and closer to 1.618 as the sequence progresses.

Nature utilizes this mathematical pattern to maximize space and energy consumption:

- **Phyllotaxis (Leaf and Seed Arrangement):** Look closely at a sunflower head or a pinecone. The seeds grow in clockwise and counterclockwise spirals. The number of spirals almost always equals two consecutive Fibonacci numbers (e.g., 34 and 55 spirals). This exact spacing, dictated by Φ , ensures that every seed gets maximum sunlight and room to grow without crowding out its neighbor.
- **Shells and Marine Life:** The chambered nautilus shell grows wider at a constant geometric rate, closely mimicking the logarithmic expansion of the Golden Spiral.

This allows the organism to grow larger without changing its fundamental shape or losing structural integrity.

- **Hurricanes and Galaxies:** On a grand scale, the spiral arms of a spinning hurricane and the vast swirling arms of spiral galaxies (like our Milky Way) naturally arrange themselves along a logarithmic spiral close to , matching the path of least resistance for moving matter and energy.

2. The Golden Ratio in Art and Architecture

Artists and architects throughout history have gravitated toward the Golden Ratio because the human eye processes images composed around it remarkably fast, subconsciously associating it with balance and beauty.

Classical Architecture

The Parthenon (Ancient Greece): While historians debate whether the Greeks used the ratio explicitly, a Golden Rectangle fits almost perfectly around the facade of the Parthenon. The spacing of its columns and the height of its structural elements heavily mirror these proportions.

Renaissance Masterpieces

Leonardo da Vinci: Da Vinci was fascinated by mathematics and illustrated a book titled *De Divina Proportione* ("On the Divine Proportion"). He intentionally incorporated the ratio into his masterpieces:

- **The Mona Lisa:** If you draw a Golden Rectangle over her face and a Golden Spiral starting from her wrist, it contours perfectly around her eye and face, guiding the viewer's gaze.
- **The Last Supper:** The architectural layout of the room, the alignment of the windows, and the positioning of Jesus and the disciples are mapped out using these geometric constraints.
- **Modern Art and Design**
- **Salvador Dalí:** In his painting *The Sacrament of the Last Supper*, Dalí explicitly painted the canvas to fit the exact dimensions of a Golden Rectangle. He also chose a massive, translucent regular dodecahedron (a 3D shape built from pentagons, which are deeply tied to the Golden Ratio) to float above the table.

Modern Branding: You see it every day in graphic design. Corporate giants like **Apple**, **Google** (in its G logo), and **Twitter/X** use intersecting circles derived from the Golden Ratio to construct their iconic logos, ensuring visual harmony and instantly recognizable symmetry.

Summary Table: Nature vs. Art

Domain	How Manifests	Primary Function
Nature (Botany)	Sunflower seeds, pinecones, leaf spacing	Efficiency: Maximizes space, sunlight, and structural density.
Nature (Physics)	Hurricanes, spiral galaxies	Dynamics: Follows the path of least resistance for moving energy.
Classical Art	Paintings (<i>Mona Lisa</i>), Architecture (Parthenon)	Aesthetics: Creates an intuitive sense of balance and focal harmony.
Modern Design	Tech logos, web layouts, product design	Clarity: Allows the human brain to scan and recognize shapes quickly.

The Golden Ratio serves as a striking bridge between abstract mathematics and the physical world—proving that what is functional in nature is often what we find most beautiful in art.

Video Lesson:

https://youtu.be/mVO2dcuR7P0?si=2k-5FT6E0ZB_vr6U

I. Activity No. 1	Practicing Learned Skills
II. Materials:	Paper and pen, Calculator (optional)
III. Instructions:	1. Prepare a ruler, compass, pencil, and graph paper. 2. Draw a square measuring 6 cm × 6 cm. 3. Locate and mark the midpoint of one side of the square. 4. Place the compass point on the marked midpoint and adjust the compass to reach the opposite corner of the square. 5. Draw an arc extending beyond the side of the square. 6. Use the intersection of the arc and the extended side to determine the length of the rectangle. 7. Complete the figure by drawing the remaining sides to form a rectangle. 8. Measure the length and width of the constructed rectangle. 9. Compute the ratio of the length to the width. 10. Compare the computed ratio with the approximate value of the Golden Ratio, 1.618. 11. Record observations and describe how the construction demonstrates the Golden Ratio
IV. Reflection:	Answer each question. 1. What pattern do you observe in the Fibonacci sequence as it continues? 2. What value do the ratios of consecutive Fibonacci numbers approach? 3. How is this value related to the golden ratio?

I. Activity No. 2	Assessing Learning Outcomes
II. Materials:	Answer sheet
III. Instructions:	Read each question carefully and choose the best answer. Write the letter of your answer. 1. What is the approximate value of the golden ratio? A. 1.414 B. 1.500 C. 1.618 D. 2.000 2. Which Greek letter is used to represent the golden ratio? A. α (alpha) B. β (beta) C. φ (phi) D. π (pi) 3. What defines a golden rectangle? A. A rectangle with equal sides B. A rectangle whose length-to-width ratio is about 1.618 C. A rectangle with four right angles D. A rectangle with parallel sides only 4. Which sequence is most closely related to the golden ratio? A. Arithmetic sequence B. Geometric sequence C. Fibonacci sequence D. Harmonic sequence 5. How is each term in the Fibonacci sequence generated? A. By multiplying the previous term by 2 B. By adding the two previous terms C. By squaring the previous term D. By subtracting the previous term 6. Which natural object commonly displays a pattern related to the golden spiral? A. Cubes B. Seashells C. Triangles D. Rectangular blocks 7. How is a golden spiral constructed using rectangles? A. By drawing circles inside triangles B. By drawing quarter circles inside golden rectangles

	C. By connecting random points D. By stacking identical squares only 8. Which field most commonly applies the golden ratio to create visually balanced designs? A. Architecture B. Painting C. Graphic design D. All of the above 9. If a rectangle has side lengths 8 and 13, what is the ratio of the longer side to the shorter side? A. 1.25 B. 1.50 C. 1.625 D. 2.00 10. If the width of a golden rectangle is 10 cm, what should be its approximate length? A. 12 cm B. 14 cm C. 16 cm D. 18 cm 11. Why is the golden ratio often considered visually pleasing in art and design? A. It creates random arrangements B. It produces balanced and harmonious proportions C. It increases object size D. It reduces detail in designs 12. How are the Fibonacci sequence and the golden ratio mathematically related? A. They are not related B. The ratio of consecutive Fibonacci numbers approaches the golden ratio C. Fibonacci numbers are multiples of the golden ratio D. Fibonacci numbers are equal to the golden ratio 13. If a rectangle has a ratio far from 1.618, what can be concluded about it? A. It is a perfect golden rectangle B. It is not a golden rectangle C. It is a square D. It forms a golden spiral 14. How does the golden angle contribute to pattern formation in nature? A. It creates random spacing B. It helps distribute elements evenly around a center C. It forms perfect squares D. It reduces growth patterns 15. Why do artists and architects use the golden ratio when designing structures or artworks? A. To make designs more complex B. To achieve visual balance and proportional harmony C. To reduce material costs D. To speed up construction
IV. Reflection:	Answer each question. 1. What is the golden ratio and why is it important? 2. How are the golden rectangle, spiral, and angle related to the golden ratio? 3. How can you apply the golden ratio in creating your own artwork?

I. Activity 3:	Extending and Reinforcing Learning				
II. Materials:	printed images or observations from surroundings				
III. Instructions:	For Beginner:				
	I. Choose the letter that matches the word to its meaning.				
	<table><tr><th>Word</th><th>Meaning</th></tr><tr><td>Golden Ratio</td><td>A. A special number about 1.618</td></tr></table>	Word	Meaning	Golden Ratio	A. A special number about 1.618
Word	Meaning				
Golden Ratio	A. A special number about 1.618				

	Spiral	B. A curve that goes around and around
	Rectangle	C. A shape with four sides

II. Complete the sentences:

1. The golden ratio is about _____.
2. A spiral looks like something that goes _____.
3. A rectangle has _____ sides.

III. Guided Drawing

1. Draw a square.
2. Inside the square, draw a curved line (like a quarter circle).
3. Repeat the curve to make a simple spiral.
(Teacher may demonstrate step-by-step.)

For Intermediate:
Explain the Connection

1. Compute the ratio of the following:
 - Example: $13 \div 8$
 - Compare your answer with ϕ (1.618).
2. Write a short explanation (3–5 sentences) describing:
 - How the golden rectangle is related to the golden ratio
 - How the golden spiral is formed
3. Complete the concept map linking:
 - Golden ratio $\rightarrow$ Golden rectangle $\rightarrow$ Golden spiral $\rightarrow$ Golden angle
 - Give **two real-life examples** where these patterns appear.

For Advanced:
Create and Illustrate

1. Use a ruler and compass to **construct a golden rectangle**:
2. Start with a square
3. Extend it using correct proportions
4. Divide the rectangle into squares and **draw a golden spiral** using arcs.
5. Draw a circle and **measure the golden angle (approximately 137.5°)**.
6. Label all parts clearly.

Reflection:	Answer each question: 1. How did your understanding of the golden ratio change as you moved from identifying basic meanings to constructing and illustrating geometric patterns? 2. What relationships did you observe among the golden ratio, golden rectangle, and golden spiral, and how did these relationships appear in your drawings or computations? 3. Where do you think patterns based on the golden ratio (such as spirals or golden rectangles) appear in real life, and why might artists, designers, or architects use them?
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*****END OF LESSON 2.1*****

*****END OF LESSON 2.1*****

Lesson 2.2. FIBONACCI SEQUENCE

Objectives:	At the end of the lesson, learners should be able to: - describe the Fibonacci sequence and describe the rule for generating its terms. - investigate and explain the historical origin of the Fibonacci sequence. - generate terms of the Fibonacci sequence. - identify occurrences of Fibonacci numbers in plants, flowers, shells, and artworks. - illustrate or present examples of Fibonacci sequences found in nature and art through drawings, diagrams, or digital media.
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Key Ideas:	The Fibonacci sequence is a sequence of numbers in which the first two terms are 0 and 1. Each succeeding term is obtained by adding the two preceding terms. In math notation, if F_n is a term in the sequence such that $n > 2$, F_n is defined as follows: $F_n = F_{n-1} + F_{n-2}$ The formula provides the pattern of the Fibonacci sequence, that is, a term ($n > 2$) is the sum of the two terms that immediately precede it. The sequence begins with 0 and 1, and the rest follows. The first 8 terms of the Fibonacci sequence is 0, 1, 1, 2, 3, 5, 8, 13 $F_1 = 0$ $F_2 = 1$ $F_3 = F_2 + F_1 = 1 + 0 = 1$ $F_4 = F_3 + F_2 = 1 + 1 = 2$ $F_5 = F_4 + F_3 = 2 + 1 = 3$ $F_6 = F_5 + F_4 = 3 + 2 = 5$ $F_7 = F_6 + F_5 = 5 + 3 = 8$ $F_8 = F_7 + F_6 = 8 + 5 = 13$																																										
Examples:	Example 1. Find the 10th and 13th term of the Fibonacci sequence. Solution: Create a table of values for n and F_n and use the pattern in finding the value of F_n. <table><tr><td>n</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td></tr><tr><td>F_n</td><td>0</td><td>1</td><td>1</td><td>2</td><td>3</td><td>5</td><td>8</td><td>13</td><td>21</td><td>34</td><td>55</td><td>89</td><td>144</td></tr></table> Therefore, $F_{10} = 34$ and $F_{13} = 144$. Example 2. Which of the following set of numbers is part of the Fibonacci sequence? A. 233, 377, 610 B. 144, 237, 381, 618 C. 987, 1597, 2584 Solution: Set A = 233, 377, 610 belongs to the Fibonacci sequence since $610 = 377 + 233$ and the earlier terms follow the starting pattern, that is, <table><tr><td>$377 - 233 = 144$</td><td>$21 - 13 = 8$</td></tr><tr><td>$233 - 144 = 89$</td><td>$13 - 8 = 5$</td></tr><tr><td>$144 - 89 = 55$</td><td>$8 - 5 = 3$</td></tr><tr><td>$89 - 55 = 34$</td><td>$5 - 3 = 2$</td></tr><tr><td>$55 - 34 = 21$</td><td>$3 - 2 = 1$</td></tr><tr><td>$34 - 21 = 13$</td><td>$2 - 1 = 1$</td></tr><tr><td></td><td>$1 - 1 = 0$</td></tr></table> Set B = 144, 237, 381, 618 does NOT belong to the Fibonacci sequence though the term was the sum of the two preceding terms, he it failed to follow the starting pattern of the Fibonacci sequence. Set C = 987, 1597, 258 is part of the Fibonacci sequence. If you subtract the first two terms, you will get a certain number. If you continue the process, you will obtain the starting pattern of the Fibonacci sequence, just like in Set A.	n	1	2	3	4	5	6	7	8	9	10	11	12	13	F_n	0	1	1	2	3	5	8	13	21	34	55	89	144	$377 - 233 = 144$	$21 - 13 = 8$	$233 - 144 = 89$	$13 - 8 = 5$	$144 - 89 = 55$	$8 - 5 = 3$	$89 - 55 = 34$	$5 - 3 = 2$	$55 - 34 = 21$	$3 - 2 = 1$	$34 - 21 = 13$	$2 - 1 = 1$		$1 - 1 = 0$
n	1	2	3	4	5	6	7	8	9	10	11	12	13																														
F_n	0	1	1	2	3	5	8	13	21	34	55	89	144																														
$377 - 233 = 144$	$21 - 13 = 8$																																										
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$144 - 89 = 55$	$8 - 5 = 3$																																										
$89 - 55 = 34$	$5 - 3 = 2$																																										
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$34 - 21 = 13$	$2 - 1 = 1$																																										
	$1 - 1 = 0$																																										
Video Lesson:	https://youtu.be/mVO2dcuR7P0?si=2k-5FT6E0ZB_vr6U																																										

I. Activity 1:	Practicing Learned Skills
II. Materials:	worksheet or paper and pen, calculator (optional)
III. Instructions:	Task 1. Fibonacci Challenge During a mathematics class, the teacher asked the LEARNERS to generate terms of the Fibonacci sequence greater than 200. The teacher reminded them that in the Fibonacci sequence, each term is obtained by adding the two preceding terms. After a few minutes, two learners presented their answers: • Maria's sequence: 233, 377, 610, 987 • Alicia's sequence: 377, 612, 989, 1601 Who generated the correct Fibonacci sequence with a value greater than 200, Maria or Alicia? Task 2. His Story of Fibonacci Perform a short skit where one LEARNER acts as a "Fibonacci expert" and explains the historical origin of the Fibonacci sequence to a group of LEARNERS from another academic strand. The expert should discuss Leonardo of Pisa, the rabbit population problem from <i>Liber Abaci</i>, how this problem led to the discovery of the Fibonacci sequence, and its significance in nature and art. Make the presentation clear, engaging, and easy to understand, using examples when possible. Task 3. Nature Pattern Hunt Identify at least five natural objects or images (from your surroundings or from the teacher-provided materials) that exhibit Fibonacci patterns or sequences. Task 4. Fibonacci Art Create a drawing, poster, structural design, or digital artwork that shows a Fibonacci pattern and/or the Golden Ratio. Include labels indicating the Fibonacci numbers and Golden Ratio used in your design. Task 5. Fibonacci Journal Create a journal entry highlighting the key concepts of the Fibonacci sequence. Each line of your entry should contain a number of words corresponding to the terms in the Fibonacci sequence. Task 6. Jingle Presentation Perform a choreographed jingle presentation summarizing the key concepts of Fibonacci sequence. You may adapt a tune for your jingle.
IV. Reflection:	Answer each question. 1. How did the different activities (sequence analysis, skit, nature hunt, art, journal, and jingle) help you understand the Fibonacci sequence in both mathematical and real-world contexts? 2. What patterns or relationships did you notice between the Fibonacci sequence and natural or artistic forms, and how did these patterns support your understanding of the concept? 3. Which activity helped you understand the Fibonacci sequence most deeply, and how did it change or strengthen your thinking about how mathematics appears in everyday life?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	Part I. Fibonacci Pattern Evaluation Instructions: 1. Examine each image carefully and determine whether it shows a Fibonacci sequence pattern or an approximation of the Golden Ratio. 2. In the explanation column, briefly describe why the image is considered a Fibonacci-related example.

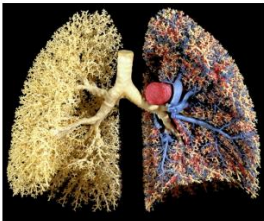
	3. Rate each image from 1 to 5 based on how clearly the pattern is represented, where 1 indicates the least clear pattern and 5 indicates a very clear pattern.			
	Observed Object	Fibonacci Pattern? (Yes/No)	Explanation (How Fibonacci or Golden Ratio is shown)	Rating (1–5)
				
				
				
Part II. Making Connections A robotics engineering LEARNER designs a self-expanding robot arm where each segment added follows the Fibonacci sequence to optimize strength and flexibility. Questions: 1. If the first two segments measure 1 unit each, what are the lengths of the next 6 segments? 2. What is the total length of the robot arm after 8 segments? 3. Why is the Fibonacci sequence useful in making mechanical structures efficient and flexible?				
IV. Reflection:	Answer each question. 1. What patterns or similarities did you notice between the natural examples in Part I and the robot arm design in Part II? 2. How did evaluating real-life images help you better understand the connection between the Fibonacci sequence and the Golden Ratio? 3. How can applying the Fibonacci sequence in designs (like the robot arm) influence innovation in real-world fields such as engineering or architecture?			

I. Activity 3:	Extending And Reinforcing Learning
II. Materials:	Printed images or observations from surroundings
IV. Instructions:	Remediation: Fibonacci Number Sort 1. On a sheet of paper, create two columns: Fibonacci Numbers and Not Fibonacci Numbers. 2. From the box of numbers, pick a number, determine whether if Fibonacci number or not and write that number in the correct column. 3. For the numbers you placed in the Fibonacci column, verify them using the rule: each number equals the sum of the two preceding numbers. 4. Compare answers with a partner and discuss any numbers where you disagreed. Enhancement: Fibonacci Design Project 1. Create a visual design (poster, digital artwork, or geometric drawing). 2. Use Fibonacci numbers to construct shapes such as but not limited to Fibonacci rectangles or spirals. 3. Label the Fibonacci numbers used in the design. 4. Highlight where the Golden Ratio appears in your design. 5. Write a short paragraph explaining how the Fibonacci sequence influenced your artwork.
IV. Reflection:	Answer each question: 1. How did sorting and verifying numbers in the remediation activity help you better understand how the Fibonacci sequence is formed?

	2. What relationship did you observe between the Fibonacci sequence and the visual patterns you created in the enhancement design? 3. How did using Fibonacci numbers in your artwork help you connect mathematical concepts to creativity and design?
*****END OF LESSON 2.2*****	

Lesson 2.3. FRACTALS	
Objectives:	At the end of the lesson, you are expected to: 1. describe fractals using mathematical terms, highlighting self-similarity, recursion, and infinite complexity. 2. identify examples of fractals in natural objects (trees, clouds, mountains, rivers) and artworks (digital art, mosaics, patterns). 3. analyze how fractal properties appear in both natural and man-made designs. 4. construct simple fractal patterns manually using geometric shapes, repetition, or recursion. 5. use technology (e.g., drawing software or fractal generators) to create more complex fractal designs. 6. apply creativity and mathematical understanding to produce an original fractal artwork suitable for the class exhibit.
Key Ideas:	Definition of Fractals A fractal is a never-ending, infinitely complex pattern that is created through repetition of a simple process over and over in a continuous feedback loop. These patterns are self-similar across different scales, meaning that smaller parts resemble the whole structure. Fractals are generated through recursion, where a rule or equation is repeatedly applied to produce increasingly detailed structures. Because of this process, fractals are often described as images of dynamic systems or “pictures of chaos,” where simple rules can generate highly complex forms. Characteristics of Fractals Fractals have several key properties that make them unique: • Self-Similarity: Smaller parts of a fractal resemble the overall structure, either exactly or in a slightly varied form. This similarity appears across different scales. • Recursion: Fractals are formed through a repeating process where a simple rule or equation is applied repeatedly. This recursive repetition generates increasingly complex patterns over time. • Infinite Complexity: Fractals can be zoomed into endlessly, with each level revealing more detail without reaching a final, simple form. Fractals in Nature and Art Fractals are extremely common in nature because many natural processes grow through repeated patterns. Examples include trees, rivers, coastlines, mountains, clouds, seashells, and hurricanes. In these systems, branching or spreading structures often follow fractal-like rules, producing similar shapes at different scales. In art, fractals appear in both traditional inspiration and digital creation. Artists use fractal patterns to create visually complex and aesthetically appealing designs that reflect infinity, symmetry, and natural order. Computer-generated fractal art, such as images based on the Mandelbrot Set, demonstrates how mathematical recursion can produce intricate and beautiful visual forms. Applications of Fractals Fractals have important applications across different fields. In mathematics and computer science, they are used to model complex systems and generate detailed images using simple equations, such as the Mandelbrot Set. In nature and environmental science, fractal models help describe irregular structures like coastlines, mountain ranges, and cloud formations.

	In technology, fractals are used in image compression, computer graphics, and digital animation to create realistic natural landscapes. In medicine, fractal geometry helps analyze structures such as blood vessels and lung tissues, which also follow branching patterns. Overall, fractals provide a powerful way to understand, simulate, and represent complex systems in both science and art
Video Lesson:	https://youtu.be/YiGBNDDgH0?si=7_7JvNA-wCksOdU2

I. Activity 1:	Practicing Learned Skills
II. Materials:	paper and pen, Calculator (optional)
III. Instructions:	Part I. Analyze each pattern below to determine whether it demonstrates a fractal, Fibonacci sequence, or Golden Ratio, and provide reasons for your classification.  (1)  (2)  (3)  (4) Part II. 1. Work with a pair. Ensure that you have ruler, protractor, pencil, colored papers, glue, $\frac{1}{4}$ Manila paper, and coloring materials. 2. Using geometric shapes, create your own original fractal and draw/paste it on the Manila paper. Illustrations should start from Iteration 0 to Iteration 8. 3. Below your drawing, write a short description on how you came up with your fractal highlighting the features of fractals. Part III. 1. Using your smartphones, tablets or laptops, access editing and/or drawing applications (MS Productivity tools, GeoGebra, etc.) online fractal generator software such as Online Fractal Creator by Visnos, Fractal Generator - Huo's Coding Lab, etc. 2. Create 2 original fractals, one for each given technology tools. LEARNERS plan their design, ensuring it demonstrates: • Self-similarity – smaller parts resemble the whole • Recursion – repeating patterns at multiple scales • Infinite complexity – visually intricate patterns 3. Show up to 8 iterations and explain the rule/process of creating the fractal. 4. Present your final output in the form of infographics. Below the fractal you created, prepare a brief description explaining the mathematics behind the fractal, including (1) Features that show self-similarity, recursion, or infinite complexity; and (2) Inspiration from nature, art, or other designs. Save your work in PNG/JPEG format.
IV. Reflection:	Answer each question. 1. How did your fractal change from Iteration 0 to Iteration 8, and what patterns did you observe as it developed? 2. What features in your created pattern reflect self-similarity, recursion, or the Golden Ratio? 3. How could your pattern be applied in real-world designs or natural systems? 4. What challenges did you encounter while creating your fractal manually or using technology, and how did you address them?

	5. How did the use or absence of technology affect the accuracy, detail, or creativity of your fractal design?
	6. How does creating your own design deepen your understanding of fractals?

I. Activity No. 2	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	Read each question carefully. Choose the letter of the correct answer. - Which of the following best describes a fractal? A. A pattern that repeats itself at different scales B. A figure with equal sides and angles C. A shape with only straight lines D. A design with no repetition - What is meant by self-similarity in fractals? A. The pattern changes completely at each level B. Smaller parts resemble the whole structure C. The pattern disappears when zoomed in D. The figure has no repeating parts - Which of the following is a natural example of a fractal? A. A flat wall B. A straight road C. A perfect square D. A tree branching pattern - A pattern is created by repeatedly applying the same rule to generate smaller and smaller shapes. Which concept of fractals does this illustrate? A. Symmetry B. Recursion C. Translation D. Rotation - Which situation best demonstrates a fractal pattern based on repeated branching? A. The branching pattern of lightning or rivers B. A tiled floor with identical squares C. A smooth, curved surface D. A single straight line - When zooming into a fractal image, new details continue to appear at every level. What property of fractals is being shown? A. Linear growth B. Uniform scaling C. Infinite complexity D. Finite measurement - A digital artist uses a simple mathematical rule repeatedly to create a highly detailed and complex design. Why is this approach effective in producing fractal art? A. It removes the need for planning B. It limits the number of design elements C. It creates random and unpredictable patterns D. It builds complex patterns from simple repeated processes - An artist creates a pattern where each smaller section looks identical to the whole image. What fractal property is most evident? A. Rotation B. Symmetry C. Translation D. Self-similarity

	9. How can understanding fractals help engineers design structures? A. By making structures less detailed B. By avoiding patterns in construction C. By copying only simple geometric shapes D. By applying repeating patterns to improve strength and efficiency 10. A coastline appears more detailed the closer you observe it, with smaller irregular patterns emerging. What does this illustrate about fractals? A. They show increasing detail at different scales B. They have fixed and simple shapes C. They only exist in mathematics D. They cannot be found in nature Part 2. Generating Fractals 1. Create a Fractal Artwork • Using paper, coloring materials, or digital tools, design a small fractal pattern. • Your artwork should clearly show repetition, self-similarity, or recursion. • You may use geometric shapes (e.g., triangles, squares, lines) or create your own design inspired by patterns in nature (e.g., leaves, branches, shells). • Ensure that your design shows at least 2–3 levels of repetition or iteration. 2. Write a Mathematical Explanation • Write a short explanation (3–5 sentences) describing: • The pattern or rule you used to create your fractal • How repetition or recursion is shown in your design • How your artwork demonstrates self-similarity 3. Presentation of Work • Submit your artwork with your written explanation. • Be prepared to briefly explain your work to the class if asked.
IV. Reflection:	Answer each question: 1. How did answering the questions and creating your own fractal artwork deepen your understanding of self-similarity, recursion, and infinite complexity? 2. In what ways can the idea of fractals help you better appreciate patterns found in nature, art, or real-life structures?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	Printed images or observations from surroundings
III. Instructions:	Beginner (Remediation Level) 1. Choose from guided fractal templates (e.g., simple repeating patterns or step-by-step fractal drawings such as triangles or trees). 2. Replicate the pattern by following clear instructions. 3. Write 2–3 sentences describing what repeats in their design. 4. Support Provided: • Visual guides and step-by-step instructions • Teacher-assisted discussion on basic fractal concepts Intermediate (Core Level) 1. Design your own fractal-inspired artwork using repeated patterns or shapes. 2. You may use manual drawing or digital tools. 3. Write a short paragraph explaining how their design shows self-similarity and repetition. 4. Challenge Level: • Requires independent planning and application of fractal concepts Advanced (Enhancement Level)

	1. Create a more complex fractal artwork (e.g., multi-level patterns, recursive designs, or nature-inspired fractals). 2. Explore digital platforms or incorporate mathematical rules in generating the design. 3. Write a detailed explanation (1–2 paragraphs) describing: • Fractal properties (self-similarity, scaling, iteration) • The process used to generate the design • Real-life connections of fractals
IV. Reflection:	Answer each question. 1. How did the idea of repetition and self-similarity appear in the different fractal designs you explored or created? 2. What differences did you notice between guided fractal patterns and independently designed fractal artworks in terms of complexity and understanding of fractal concepts? 3. How did creating or replicating fractal patterns help you see connections between mathematics and natural or real-world structures?

*****END OF LESSON 2.3*****

Unit 3. INTRODUCTION TO MATRICES

Lesson 3. 1. OPERATIONS ON MATRICES	
Objectives:	At the end of the lesson, you are expected to: 1. define a matrix and identify its components, including elements, rows, columns, and dimensions. 2. calculate the sum, difference, and product of matrices including scalar multiplication. 3. apply the rule of transposition to any given matrix. 4. demonstrate understanding of matrix operations in real-life problems.
Key Ideas: and Examples:	Understanding Key Concepts A matrix is a rectangular arrangement of numbers, symbols, or expressions in rows and columns. It is used to organize data and can help solve real-life problems in mathematics, science, and other fields. Components of a Matrix • Element – each individual number or symbol in a matrix. • Rows – horizontal lines of elements. • Columns – vertical lines of elements. • Dimension – indicates how many rows and columns the matrix has. The dimension of a matrix is written as $i \times j$, where i is the number of rows and j is the number of columns. Operations of Matrices 1. Matrix addition is performed by adding the corresponding elements of two matrices that have the same dimension (same number of rows and columns). Rule: If $A=[a_{ij}]$ and $B=[b_{ij}]$ are matrices of the same dimension, then $A+B=a_{ij}+ [b_{ij}]$ Take note: Matrices must have the same dimensions to be added. 2. Matrix subtraction is performed by subtracting the corresponding elements of two matrices with the same dimension. Rule: If $A=[a_{ij}]$ and $B=[b_{ij}]$ are matrices of the same dimension, then $A-B=a_{ij}- [b_{ij}]$ Take note: Matrices must have the same dimensions to be subtracted. 3. Matrix Multiplication is the process of multiplying two matrices to get a new matrix. For two matrices A and B to be multiplied, the number of columns in A must be equal to the number of rows in B. Rule: $C = A \times B$ where each element c_{ij} of C is calculated as: $c_{ij} = (\text{row } i \text{ of } A) \times (\text{column } j \text{ of } B)$

4. Scalar Multiplication means multiplying every element of a matrix by a number (scalar).

Rule: If k is a number and A is matrix,

$$kA = a_{ij}$$

5. Transpose of a Matrix is obtained by interchanging its rows and columns.

Rule: If matrix A is given, its transpose is written as: A^T

Components of a Matrix

Examples:

1) 2×2 Matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

- Elements: 1, 2, 3, 4
- Rows: [1, 2], [3, 4]
- Columns: [1, 3], [2, 4]
- Dimension: 2×2 (2 rows, 2 columns)

2) 3×3 Matrix

$$B = \begin{bmatrix} 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \end{bmatrix}$$

- Elements: 5, 6, 7, 8, 9, 10, 11, 12, 13
- Rows: [5, 6, 7], [8, 9, 10], [11, 12, 13]
- Columns: [5, 8, 11], [6, 9, 12], [7, 10, 13]
- Dimension: 3×3 (3 rows, 3 columns)

3) 2×3 Matrix

$$C = \begin{bmatrix} 2 & 7 & 1 & 4 & 9 & 6 \end{bmatrix}$$

- Elements: 1, 2, 4, 6, 7, 9
- Rows: [2, 7, 1], [4, 9, 6]
- Columns: [2, 4], [7, 9], [1, 6]
- Dimension: 2×3 (2 rows, 3 columns)

Operations of Matrices

Matrix Addition

1. If $A = \begin{bmatrix} 2 & 4 & 1 & 3 \end{bmatrix}$ & $B = \begin{bmatrix} 5 & 1 & 2 & 6 \end{bmatrix}$, compute $A + B$.

$$A+B = \begin{bmatrix} 2+5 & 4+1 & 1+2 & 3+6 \end{bmatrix} = \begin{bmatrix} 7 & 5 & 3 & 9 \end{bmatrix}$$

2. If $C = \begin{bmatrix} 1 & 0 & 2 & 3 & 1 & 4 & 5 & 7 & 0 \end{bmatrix}$ & $D = \begin{bmatrix} 8 & 3 & 5 & 1 & 6 & 4 & 0 & 2 & 7 \end{bmatrix}$, compute $C + D$.

$$C+D = \begin{bmatrix} 1+8 & 0+3 & 2+5 & 3+1 & 1+6 & 4+4 & 5+0 & 7+2 & 0+7 \end{bmatrix} = \begin{bmatrix} 9 & 3 & 7 & 4 & 7 & 8 & 5 & 9 & 7 \end{bmatrix}$$

Matrix Subtraction

1. If $A = \begin{bmatrix} 2 & 4 & 1 & 3 \end{bmatrix}$ & $B = \begin{bmatrix} 5 & 1 & 2 & 6 \end{bmatrix}$, compute $A - B$.

$$A-B = \begin{bmatrix} 2-5 & 4-1 & 1-2 & 3-6 \end{bmatrix} = \begin{bmatrix} -3 & 3 & -1 & -3 \end{bmatrix}$$

2. If $C = \begin{bmatrix} 1 & 0 & 2 & 3 & 1 & 4 & 5 & 7 & 0 \end{bmatrix}$ & $D = \begin{bmatrix} 8 & 3 & 5 & 1 & 6 & 4 & 0 & 2 & 7 \end{bmatrix}$, compute $C - D$.

$$C-D = \begin{bmatrix} 1-8 & 0-3 & 2-5 & 3-1 & 1-6 & 4-4 & 5-0 & 7-2 & 0-7 \end{bmatrix} = \begin{bmatrix} -7 & -3 & -3 & 2 & -5 & 0 & 5 & 5 & -7 \end{bmatrix}$$

Matrix Multiplication

1. If $A = \begin{bmatrix} 2 & 4 & 1 & 3 \end{bmatrix}$ & $B = \begin{bmatrix} 5 & 1 & 2 & 6 \end{bmatrix}$, compute $A \times B$.

number of columns of matrix $A = 2$

number of rows of matrix $B = 2$

$$AB = \begin{bmatrix} 25+(4 \times 2) & 2 \times 1+(4 \times 6) & 1 \times 5+(3 \times 2) & 1 \times 1+(3 \times 6) \end{bmatrix} = \begin{bmatrix} 18 & 26 & 11 & 19 \end{bmatrix}$$

2. If $C = \begin{bmatrix} 2 & 7 & 1 & 4 & 9 & 6 \end{bmatrix}$ & $D = \begin{bmatrix} 4 & 1 & 0 & 3 & 5 & 2 \end{bmatrix}$, compute $C \times D$.

number of columns of matrix $C = 3$

number of rows of matrix $D = 3$

$$CD = \begin{bmatrix} 24+70+15 & 21+73+12 & 44+90+65 & 41+93+62 \end{bmatrix}$$

$$CD = \begin{bmatrix} 13 & 25 & 46 & 43 \end{bmatrix}$$

Scalar Multiplication

1. If $A = \begin{bmatrix} 2 & 4 & 1 & 3 \end{bmatrix}$, compute $3A$.

$$3A = 3 \begin{bmatrix} 2 & 4 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 6 & 12 & 3 & 9 \end{bmatrix}$$

2. If $B = \begin{bmatrix} 5 & 1 & 2 & 6 \end{bmatrix}$, compute $-2B$.

$$-2B = -2 \begin{bmatrix} 5 & 1 & 2 & 6 \end{bmatrix} = \begin{bmatrix} -10 & -2 & -4 & -12 \end{bmatrix}$$

Transpose of a Matrix

1. If $A = \begin{bmatrix} 1 & -3 & 7 & 5 \end{bmatrix}$, find A^T .

$$A^T = \begin{bmatrix} 1 & 7 & -3 & 5 \end{bmatrix}$$

2. If $B = \begin{bmatrix} 1 & 0 & 2 & 3 & 1 & 4 & 5 & 7 & 0 \end{bmatrix}$, find B^T .

$$B^T = \begin{bmatrix} 1 & 3 & 5 & 0 & 1 & 7 & 2 & 4 & 0 \end{bmatrix}$$

Video Lessons:

Video Lessons:

1. Components of a Matrix

- <https://www.youtube.com/watch?v=F37rd-BpVOo&t=5s>
- <https://www.youtube.com/watch?v=JMjbPh1Mjn8>

2. Operations of Matrices

- https://www.youtube.com/watch?v=ZCmVpGv6_1g
- https://www.youtube.com/watch?v=7jb_AO_hRc8
- <https://www.youtube.com/watch?v=4lHyTQH1iS8>
- <https://www.youtube.com/watch?v=o6tGHLkZvVM>
- https://www.youtube.com/watch?v=g_Rz94DXvNo

I. Activity 1:

Practicing Learned Skills

II. Materials:

Printed worksheets, paper, Pen
Calculators (optional)
Whiteboard/marker or projector
Role cards or labels (Leader, Recorder, Checker, Presenter)
Reference sheet on matrix rules

III. Instructions:

Option 1: Guided Practice and Independent Activity

Part I. Guided Practice

Instructions:

1. Work with a partner or small group.
2. Carefully study the matrices given in each problem.
3. Follow the step-by-step guide provided by your teacher.
4. Discuss your answers with your group before writing the final solution.
5. Show complete solutions for each operation.

Given the matrices:

$$A = \begin{bmatrix} 2 & 1 & 3 & 0 & 4 & 5 & 6 & 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 & 2 & 4 & 0 & 1 & 5 & 2 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 6 & 1 & 3 & 5 & 8 \end{bmatrix}$$

Compute the following operations (where possible):

1. $A + B$
2. $B - A$
3. $A \times B$
4. $3C$
5. C^T

Part II. Independent Learning

Instructions:

1. Work individually to complete the activity.
2. Carefully study the matrices provided in each problem.
3. Apply the correct matrix operation.
4. Show your complete solution and final matrix answer.
5. Write your answers in your notebook.
6. Check that the dimensions of matrices allow the operation before solving.

$$A = \begin{bmatrix} 1 & 3 & -2 & 4 & 0 & 7 & 5 & -1 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 3 & 2 & 4 & 0 & 1 & 5 & 2 & 3 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 6 & 5 & 2 & -5 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 4 & 5 & -3 & -2 & 1 \end{bmatrix}$$

Compute the following operations (where possible):

1. $A + B$
2. $C - D$
3. $A \times D$
4. $3B$

	5. D^T Option 2: The Collaborative Matrix Challenge Instructions: - Form groups of 4-5 learners. - Each group will solve the matrix operation problems given below. - Assign roles within the group: - Leader – guides the discussion. - Recorder – writes the group’s solution. - Checker – verifies the computations. - Presenter – explains the solution to the class. - Show the complete solution and steps for each problem. - Each group will present one assigned problem and explain how the matrix operation was performed. - Explain why the operation is possible (or not possible) and how each element of the resulting matrix was obtained. Let $A = \begin{bmatrix} 4 & 2 & 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & -1 & 2 & 1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 5 & 1 & 6 & 0 \end{bmatrix}$, $D = \begin{bmatrix} 1 & -1 & 2 & 0 & 3 & 1 \end{bmatrix}$, $E = \begin{bmatrix} 1 & 2 & 1 & 0 & 1 & 0 \end{bmatrix}$ Compute the following operations (where possible): $2A - B$ $4C$ $2E^T$ $B + D$ $3A^T - 2C$ $(A + C)^T$ $2B - E$ $A - D$ $(B - 3E)^T$ $B \times D$
IV. Reflection:	Option 1: Answer each question. - What steps did you follow to ensure your matrix solutions were accurate? - How does scalar multiplication differ from matrix multiplication? - Which matrix operation did you find most challenging, and why? - How can you verify that your final matrix answer is correct? - What real-world situations can you think of where matrix operations are applied? Option 2: Answer each question: - Why is it necessary to check matrix dimensions before performing operations? - Which matrix operation was easiest for your group, and which was most difficult? Why? - What strategies did your group use to ensure accuracy in matrix operations? - What challenges did your group face in collaborating, and how were they resolved? - How do matrix operations connect to real-world applications (e.g., computer graphics, economics, data analysis)?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Printed worksheet or test paper Notebook or scratch paper Pens/pencils Calculator (optional)
III. Instructions:	Read each question carefully. Choose the letter of the correct answer. 1. Which of the following represents a 2×3 matrix? A. $\begin{bmatrix} 3 & -5 & 4 & 2 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 0 & -2 & 5 & -3 & 6 \end{bmatrix}$

- C. 1 2 3 4 5 6 -1 0 2
D. 7
2. In a matrix $A = \begin{bmatrix} 5 & 8 & 2 & 7 \end{bmatrix}$, what is the element at row 2, column 1?
A. 5
B. 2
C. 7
D. 8
3. What is the transpose of $A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \end{bmatrix}$?
A. 1 4 2 5 3 6
B. 1 2 3 4 5 6
C. 6 5 4 3 2 1
D. 1 2 3 4 5 6
4. If $A = \begin{bmatrix} 2 & 3 & 4 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 & 2 \end{bmatrix}$, what is $A+B$?
A. 3 3 4 7
B. 2 3 4 5
C. 2 0 0 2
D. 1 3 4 7
5. Which of the following is **NOT** possible?
A. Adding two 2×3 matrices
B. Multiplying a 2×3 matrix by a 3×2 matrix
C. Multiplying a 2×3 matrix by a 2×2 matrix
D. Transposing a 3×3 matrix
6. If $A = \begin{bmatrix} -2 & 3 & 0 & 4 \end{bmatrix}$, what is $3A$?
A. -6 9 0 12
B. -2 3 0 4
C. 3 9 0 12
D. -6 12 0 9
7. Find the product of $A = \begin{bmatrix} 1 & -2 & 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & -1 & 3 \end{bmatrix}$.
A. 0 -5 6 3
B. 1 -5 6 3
C. 4 7 6 3
D. 0 -5 3 6
8. The element in the second row, third column of A corresponds to which element in A?
A. Row 2, Column 3
B. Row 2, Column 2
C. Row 3, Column 3
D. Row 3, Column 2
9. Which of the following correctly describes the dimension of a matrix?
A. Number of rows only
B. Number of columns only
C. Number of rows $\times$ Number of columns
D. Sum of all elements
10. If $A = \begin{bmatrix} 2 & -4 & 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 2 & 1 & -1 \end{bmatrix}$, what is $A-B$?
A. 5 -6 4 2
B. -1 -2 6 0
C. 5 -2 4 2
D. 2 -4 5 1
11. Two stores recorded their weekly sales (in thousands), what is $A + B$?
 $A = \begin{bmatrix} 20 & 25 \end{bmatrix}$, $B = \begin{bmatrix} 15 & 10 \end{bmatrix}$
A. 35 35
B. 5 15
C. 300 250
D. 20 10
12. A delivery company charges ₱50 per package. If matrix represents packages delivered, what is $50A$?
 $A = \begin{bmatrix} 3 & 5 & 2 & 4 \end{bmatrix}$
A. 100 200 150 250
B. 200 100 250 150
C. 250 150 200 100
D. 150 250 100 200

	13. A store records profit (in thousands of pesos). What does $A - B$ BEST represent? Branch A: $A = \begin{bmatrix} 8 & 5 & 6 & 4 \end{bmatrix}$ Branch B: $B = \begin{bmatrix} 5 & 7 & 3 & 6 \end{bmatrix}$ A. Total combined profit of both branches B. Difference in profit between Branch A and Branch B C. Average profit of both branches D. Product of profits of both branches 14. A company increases all costs by 20%. Original cost matrix. Which operation correctly models the new cost? $A = \begin{bmatrix} 50 & 100 & 150 & 200 \end{bmatrix}$ A. $A + 20$ B. $0.2A$ C. $1.2A$ D. $A - 20$ 15. Which situation can be represented using matrix multiplication? A. Counting total students in class B. Adding two matrices of sales C. Listing student names D. Computing total cost from quantity and price matrices
IV. Reflection:	Answer each question. - What is a matrix and how are its components such as elements, rows, columns, and dimensions identified? - How do you perform addition, subtraction, and scalar multiplication on matrices? - How is matrix multiplication carried out and what conditions must be satisfied? - What is the rule of transposition and how does it change a matrix? - How can matrix operations be applied in solving real-life problems?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	Printed worksheet or test paper Notebook or scratch paper Pens/pencils Calculator (optional)
III. Instructions:	Remedial Activity: Solve the following matrix operation. Show your complete solution for each item. Write your answers in your notebook. Given the following matrices: $A = \begin{bmatrix} 5 & -2 \\ 3 & 0 \end{bmatrix}, B = \begin{bmatrix} 4 & 1 \\ -3 & -5 \end{bmatrix}, C = \begin{bmatrix} 2 & 0 \\ 4 & 7 \end{bmatrix}$ Find: $A + C$ $B - C$ $3B$ $A \times B$ A^T Enrichment Activity: Solve each matrix operation to discover the missing matrix. Explain the steps used to obtain your answer. Write your answer in your notebook. - If $A = \begin{bmatrix} 3 & 5 \\ 6 & 1 \end{bmatrix}$ and $A + B = \begin{bmatrix} 0 & 4 \\ 5 & 2 \end{bmatrix}$ find matrix B. - If $2A = \begin{bmatrix} 8 & 4 \\ -6 & 10 \end{bmatrix}$, find matrix A. - If $B^T = \begin{bmatrix} 3 & -2 \\ 5 & -4 \\ 6 & 9 \end{bmatrix}$, find matrix B. - If $C = \begin{bmatrix} 4 & -2 \\ 0 & 3 \end{bmatrix}$ and $C - D = \begin{bmatrix} 1 & 5 \\ -2 & 7 \end{bmatrix}$ find matrix D.

	5. $3C = \begin{bmatrix} 12 & 15 \\ -9 & 3 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 5 \\ 1 & -1 \end{bmatrix}$, find $C + D^T$.
IV. Reflection:	Answer each question. 1. How did performing different matrix operations (addition, subtraction, scalar multiplication, and multiplication) help you understand how matrices represent and organize information? 2. What patterns or rules did you notice when working with matrix operations, especially in terms of dimensions and element-wise operations? 3. How did the process of solving for missing matrices in the enrichment activity deepen your understanding of inverse thinking and problem-solving in mathematics?
*****END OF LESSON 3.1*****	

Lesson 3.2. PROPERTIES OF MATRICES	
Objectives:	At the end of the lesson, you are expected to: 1. Identify and describe the properties of matrix operations such as commutative, associative, distributive, identity, and zero matrix properties. 2. Apply the properties of matrix operations to solve matrix expressions correctly. 3. Appreciate the importance of mathematical rules in ensuring accuracy when performing matrix operations.
Key Ideas and Examples:	Properties of Matrices Matrix Addition 1. Commutative Property of Addition - the order of addition does not affect the result. $\mathbf{A + B = B + A}$ 2. Associative Property of Addition - how matrices are grouped in addition does not affect the result. $\mathbf{(A + B) + C = A + (B + C)}$ 3. Distributive Property - a scalar or matrix can be distributed over addition. $\mathbf{k(A + B) = kA + kB}$ 4. Zero Matrix Property - adding a zero matrix does not change the original matrix. $\mathbf{A + O = A}$ Matrix multiplication is different from addition because the order of multiplication matters, and not all the rules from addition apply. 1. Non-Commutative Property - the order of multiplication matters; $A \times B \neq B \times A$ in most cases. 2. Associative Property - grouping does not affect the result in multiplication. $\mathbf{(A \times B) \times C = A \times (B \times C)}$ 3. Distributive Property - multiplication distributes over addition. $\mathbf{A \times (B + C) = A \times B + A \times C}$ 4. Identity Matrix Property - multiplying a matrix by an identity matrix does not change the matrix. $\mathbf{A \times I = I \times A = A}$ 5. Multiplication by Zero Matrix - multiplying any matrix by a zero matrix results in a zero matrix. $\mathbf{A \times 0 = 0, 0 \times A = 0}$ Transpose of a Matrix Properties

1. Transpose of a Transpose - taking the transpose twice returns the original matrix.

$$A^{TT}=A$$

2. Transpose of a Sum - the transpose of a sum equals the sum of the transpose.

$$(A+B)^T=A^T+B^T$$

3. Transpose of a Scalar Multiple - transpose of a matrix multiplied by a scalar equals the scalar multiplied by the transpose.

$$kA^T=kA^T$$

4. Transpose of a Product - the transpose of a product reverses the order of multiplication.

$$(AB)^T=B^TA^T$$

5. Symmetric Matrices - A matrix is symmetric if it is equal to its transpose.

$$A=A^T$$

Properties of Matrices

Matrix Addition

1. Commutative Property of Addition

$$A=2 \ 3 \ 1 \ 4, B=5 \ 1 \ 0 \ 2$$

$$A+B=2+5 \ 3+1 \ 1+0 \ 4+2 =7 \ 4 \ 1 \ 6$$

$$B+A=5+2 \ 1+3 \ 0+1 \ 2+4 =7 \ 4 \ 1 \ 6$$

2. Associative Property of Addition

$$A=2 \ 3 \ 1 \ 4, B=5 \ 1 \ 0 \ 2, C=1 \ 0 \ 2 \ 1$$

$$A+B+C=2 \ 3 \ 1 \ 4 +6 \ 1 \ 2 \ 3 =8 \ 4 \ 3 \ 7$$

3. Distributive Property

Let $k=2$

$$A+B=7 \ 4 \ 1 \ 6$$

$$2A+B=27 \ 4 \ 1 \ 6 =14 \ 8 \ 2 \ 12$$

$$2A+2B=22 \ 3 \ 1 \ 4 +25 \ 1 \ 0 \ 2 =4 \ 6 \ 2 \ 8 +10 \ 2 \ 0 \ 4 =14 \ 8 \ 2 \ 12$$

4. Zero Matrix Property

$$A+0=3 \ 2 \ 1 \ 5 +0 \ 0 \ 0 \ 0 =3 \ 2 \ 1 \ 5$$

Matrix multiplication

1. Non-Commutative Property

$$A=1 \ 2 \ 0 \ 1, B=3 \ 0 \ 1 \ 2$$

$$A \times B=1(3)+(2)(1 \ 1)(0)+(2)(2 \ 0)(3)+(1)(1 \ 0)(0)+(1)(2 \ 0)=5 \ 4 \ 1 \ 2$$

$$B \times A=3(1)+(0)(0 \ 3)(2)+(0)(1 \ 1)(1)+(2)(0 \ 1)(2)+(2)(1 \ 2)=3 \ 6 \ 1 \ 4$$

2. Associative Property

$$A=1 \ 0 \ 2 \ 1, B=0 \ 1 \ 1 \ 2, C=2 \ 0 \ 1 \ 3$$

Compute $(A \times B) \times C$ and $A \times (B \times C)$

- Both will result in the same matrix:

$$(A \times B) \times C=A \times (B \times C)=1 \ 3 \ 5 \ 8$$

3. Distributive Property

$$A=1 \ 2 \ 0 \ 1, B=3 \ 0 \ 1 \ 2, C=0 \ 1 \ 2 \ 1$$

$$B+C=3 \ 1 \ 3 \ 3$$

$$A \times (B+C)=1 \ 2 \ 0 \ 1 \ 3 \ 1 \ 3 \ 3 =9 \ 7 \ 3 \ 3$$

$$A \times B+A \times C=5 \ 4 \ 1 \ 2 +4 \ 3 \ 2 \ 1 =9 \ 7 \ 3 \ 3$$

4. Identity Matrix Property

$$I=1 \ 0 \ 0 \ 1, A=2 \ 3 \ 1 \ 4$$

$$A \times I=2 \ 3 \ 1 \ 4, I \times A=2 \ 3 \ 1 \ 4$$

5. Multiplication by Zero Matrix

	$0=0\ 0\ 0\ 0\ 0, A=2\ 3\ 1\ 4$ Transpose of a Matrix 1. Transpose of a Transpose $A=1\ 2\ 3\ 4$ $AT=1\ 3\ 2\ 4, (AT)T=1\ 2\ 3\ 4=A$ 2. Transpose of a Sum $A=1\ 2\ 3\ 4, B=0\ 5\ 6\ 1$ $A+B=1\ 7\ 9\ 5, (A+B)T=1\ 9\ 7\ 5$ $AT+BT=1\ 3\ 2\ 4+0\ 6\ 5\ 1=1\ 9\ 7\ 5$ 3. Transpose of a Scalar Multiple $A=2\ 3\ 1\ 4, k=3$ $3A=6\ 9\ 3\ 12, (3A)T=6\ 3\ 9\ 12$ $3AT=32\ 1\ 3\ 4=6\ 3\ 9\ 12$ 4. Transpose of a Product $A=1\ 2\ 0\ 1, B=3\ 0\ 1\ 2$ $AB=1\cdot3+2\cdot1\ 1\cdot0+2\cdot2\ 0\cdot3+1\cdot1\ 0\cdot0+1\cdot2=5\ 4\ 1\ 2$ $(AB)T=5\ 1\ 4\ 2, BTAT=3\ 1\ 0\ 2\ 1\ 0\ 2\ 1=5\ 1\ 4\ 2$ 5. Symmetric Matrices $A=2\ 3\ 3\ 4, AT=2\ 3\ 3\ 4$
Video Lesson:	Video Lessons: 1. Components of a Matrix https://www.youtube.com/watch?v=F37rd-BpVOo&t=5s https://www.youtube.com/watch?v=JMjbPh1Mjn8 2. Operations of Matrices https://www.youtube.com/watch?v=ZCmVpGv6_1q https://www.youtube.com/watch?v=7jb_AO_hRc8 https://www.youtube.com/watch?v=4IHytQH1iS8 https://www.youtube.com/watch?v=o6tGHLkZuVM https://www.youtube.com/watch?v=g_Rz94DXuNo

I. Activity 1:	Practicing Learned Skills																											
II. Materials:	Printed worksheets Notebook or paper Pen Calculators (optional) Whiteboard/marker or projector Role cards or labels (Leader, Recorder, Checker, Presenter) Reference sheet																											
III. Instructions:	Option 1: Guided Practice and Independent Learning Situation: A school recorded the number of notebooks and pens distributed to students in two sections over two days. <table><tr><th>Matrix A (Day 1)</th><th>Notebooks</th><th>Pens</th></tr><tr><td>Section 1</td><td>12</td><td>8</td></tr><tr><td>Section 2</td><td>10</td><td>6</td></tr></table> <table><tr><th>Matrix B (Day 2)</th><th>Notebooks</th><th>Pens</th></tr><tr><td>Section 1</td><td>5</td><td>4</td></tr><tr><td>Section 2</td><td>3</td><td>2</td></tr></table> <table><tr><th>Matrix C (extra allocation)</th><th>Notebooks</th><th>Pens</th></tr><tr><td>Section 1</td><td>5</td><td>4</td></tr><tr><td>Section 2</td><td>3</td><td>2</td></tr></table> Part I. Guided Practice Instructions:	Matrix A (Day 1)	Notebooks	Pens	Section 1	12	8	Section 2	10	6	Matrix B (Day 2)	Notebooks	Pens	Section 1	5	4	Section 2	3	2	Matrix C (extra allocation)	Notebooks	Pens	Section 1	5	4	Section 2	3	2
Matrix A (Day 1)	Notebooks	Pens																										
Section 1	12	8																										
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Section 1	5	4																										
Section 2	3	2																										
Matrix C (extra allocation)	Notebooks	Pens																										
Section 1	5	4																										
Section 2	3	2																										

1. Work with a partner and examine the given matrices carefully.
2. Compute each required matrix operation step-by-step.
3. Write your solutions clearly in matrix form.
4. Compare your answers with your partner and discuss your observations.

Compute:

1. $A+B$ and $B+A$
2. $A+B+C$ and $A+(B+C)$
3. $2A+B$ and $2A+2B$

Part II. Independent Learning

Instructions:

1. Work independently and solve each task carefully.
2. Show all your solutions step-by-step.
3. Check your answers using the properties learned.
4. Reflect on your results and answer the questions.

Compute:

1. AI , let $I = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}$
2. $A+O$, let $O = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$
3. $(A+B) - B$

Option 2: Collaborative Activity

- a. Divide the class into groups of 4-5 members.
- b. Each group will solve the matrix operation problems given below.
- c. Assign roles within the group:
 - Leader – guides the discussion.
 - Recorder – writes the group's solution.
 - Checker – verifies the computations.
 - Presenter – explains the solution to the class.
- d. Show the complete solution and steps for each problem.
- d. Each group will present one of their answers to the problem and process questions.

Part I. Properties of Matrix Addition

A small café records the number of drinks sold in two branches over two days.

Each matrix represents sales:

Branch A:	Coffee	Tea
Monday	20	15
Tuesday	18	22

Branch B:	Coffee	Tea
Monday	12	18
Tuesday	14	10

Instructions:

1. Convert the table into matrices and **perform the following operations:**
 - a. Find $A+B$ and $B+A$.
 - b. Introduce a zero matrix $O = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$. Compute $A+O$.
 - c. Add a third matrix $C = \begin{bmatrix} 5 & 3 & 2 & 4 \end{bmatrix}$ in different groupings: $A+B+C$ and $A+(B+C)$.
2. Record your answers and note which property each result illustrates.

Part II. Properties of Matrix Multiplication

A bakery tracks production in two ovens over two days. Each matrix represents baked goods:

$$A = \begin{bmatrix} 2 & 3 & 1 & 4 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 2 & 1 \end{bmatrix}$$

Instructions:

1. Perform the following operations:
 - a. Multiply $A \times B$ and $B \times A$.
 - b. Multiply $A \times I$ where I is the identity matrix.

	c. Multiply by zero matrix 0. d. Verify distributive property: Compute $A \times B + C$ and compare with $A \times (B + C)$ for $C = \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$ 2. Record results and identify which property each operation demonstrates. Part III. Properties of Matrix Transpose A school tracks student scores in two subjects over two exams: $A = \begin{bmatrix} 85 & 90 & 78 & 88 \end{bmatrix}$ Instructions: - Perform the following operations: - Compute AT. - Compute $(AT)^T$. - Introduce another matrix $B = \begin{bmatrix} 2 & 1 & 3 & 4 \end{bmatrix}$ and compute $(B)^T$ and compare with $AT + BT$. - Compute kAT for $k=2$ and compare with $(kA)^T$. - Compute $(AB)^T$ and compare with $BTAT$. - Record your answers and identify which property each result demonstrates.
IV. Reflection:	Option 1. Guided Practice: Processing Questions - What do you observe when comparing $A+B$ and $B+A$? - Are $A+B+C$ and $A+B+C$ equal? - How does $2A+B$ compare with $2A+2B$? - How do these properties help simplify computations? Independent Learning: Processing Questions - What happens when a matrix is multiplied by the identity matrix? - What happens when a matrix is added to a zero matrix? - What do you observe in the result of $(A+B)-B$? - Why are these properties useful in real-life situations? Option 2. Collaborative Activity: Processing Questions - How does verifying $A+B=B+A$ help us understand the difference between commutative properties in addition versus multiplication of matrices? - Why is the zero matrix considered the additive identity, and how does this concept connect to identities in other mathematical structures? - In what way does the associative property of matrix addition support flexibility in solving real-world problems involving grouped data? - Why does multiplying by the identity matrix preserve the original matrix, and how does this reflect the role of identity elements in algebraic systems? - How does proving $(AB)^T = B^T A^T$ deepen our understanding of how transposition interacts with multiplication order?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	Printed worksheet or test paper Notebook or scratch paper Pens/pencils Calculator (optional)
III. Instructions:	Part I – Multiple Choice Instruction: Read each question carefully and choose the correct answer from the given options. - Which property is illustrated by the equation $A+B=B+A$? - Associative Property - Commutative Property - Identity Property - Distributive Property - Which of the following represents the associative property of matrix addition?

- A. $A+B=B+A$
 B. $A+B)+C=A+(B+C$
 C. $AB+C=AB+AC$
 D. $A+0=A$
3. If A is any matrix and O is the zero matrix of the same order, then $A+O=A$. Which property is this?
 A. Zero Matrix Property
 B. Commutative Property
 C. Associative Property
 D. Identity Property
4. Which matrix property states that multiplying a matrix by the identity matrix results in the same matrix?
 A. Zero Property
 B. Distributive Property
 C. Identity Property
 D. Associative Property
5. Which statement about matrix multiplication is TRUE?
 A. Matrix multiplication is always commutative.
 B. Matrix multiplication is associative.
 C. Matrix multiplication always produces the same matrix size.
 D. Matrix multiplication has no identity element.
6. Which property is illustrated by the equation $AB+C=AB+AC$?
 A. Commutative Property
 B. Associative Property
 C. Distributive Property
 D. Identity Property
7. If A is any matrix, what is the result of multiplying it by the zero matrix?
 A. Identity matrix
 B. Matrix A
 C. Zero matrix
 D. Transpose of A
8. What property is illustrated by $(A^T)^T=A$?
 A. Double Transpose Property
 B. Identity Property
 C. Associative Property
 D. Distributive Property
9. Which of the following is the transpose property of matrix addition?
 A. $(A+B)^T=AT+BT$
 B. $(A+B)^T=AB$
 C. $(A+B)^T=BTAT$
10. Which of the following describes the transpose of a product of matrices?
 A. $(ATBT)$
 B. $(BTAT)$
 C. (AB)
 D. $(AB)^T=A+B$

Part II – Problem Solving

11-20. Analyzing School Resource Allocation Using Matrix Properties.

Table 1: Classroom Resources (Matrix A)

Grade Level	Projectors	Laptops
Grade 7	2	3
Grade 8	4	1

Table 2: Newly Donated Resources (Matrix B)

Grade Level	Projectors	Laptops
Grade 7	1	2
Grade 8	0	3

	Table 3: Cost per Unit (Matrix C)					
	<table> <tr> <th>Equipment</th><th>Cost (P)</th></tr> <tr> <td>Projector</td><td>10,000</td></tr> <tr> <td>Laptop</td><td>25,000</td></tr> </table>	Equipment	Cost (P)	Projector	10,000	Laptop
Equipment	Cost (P)					
Projector	10,000					
Laptop	25,000					
	11. What does matrix A represent in the given school context? 12. Compute $A+B$. What does the result represent? 13. Is $A+B=B+A$? Explain briefly. 14. Find $2A$. What does it represent? 15. Verify if $2(A+B)=2A+2B$. 16. What happens if matrix A is multiplied by an identity matrix? 17. If a zero matrix is added to B, what is the result? 18. Does matrix multiplication follow the commutative property? Justify using concept understanding. 19. Compute total cost of resources for Grade 7 using $(A+B) \times C$. 20. Which grade level requires more total budget and why?					
IV. Reflection:	Answer each question. 1. Why does verifying $A+B=B+A$ demonstrate the commutative property of matrix addition? 2. How does the equation $A+B)+C=A+(B+C)$ illustrate the associative property? 3. What does adding the zero matrix to any matrix show about identity in addition? 4. Why does multiplying a matrix by the identity matrix preserve its value? 5. How does proving $(AB)^T = B^T A^T$ deepen understanding of transpose in multiplication?					

I. Activity 3.	Extending and Reinforcing Learning
II. Materials:	Printed worksheet or test paper Notebook or scratch paper Pens/pencils Calculator (optional)
III. Instructions:	Remedial Activity a. Study each problem carefully. b. Perform the matrix operation when necessary. c. Identify the matrix property illustrated in each item. d. Write your answer in your notebook. 1. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. Find: a. $A + B$ & b. $B + A$ 2. Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and the identity matrix $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, Find AI. 3. Let $A = \begin{bmatrix} 4 & 2 \\ 1 & 5 \end{bmatrix}$ and $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ find: $A + O$ 4. Let $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ & $C = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$. Compute $A(B + C)$ 5. Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$. Find the transpose A^T. Enrichment Activity 1. Analyze each problem carefully before solving. 2. Apply appropriate properties of matrix addition, multiplication, and transpose. 3. Show your complete reasoning or proof where necessary. 4. Explain which matrix property supports your answer. 1. Investigating Non-Commutativity Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$

	a. Compute AB. b. Compute BA. c. Compare the results and explain what this shows about the commutative property of matrix multiplication. 2. Proving the Distributive Property Let A, B, and C be matrices of appropriate dimensions. a. Compute $A(B + C)$ using the matrices below: $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$ b. Compute $AB + AC$. c. Explain why the results confirm the distributive property of matrix multiplication. 3. Identity Matrix Investigation Suppose A is a 3×3 matrix and I is the 3×3 identity matrix. a. Predict the result of AI and IA. b. Explain why the identity matrix behaves like the number 1 in matrix multiplication. c. Describe why this property is useful in solving matrix equations. 4. Exploring Transpose of a Product Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} B = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$ a. Compute AB. b. Find $(AB)^T$. c. Compute $B^T A^T$. d. Compare the results and explain the transpose property of matrix multiplication. 5. Real-World Matrix Reasoning A school cafeteria tracks sales of two food items across two days using matrix $S = \begin{bmatrix} 50 & 30 \\ 40 & 20 \end{bmatrix}$ Rows represent days and columns represent food items. a. Interpret the meaning of the matrix entries. b. Find S^T. c. Explain how the transpose changes the interpretation of the data. d. Suggest how school administrators could use this information for decision-making.
IV. Reflection:	Answer each question. - Which row operations or elimination steps made it easier for you to simplify the system, and why? - What difficulties did you experience while solving for the variables, and what strategies did you use to check if your answers are correct? - In what real-life situations (such as pricing in a canteen or budgeting) can systems of linear equations be useful, and how do they help in making decisions?
*****END OF LESSON 3.2*****	

Lesson 3.3. SOLVING SYSTEMS OF LINEAR EQUATIONS USING MATRICES AND ROW OPERATIONS	
Objectives:	By the end of the lesson, you are expected to: - represent a system of linear equations as an augmented matrix accurately; - illustrate and perform all basic matrix row operations: row swapping, scalar multiplication, and row addition; and

	3. apply row operations to reduce matrices to their reduced row echelon form (RREF) and solve systems of linear equations.
Key Ideas and Example	In this activity, you will master the art of translating complex algebraic systems into a structured matrix format, allowing you to solve equations with greater precision and speed. By accurately representing a system of linear equations as an augmented matrix, you effectively strip away the variables to focus on the numerical relationships that define the system. You will learn to navigate these matrices using the three fundamental tools of linear algebra: row swapping to organize your work, scalar multiplication to simplify values, and row addition to eliminate variables. As you apply these operations to reach Reduced Row Echelon Form (RREF), you aren't just moving numbers; you are systematically uncovering the "identity" of the system to determine whether it possesses a unique solution, infinite possibilities, or no solution at all. Matrix Row Operations A row operation is an action applied to the rows of a matrix that transforms it without changing the solution set of the corresponding system of linear equations. - Row Swapping ($R_i R_j$) – exchanging two rows of a matrix $\begin{array}{cccccccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \quad R_1 R_2 \quad \begin{array}{cccccccc} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}$ - Row Multiplication (kR_i) – multiplying a non-zero scalar k to all the elements within the row $\begin{array}{cccccccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \quad 2R_1 R_1 \quad \begin{array}{cccccccc} 2 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \end{array}$ - Row Addition ($kR_i + R_j R_j$) - Add a scalar multiple of one row to another row, and replace the later row with that sum $\begin{array}{cccccccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \quad 3R_1 + R_3 R_3 \quad \begin{array}{cccccccc} 4 & 0 & 0 & 4 & 0 & 0 & 3 & 1 \end{array}$ Elementary Matrix An elementary matrix is a matrix formed by applying exactly one row operation on identity matrix. Reduced Row Echelon Form (RREF) A matrix is in reduced row echelon form if the entries along the main diagonal are all 1 and the rest of the entries are all 0. Sequence of row operations are applied to reduce a matrix to its reduced row echelon form. Example: Reduce the following matrices. 1. $\begin{array}{cccc} 1 & 1 & 1 & -1 \end{array}$ Solution: $\begin{array}{cccc} 1 & 1 & 1 & -1 \end{array} \quad -1R_1 + R_2 R_2 \quad \begin{array}{cccc} 1 & 1 & 0 & -1 \end{array}$ $\begin{array}{cccc} 1 & 1 & 0 & -1 \end{array} \quad R_2 + R_1 R_1 \quad \begin{array}{cccc} 2 & 2 & 0 & -2 \end{array}$ $\begin{array}{cccc} 2 & 2 & 0 & -2 \end{array} \quad -1R_2 R_2 \quad \begin{array}{cccc} 1 & 1 & 0 & -1 \end{array}$ 2. $\begin{array}{cccc} 2 & 3 & 4 & 5 \end{array}$ Solution: $\begin{array}{cccc} 2 & 3 & 4 & 5 \end{array} \quad -2R_1 + R_2 R_2 \quad \begin{array}{cccc} 2 & 3 & 0 & -1 \end{array}$ $\begin{array}{cccc} 2 & 3 & 0 & -1 \end{array} \quad 3R_2 + R_1 R_1 \quad \begin{array}{cccc} 5 & 9 & 0 & -4 \end{array}$ $\begin{array}{cccc} 5 & 9 & 0 & -4 \end{array} \quad -5R_1 R_1 \quad \begin{array}{cccc} 1 & 1 & 0 & -1 \end{array}$

$$\begin{array}{ccccccc} 1 & 0 & 0 & -1 & -1 & 2 & 2 \\ R_2 & R_2 & & & & & \\ 1 & 0 & 0 & 1 & & & \end{array}$$

3. $\begin{array}{ccccccc} 2 & 4 & 6 & 1 & 3 & 5 & 3 \\ 5 & 3 & 5 & 8 & & & \end{array}$

Solution:

$$\begin{array}{ccccccc} 2 & 4 & 6 & 1 & 3 & 5 & 3 \\ 5 & 3 & 5 & 8 & & & \\ R_2 & R_1 & & & & & \\ 1 & 3 & 5 & 2 & 4 & 6 & 3 \\ 5 & 8 & & & & & \\ 1 & 3 & 5 & 0 & -2 & -4 & 3 \\ 5 & 8 & & & & & \\ -3R_1 + R_3 & R_3 & & & & & \\ 1 & 3 & 5 & 0 & -2 & -4 & 0 \\ -4 & -7 & & & & & \\ -2R_2 + R_3 & R_3 & & & & & \\ 1 & 3 & 5 & 0 & -2 & -4 & 0 \\ 0 & 0 & 1 & & & & \\ 1 & 3 & 5 & 0 & -2 & -4 & 0 \\ 0 & 0 & 1 & & & & \\ -12R_2 & R_2 & & & & & \\ 1 & 3 & 5 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & & & & \\ -3R_2 + R_1 & R_1 & & & & & \\ 1 & 0 & -1 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & & & & \\ R_3 + R_1 & R_1 & & & & & \\ 1 & 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & & & & \\ -2R_3 + R_2 & R_2 & & & & & \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & & & & \end{array}$$

Augmented Matrix

An augmented matrix is the matrix formed by appending the constant matrix B to the coefficient matrix A of a system of linear equations. It is written as $[A|B]$.

Thus, the augmented matrix combines the coefficient matrix A and the constant matrix B into a single matrix, which allows us to apply elementary row operations to solve the system.

Example: Convert the following system of linear equation into a augmented matrix.

1. $\begin{cases} x+y=5 \\ 2x+3y=11 \end{cases}$

Solution:

Matrix Representation

$$\text{Mat } A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}, \text{ Mat } X = \begin{bmatrix} x \\ y \end{bmatrix}, \text{ Mat } B = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Augmented Matrix

$$B = \begin{bmatrix} 5 & 11 \end{bmatrix}$$

2. $\begin{cases} x+y+z=6 \\ 2x+3y+z=13 \\ x+3y+2z=11 \end{cases}$

Solution:

Matrix Representation

$$\text{Mat } A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 3 & 2 \end{bmatrix}, \text{ Mat } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \text{ Mat } B = \begin{bmatrix} 6 \\ 13 \\ 11 \end{bmatrix}$$

Augmented Matrix

$$B = \begin{bmatrix} 6 & 13 & 11 \end{bmatrix}$$

Solving Systems of Linear Equation through Matrices

Solving a system of linear equations using matrices and reduced row echelon form (RREF) is the process of representing the system as an augmented matrix and

applying row operations to transform the matrix into reduced row echelon form, from which the values of the variables can be determined directly.

Steps on solving linear equations using matrices:

- Step 1: Identify Matrix A, Matrix X, and Matrix B based on $AX=B$.
- Step 2: Form the augmented matrix $A|B$.
- Step 3: Apply sequence of row operations (row swapping, scalar multiplication, and row addition) to reduced it to reduced row echelon form.
- Step 4: The solution will be within the final matrix after it was reduced.

Example: Find the solution of the following systems of linear equation using reducing matrices into reduced row echelon form.

1. $\begin{cases} x+y=5 \\ 2x+3y=11 \end{cases}$

- Step 1: Identify Matrix A, Matrix X, and Matrix B based on $AX=B$.

$$\text{Mat A} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}, \text{Mat X} = \begin{bmatrix} x \\ y \end{bmatrix}, \text{Mat B} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

- Step 2: Form the augmented matrix $A|B$.

$$B = \begin{bmatrix} 5 & 11 \end{bmatrix}$$

- Step 3: Apply sequence of row operations (row swapping, scalar multiplication, and row addition) to reduced it to reduced row echelon form.

$$\begin{array}{ccc} 5 & 11 & -2R_1+R_2 \\ 5 & 1 & -1R_2+R_1 \end{array} \begin{array}{c} R_2 \\ R_1 \end{array}$$

- Step 4: The solution will be within the final matrix after it was reduced.

$$\text{Reduced B} = \begin{bmatrix} 4 & 1 \end{bmatrix}$$

$$x=4, y=1$$

the first column is for the first variable (x) and the second column is for the second variable (y).

2. $\begin{cases} x+y+z=6 \\ 2x+3y+z=13 \\ x+3y+2z=11 \end{cases}$

- Step 1: Identify Matrix A, Matrix X, and Matrix B based on $AX=B$.

$$\text{Mat A} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 3 & 2 \end{bmatrix}, \text{Mat X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \text{Mat B} = \begin{bmatrix} 6 \\ 13 \\ 11 \end{bmatrix}$$

- Step 2: Form the augmented matrix $A|B$.

$$B = \begin{bmatrix} 6 & 13 & 11 \end{bmatrix}$$

- Step 3: Apply sequence of row operations (row swapping, scalar multiplication, and row addition) to reduced it to reduced row echelon form.

$$\begin{array}{ccc} 6 & 13 & 11 \\ 6 & 1 & 11 \end{array} \begin{array}{c} R_2 \\ -2R_1+R_2 \end{array}$$

$$\begin{array}{ccc} 6 & 1 & 11 \\ 6 & 1 & 5 \end{array} \begin{array}{c} R_2 \\ -1R_1+R_3 \end{array}$$

$$\begin{array}{ccc} 6 & 1 & 5 \\ 5 & 1 & 5 \end{array} \begin{array}{c} R_2 \\ -1R_2+R_1 \end{array}$$

$$\begin{array}{ccc} 5 & 1 & 5 \\ 5 & 1 & 3 \end{array} \begin{array}{c} R_2 \\ -2R_2+R_3 \end{array}$$

$$\begin{array}{ccc} 5 & 1 & 3 \\ 5 & 1 & 1 \end{array} \begin{array}{c} R_2 \\ 13R_3 \end{array}$$

$$\begin{array}{ccc} 5 & 1 & 1 \\ 5 & 2 & 1 \end{array} \begin{array}{c} R_3 \\ R_3+R_2 \end{array}$$

	$\begin{bmatrix} 5 & 2 & 1 \\ -2 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 5 & 2 & 1 \\ -2 & 3 & 1 \\ 3 & 2 & 1 \end{bmatrix}$ Step 4: The solution will be within the final matrix after it was reduced. $\text{Reduced B} = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $x=3, y=2, z=1$
Video Lesson	- System of Equations – Setting up the Augmented Matrix: https://www.youtube.com/watch?v=zQDsz4yn9Do - Elementary Row Operations (Linear Algebra): https://www.youtube.com/watch?v=69Z14u4MJzA - Solving Systems with RREF: https://www.youtube.com/watch?v=sJlMxfjhLi4

I. Activity 1:	Practicing Learned Skills
II. Materials:	- Worksheet or Notebook - Scratch Paper - Pencil and Eraser - Non-Scientific Calculator
III. Instructions:	Solve the following systems of linear equation using matrices. $\begin{cases} 2x+y=7 \\ x+2y=8 \end{cases}$ $\begin{cases} 3x+y=10 \\ 2x-y=1 \end{cases}$ $\begin{cases} x+y+z=6 \\ x+2y+z=9 \\ 2x+y+3z=14 \end{cases}$ $\begin{cases} x+2y+z=8 \\ 2x+y+z=9 \\ x+y+2z=7 \end{cases}$
IV. Reflection:	Answer each question: - When moving from a 2×2 system (Problems 1 & 2) to a 3×3 system (Problems 3 & 4), why is it important to follow to secure a 1 on the diagonal before creating 0s in that column rather than fixing numbers randomly? - The lesson notes suggest that the most common trap is trying to combine too many steps at once. How did writing down your specific row operation notation for every step help you identify or prevent errors?

I. Activity 2:	Assessing Learning Outcomes
II. Materials:	- Worksheet or Notebook - Scratch Paper - Pencil and Eraser - Non-Scientific Calculator
III. Instructions:	I. Represent the following system of linear equations as an augmented matrix $\begin{cases} 2x+3y=12 \\ x-y=1 \end{cases}$ $\begin{cases} x+y+z=6 \\ 2x-y+z=5 \\ x+2y-z=1 \end{cases}$ II. Apply the provided sequence of row operations on the following matrices. $\begin{bmatrix} 1 & 2 & 3 & 4 & 8 & 12 & 3 & 6 & 9 \end{bmatrix}$ Perform the following sequence of row operations: $R_1 R_3$ $14R_2 R_2$ $-1R_2 + R_3 R_3$ $\begin{bmatrix} 2 & 2 & 4 & 6 & 11 & 5 & 11 & 0 & 15 & 6 & 12 & 18 \end{bmatrix}$ Perform the following sequence of row operations: $15R_2 R_2$ $13R_3 + R_1 R_1$ $R_1 R_2$

	III. Solve the provided system of linear equation using matrix. 1. $\begin{cases} x+2y+z=23 \\ 2x+y+3z=41 \\ x+3y+2z=38 \end{cases}$ IV. Analyze the provided problem. Translate the situations into systems of linear equation and find its solution using matrices. Zeke buys his bread everyday at a nearby bakery. The bakery sells ensaymada, Spanish bread, and monay. - On Monday, Zeke bought 2 Monay, 1 ensaymada, and 3 Spanish bread for ₱85. - On Tuesday, Zeke bought 1 monay, 3 ensaymada, and 2 Spanish bread for ₱100. - On Wednesday, Zeke bought 3 monay, 2 ensaymada, and 1 Spanish bread for ₱85. How much does each bread cost per piece?
IV. Reflection:	Answer each question: - Looking back at the entire process, how does the augmented matrix act as a bridge between a word problem and a solved system? Why is it more reliable to perform row operations on a grid of numbers than to use traditional substitution for a 3-variable problem? - When you remove the variables and the equals signs to create an augmented matrix, how does this "grid" format change the way you approach solving the problem compared to writing out full algebraic equations? In your experience with this activity, did this structured layout make it easier or harder to keep track of your calculations?

I. Activity 1:	Extending and Reinforcing Learning
II. Materials:	- Worksheet or Notebook - Scratch Paper - Pencil and Eraser - Non-Scientific Calculator
III. Instructions:	Solve each problem. 1. $\begin{cases} x + y + z = 8 \\ 2x + 3y - z = 28 \\ 3x - y + 2z = 2 \end{cases}$ Solve for x, y, and z using row operations to reduce the augmented matrix to RREF. 2. At a school fiesta, a canteen sells Lumpia, Buko Roll, and Buko Juice. - On Monday, the canteen sold 2 Lumpia, 1 Buko Roll, and 3 Buko Juice for ₱85. - On Tuesday, the canteen sold 1 Lumpia, 3 Buko Roll, and 2 Buko Juice for ₱100. - On Wednesday, the canteen sold 3 Lumpia, 2 Buko Roll, and 1 Buko Juice for ₱115. How much does each item cost per piece?
IV. Reflection:	Answer each question: - What steps in the row operations or elimination process helped you simplify the system most efficiently? - What challenges did you encounter in solving for the variables, and how did you verify that your final answers were correct? - How can systems of linear equations be applied to real-life situations like pricing in a school canteen or budgeting problems?
*****END OF LESSON 3.3*****	

Lesson 3.4. MATRIX INVERSES AND LINEAR CLASSIFICATIONS	
Objectives:	- Define the inverse of a square matrix and state the identity property $AA^{-1}=I_n$. - Calculate the inverse of matrices using the augmented matrix method $[A I]$ through elementary row operations.

	3. Classify systems of linear equations as independent, dependent consistent, or inconsistent based on the results of row reduction and the inverse of the matrix.
Key Ideas and Example	The key idea of this topic is to understand that the inverse of a square matrix serves as a vital mathematical tool for solving systems of equations and analyzing their consistency. You will start by exploring the identity property, where multiplying a matrix by its inverse results in the Identity Matrix I, which functions much like the number "1" in standard algebra. To find this inverse, you will master the augmented matrix method $[A I]$, a systematic process where you apply elementary row operations to transform your original matrix into the identity form, thereby generating the inverse on the other side. Beyond just performing calculations, you will learn to use these results to classify systems, identifying whether they are independent, dependent consistent, or inconsistent, giving you a deeper logical perspective on how different linear relationships behave and resolve. Inverse of a Matrix A square matrix A is said to have an inverse if there exists a square matrix B of the same order such that $AB=BA=I$ where I is the identity matrix of the same order. The matrix B is called the inverse of matrix A and is denoted by A^{-1}. If such a matrix does not exist, A is called singular; if it exists, A is called invertible or non-singular. Example: - The inverse of matrix $A = \begin{bmatrix} 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 \end{bmatrix}$ is the matrix $A^{-1} = \begin{bmatrix} 3 & -1 & -5 & 2 \\ 3 & -1 & -5 & 2 \end{bmatrix}$. - The inverse of matrix $A = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$ is the matrix $A^{-1} = \begin{bmatrix} 1 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 1 \\ 1 & -1 & 1 & 0 & 1 & -1 & 0 & 0 & 1 \end{bmatrix}$. Finding the Inverse using the Augmented Matrix Steps on finding the Inverse of a Matrix using the Augmented Matrix: - Write the augmented matrix $[A I]$, where I is the identity matrix of the same order as A. - Use row operations to transform the left side of the augmented matrix into the identity matrix. - After the left side becomes I, the right side of the augmented matrix will be A^{-1}. If the left side cannot be transformed into I, the matrix is non-invertible. Example: Find the inverse of the following matrices using augmented matrix. $A = \begin{bmatrix} 2 & 1 & 5 & 3 \end{bmatrix}$ Step 1: Write the augmented matrix $[A I]$, where I is the identity matrix of the same order as A. $\begin{bmatrix} 2 & 1 & 5 & 3 & 1 & 0 & 0 & 1 \end{bmatrix}$ Step 2: Use row operations to transform the left side of the augmented matrix into the identity matrix. $\begin{array}{cccc cccc} 1 & 0 & 0 & 1 & 12R1R1 & 12 & 0 & 0 & 1 \\ 12 & 0 & 0 & 1 & -5R1+R2R2 & 12 & 0 & -52 & 1 \\ 12 & 0 & -52 & 1 & 2R2R2 & 12 & 0 & -5 & 2 \\ 12 & 0 & -5 & 2 & -12R2+R1R1 & 3 & -1 & -5 & 2 \end{array}$

Step 3: After the left side becomes I, the right side of the augmented matrix will be A-1.

$$2. \quad \begin{array}{cccccccc} & & & & & 3 & -1 & -5 & 2 \\ & & & & & & & & \rightarrow & A-1=3 & -1 & -5 & 2 \\ 1 & 2 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & & & & & \end{array}$$

Step 1: Write the augmented matrix $A|I$, where I is the identity matrix of the same order as A .

1 0 0 0 1 0 0 0 1

Step 2: Use row operations to transform the left side of the augmented matrix into the identity matrix.

$$1\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 1 \quad -2R_2+R_1R_1 \quad 1\ -2\ 0\ 0\ 1\ 0\ 0\ 0\ 1$$

$$1 \ -2 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \quad R3+R1R1 \quad 1 \ -2 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1$$

$$1 \ -2 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \quad -1R_3+R_2R_2 \quad 1 \ -2 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 1$$

Step 3: After the left side becomes I, the right side of the augmented matrix will be A-1.

$$1 \ -2 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 1 \quad \rightarrow \quad A^{-1} = 1 \ -2 \ 1 \ 0 \ 1 \ -1 \ 0 \ 0 \ 1$$

Checking for Classification of Systems of Linear Equations using Matrix Inverses

1. If the inverse exist, then the system is and **Independent System (Unique Solution)**
2. If the inverse does not exist, then the system of linear equation is either **Dependent Consistent System (Infinitely Many Solutions)** or an **Inconsistent System (No Solution)**.

If the inverse does not exist, use the row operations to determine whether it is
Dependent Consistent system or Inconsistent System.

Example:

Independent System (Unique Solution)

$$\begin{cases} x+y+z=6 \\ 2x-3y+z=3 \\ x+2y-2z=3 \end{cases}$$

Augmented Matrix:

633

Reduced Augmented Matrix:

2 1 3

Since the coefficient matrix within the augmented matrix is reduced to its identity, therefore the system is Independent or has a unique solution.

Dependent Consistent System (Infinitely Many Solutions)

$$\begin{cases} x+y+z=3 \\ 2x+2y+2z=6 \\ x-y+z=1 \end{cases}$$

Augmented Matrix:

3 6 1

Reduced Augmented Matrix:

3 1 0

Since there is a row that has all zero entries ($0=0$), then the system of linear equation is dependent consistent or has infinitely many solutions.

Inconsistent System (No Solution)

	$\begin{cases} x+y+z=4 \\ 2x+y+z=6 \\ x+2y+z=7 \end{cases}$ Augmented Matrix: $\begin{bmatrix} 1 & 1 & 1 & 4 \\ 2 & 1 & 1 & 6 \\ 1 & 2 & 1 & 7 \end{bmatrix}$ Reduced Augmented Matrix: $\begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & -1 & -1 & 2 \\ 0 & 1 & 0 & 3 \end{bmatrix}$ Since there is a row that has all zero entries on the left and mapped to a non – zero value on the right ($0=C$, where c is non-zero), then the system of linear equation is an inconsistent system or has no solutions.
Video Lessons	- Inverse Matrices and their Properties: https://youtu.be/kWorj5BBy9k - Finding the Inverse of a 2×2 matrix: https://www.youtube.com/watch?v=rqryXJhL-k - Inverse of a Matrix using Augmented Matrix: https://www.youtube.com/watch?v=NXCz9dw5Ce0 - A unique solution, No Solution, or Infinitely Many Solution: https://youtu.be/18_oG-cD9Ck

I. Activity 1:	Practicing Learned Skills			
II. Materials:	• Worksheet or Notebook • Scratch Paper • Pencil and Eraser • Non-Scientific Calculator			
III. Instructions:	Complete the table below by analyzing each system of linear equations. Determine if the coefficient matrix has an inverse and, if so, provide it. Reduce the augmented matrix to Reduced Row Echelon Form and classify the system as consistent (independent or dependent) or inconsistent, providing a brief justification for your answer.			
	System of Linear Equations	Does the inverse exist? If yes, what is the inverse?	Reduced Augmented Matrix B	How do you classify the given system of linear equation? Why?
	$\begin{cases} x+y+z=6 \\ y+2z=5 \\ 2z=4 \end{cases}$			
	$\begin{cases} x+y+z=3 \\ y+z=2 \\ y+z=2 \end{cases}$			
	$\begin{cases} x+y+z=4 \\ y+2z=3 \\ y+2z=8 \end{cases}$			
IV. Reflection:	Answer each question. 1. When performing row reduction, you might encounter a row of all zeros. • If that row looks like $[0 \ 0 \ \dots \ 0]$, why does this signify a dependent consistent system (infinite solutions)? • If it looks like $[0 \ 0 \ \dots \ 0 \mid c]$ where $c \neq 0$, why does this classify the system as inconsistent (no solution)? 2. How does the existence (or non-existence) of a matrix inverse help you predict which classification the system will fall into?			

I. Activity 2.	Assessing Learning Outcomes
II. Materials:	- Worksheet or Notebook - Scratch Paper - Pencil and Eraser - Non-Scientific Calculator

III. Instructions:	For each system provided, construct the augmented matrix, perform the necessary row operations to reach the reduced form, and complete the diagnostic table.					
	System 1:					
	$\{x+y+z=12 \ y+2z=10 \ 2z=6$					
	System 2:					
	$\{x+y+z=5 \ y+z=3 \ 2y+2z=6$					
	System 3:					
$\{x+y+z=7 \ 3y+3z=12 \ y+z=9$						
Evaluation Table:						
	System	Reduced Augmented Matrix	Does the inverse exist?	If the inverse exist, what is the inverse?	System Classification	Logic Check: Why fo you classify the system as Independent, Dependent Consistent, or Inconsistent?
	1					
	2					
	3					
IV. Reflection:	Answer each question.					
	1. In System 2, why did you stop trying to find an inverse? What specific part of the matrix told you it was non-invertible?					
	2. Compare the final rows of System 2 and System 3. Both have zeros on the left. Why is one 'Infinite' and the other 'Impossible'?					
	3. If I change the constant in System 3 from '9' to '4', does the matrix suddenly become invertible? Why or why not?					

I. Activity 3.	Extending and Reinforcing Learning
II. Materials:	• Worksheet or Notebook • Scratch Paper • Pencil and Eraser • Non-Scientific Calculator
III. Instructions:	Look at the final row of the augmented matrix, assuming that the $a_{11} = 1$ and $a_{22} = 1$, for each case. Complete the following and answer the "Why" question. • Case 1 Final Row Result: $[0 \ 0 \ 1 \ \ 5]$ 1. Can you find the value of the variable? 2. Does the inverse exist? 3. Status/Classification: 4. Why? • Case 2 Final Row Result: $[0 \ 0 \ 0 \ \ 0]$ 1. Can you find the value of the variable? 2. Does the inverse exist? 3. Status/Classification: 4. Why? • Case 3 Final Row Result: $[0 \ 0 \ 0 \ \ 2]$ 1. Can you find the value of the variable? 2. Does the inverse exist? 3. Status/Classification: Why?
IV. Reflection:	Answer each question. 1. How does the final row of an augmented matrix help you determine whether a system has one solution, infinitely many solutions, or no solution?

	2. What is the relationship between the existence of an inverse matrix and the type of solution a system of equations has? 3. Why is it important to identify whether a system is consistent or inconsistent before concluding your final answer?
*****END OF LESSON 3.4*****	

Lesson 3.5 DETERMINANTS OF A 2X2 MATRIX	
Objectives:	By the end of the discussion, you should be able to: - define the determinant of a 2×2 matrix; - compute the determinant correctly; and - apply the concept of determinants in real-world problems
Key Ideas and Examples:	2x2 Matrix and Its Determinants and Properties Interpreting the Determinant a. If the determinant is positive - The matrix preserves orientation. - The transformation does not “flip” the shape. b. If the determinant is negative - The matrix reverses orientation (like a reflection or flip). - The shape is turned “inside out.” c. If the determinant is zero - The rows or columns are dependent (one is a multiple of the other). - The matrix is not invertible. - The transformation collapses the shape into a line or a point. Important Properties of 2×2 Determinants These properties connect directly to the patterns discovered in the earlier activity. a. Identical rows or columns → determinant = 0 Because there is no unique direction (rows/columns overlap), the area collapses. b. Proportional rows or columns → determinant = 0 Example: [4] and [8] One row is double the other → dependent. c. Swapping two rows or columns flips the sign $\det(A_{\text{swapped}}) = -\det(A)$ d. Multiplying a row by a constant multiplies the determinant by that constant $\det(kR_1) = k \cdot \det(A)$ e. A diagonal 2×2 matrix has determinant equal to the product of diagonal entries $\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \rightarrow \det = ad$ Examples: Compute the determinant of each matrix $A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 7 & -1 \\ 0 & 6 \end{bmatrix}$ $C = \begin{bmatrix} -3 & 2 \\ 5 & -1 \end{bmatrix}$ $D = \begin{bmatrix} 0 & 8 \\ -4 & 3 \end{bmatrix}$ $E = \begin{bmatrix} 6 & 0 \\ -2 & 5 \end{bmatrix}$ $F = \begin{bmatrix} -1 & 4 \\ 3 & -2 \end{bmatrix}$

	7.	$G = \begin{bmatrix} 9 & 2 \\ -3 & 1 \end{bmatrix}$
	8.	$H = \begin{bmatrix} 4 & -5 \\ 7 & 2 \end{bmatrix}$
	Solutions:	
	2.	$\det(A) = 3(4) - 5(2) = 12 - 10 = 2$
	3.	$\det(B) = 7(6) - (-1)(0) = 42 - 0 = 42$
	4.	$\det(C) = (-3)(-1) - 2(5) = 3 - 10 = -7$
	5.	$\det(D) = 0(3) - 8(-4) = 32$
	6.	$\det(E) = 6(5) - 0(-2) = 30$
	7.	$\det(F) = (-1)(-2) - 4(3) = 2 - 12 = -10$
	8.	$\det(G) = 9(1) - 2(-3) = 9 + 6 = 15$
		$\det(H) = 4(2) - (-5)(7) = 8 + 35 = 43$
Video Lesson:		

I. Activity 1.	Practicing Learned Skills
II. Materials:	worksheet or paper, Non-Scientific Calculator
III. Instructions:	Compute the determinant of each 2×2 matrix, then interpret the results 1. $A = \begin{bmatrix} 4 & 1 \\ -2 & 5 \end{bmatrix}$ 2. $B = \begin{bmatrix} 7 & -3 \\ 0 & 2 \end{bmatrix}$ 3. $C = \begin{bmatrix} -6 & 4 \\ 3 & -1 \end{bmatrix}$ 4. $D = \begin{bmatrix} 5 & 2 \\ 10 & 4 \end{bmatrix}$ 5. $E = \begin{bmatrix} 9 & -2 \\ 4 & 1 \end{bmatrix}$ 6. $F = \begin{bmatrix} 0 & 7 \\ -3 & 5 \end{bmatrix}$ 7. $G = \begin{bmatrix} 2 & 5 \\ 8 & -1 \end{bmatrix}$ 8. $H = \begin{bmatrix} -4 & 3 \\ 6 & 2 \end{bmatrix}$
V. Reflection:	Answer each question. 1. How does the sign and value of the determinant help you understand what happens to a geometric shape after a transformation?

	2. What relationship do you observe between a determinant equal to zero and the invertibility of a matrix? How can understanding orientation (preserved or reversed) help you interpret transformations in real-world contexts such as graphics or mapping?
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I. Activity 2:	Assessing Learning Outcomes
II. Materials:	• Worksheet or Notebook • Scratch Paper • Pencil and Eraser • Non-Scientific Calculator
III. Instructions:	Part I – Compute the Determinant Find the determinant of each matrix. 1. $A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$ 2. $B = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$ 3. $C = \begin{bmatrix} 7 & 0 \\ -3 & 6 \end{bmatrix}$
IV. Reflection:	Answer the following in one sentence: 1. When does the determinant of a 2×2 matrix become zero? 2. What happens to the determinant when two rows are swapped? 3. Why is computing the determinant useful in understanding matrices?

I. Activity 3.	Extending and Reinforcing Learning
II. Materials:	worksheet or paper, paper, Non-Scientific Calculator
III. Instructions:	1. Study the three cases below and describe what each determinant value tells you about the transformation. a. $\det(A) > 0$ b. $\det(A) < 0$ c. $\det(A) = 0$ 2. Examine the following matrices and compute their determinants. $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 8 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ for each matrix, interpret the result: • Does the transformation preserve or reverse orientation? • Is the matrix invertible? • What happens to the shape (parallelogram) formed by its column vectors?
IV. Reflection:	Answer each question. 3. How does the sign and value of the determinant help you understand what happens to a geometric shape after a transformation? 4. What relationship do you observe between a determinant equal to zero and the invertibility of a matrix? 5. How can understanding orientation (preserved or reversed) help you interpret transformations in real-world contexts such as graphics or mapping?

*****END OF LESSON 3.5***

Lesson 3.6. DETERMINANT OF A 3X3 MATRIX	
Objectives:	At the end of the lesson, you should be able to: 1. compute determinants of 3×3 matrices using Sarrus' Rule, 2. apply the cofactor (minor) expansion along any row or column to find the determinant systematically; and 3. solve real-world problems involving 3×3 matrices.
Key Ideas:	Cofactor Expansions A recursive formula must have a starting point. For cofactor expansions, the starting point is the case of 1×1 matrices. The definition of determinant directly implies that 1×1 matrices. To describe cofactor expansions, we need to introduce some notation.

$$\det(a) = a$$

Definition : Minor and Cofactor

Let be an $n \times n$ matrix.

1. The (i,j) **minor**, denoted is A_{ij} , is the $(n-1) \times (n-1)$ matrix obtained from A by deleting the i th row and the j th column.
2. The (i,j) **cofactor** C_{ij} is defined in terms of the minor by

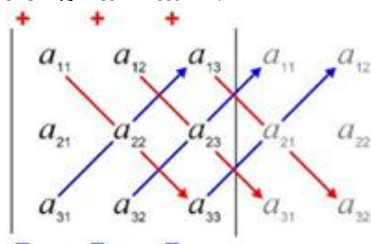
$$C_{ij} = (-1)^{i+j} \det(A_{ij})$$

Note that the signs of the cofactors follow a “checkerboard pattern.” Namely, $(-1)^{i+j}$ is pictured in this matrix:

$$[+ \ - \ + \ - \ + \ - \ + \ - \ +] + \ - \ + \ - \ + \ - \ + \ - \ +$$

Method 2: Sarrus' Rule (The Shortcut Method for 3x3)

3. Explain the setup: Show learners how to rewrite the first two columns of the matrix to the right of the original matrix.



4. Demonstrate the diagonal products:
5. Forward Diagonals (Add): Trace the three downward diagonal paths with colored markers and explain that the products of the elements along these paths are added.
6. Backward Diagonals (Subtract): Trace the three upward diagonal paths with a different color and explain that the products of the elements along these paths are subtracted.
7. State the caveat: Explicitly state that this rule only works for 3x3 matrices

Guided Activity:

Find the determinant of:

$$1. \begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 1 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 1 \end{bmatrix} \text{ using Method 1}$$

Solution

We make the somewhat arbitrary choice to expand along the first row. The minors and cofactors

$$A_{11} = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} \quad C_{11} = + \det \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} = 4$$

$$A_{12} = \begin{pmatrix} -1 & 1 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -2 & 3 \end{pmatrix} \quad C_{12} = - \det \begin{pmatrix} -1 & 1 \\ -2 & 3 \end{pmatrix} = 1$$

$$A_{13} = \begin{pmatrix} -1 & 2 \\ -2 & 2 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ -2 & 2 \end{pmatrix} \quad C_{13} = + \det \begin{pmatrix} -1 & 2 \\ -2 & 2 \end{pmatrix} = 2.$$

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} =$$

Step 1: Rewrite the first two columns after the matrix

$$\begin{bmatrix} 2 & 1 & 3 \\ -1 & 2 & 1 \\ -2 & 2 & 3 \end{bmatrix} \quad \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ -2 & 2 \end{bmatrix}$$

Step 2: Multiply the diagonals going down (left to right)

	• $(2)(2)(3)=12$ • $(1)(1)(-2)=-2$ • $(3)(-1)(2)=-6$ Sum of downward diagonals = $12+(-2)+(-6)=4$ $12 + (-2) + (-6) = 4$ Step 3: Multiply the diagonals going up (right to left) • $(3)(2)(-2)=-12$ • $(2)(1)(2)=4$ • $(3)(-1)(1)=-3$ Sum of upward diagonals = $-12 + 4 + (-3) = -11$ Step 4: Subtract $\text{Det}(A)=(\text{downward sum})-(\text{upward sum})$ $\text{Det}(A)=4 - (-11)=15$
Video Lesson:	https://youtu.be/aiBgjz5xbyg?si=qGApP3L025iWxLbe

I. Activity 1:	Practicing Learned Skills		
II. Materials:	worksheet or paper, paper, Non-Scientific Calculator		
III. Instructions:	A. Compute Using Sarrus' Rule		
	Instruction:		
	For each 3×3 matrix below, compute the determinant using Sarrus' Rule only . Show complete solutions.		
	1. $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 0 \end{bmatrix}$		
	2. $B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 4 \\ 2 & 5 & 1 \end{bmatrix}$		
	B. Compute Using Cofactor Expansion		
Instructions:			
Compute the determinant of the same matrices but this time using cofactor expansion .			
Choose any row or column. Show your full solution.			
3. $C = \begin{bmatrix} 4 & -2 & 1 \\ 3 & 0 & -1 \\ 5 & 2 & -3 \end{bmatrix}$			
4. $D = \begin{bmatrix} 6 & 1 & -2 \\ 0 & -3 & 4 \\ 2 & 5 & -1 \end{bmatrix}$			
Processing Questions:			
1. ?			
Final Check			
Write the final determinant values you obtained for each matrix (they should match):			
Matrix	Sarrus' Rule	Cofactor Expansion	
A			

	B	_____	_____
	C	_____	_____
	D	_____	_____
IV. Reflection:	Answer in 2–3 sentences each. - Which method (Sarrus or Cofactor Expansion) was faster for you? Why? - Which method helped you avoid mistakes? Why? - In what situations do you think Sarrus' Rule is more useful? Cofactor expansion		

I. Activity 2	Assessing Learning Outcomes
II. Materials:	worksheet or paper, paper, Non-Scientific Calculator
III. Instructions:	PART I. Compute the determinant of each matrix using either Sarrus' Rule or Cofactor Expansion. Show complete solutions. 1. $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 0 \end{bmatrix}$ 2. $B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -1 & 4 \\ 2 & 5 & 1 \end{bmatrix}$ PART II. Answer each in one complete sentence. - When does the determinant of a 3×3 matrix become zero? - How does swapping two rows affect the determinant? - Why is the determinant important when analyzing a 3×3 data set? PART III. Interpret the determinant of the matrix below. A simplified 3×3 sales data for three items over three days: $S = \begin{bmatrix} 20 & 25 & 22 \\ 10 & 12 & 13 \\ 15 & 18 & 17 \end{bmatrix}$ Tasks: - Compute det(S) - Explain what the determinant reveals about the sales pattern. - Why is this information useful in business?
V. Reflection:	Answer each question. - How did computing the determinant help you determine whether the information represented by a matrix is independent, dependent, or related in some way? - What challenges did you encounter when using Sarrus' Rule or cofactor expansion, and how did you ensure the accuracy of your computations? - How can the determinant of a matrix be useful in analyzing real-world data, such as sales records, business performance, or other measurable trends?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	pen and paper,
III. Instructions:	- Examine each given 3×3 matrix carefully. - For each matrix: - Decide whether Sarrus' Rule or Cofactor Expansion is the better method. - Write a short justification (1–2 sentences). - Compute the determinant using your chosen method. - Check your answer and be ready to explain your reasoning to the class. Matrix A $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 0 \\ 1 & 5 & 6 \end{bmatrix}$ Guide Question/s:

	- Which method is more efficient? Why? Expected Choice: Cofactor Expansion Reason: Presence of zeros simplifies expansion. Determinant: $= 2(18) - (-1)(24) = 36 + 24 = 60$ Matrix B $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \\ 2 & 1 & 6 \end{bmatrix}$ Guide Questions: - Which method is more convenient? - What makes this matrix suitable for that method? Expected Choice: Sarrus' Rule Reason: No excessive zeros; direct diagonal multiplication is efficient. Determinant: $= (1)(-1)(0) + (2)(4)(2) + (3)(0)(1) - [3(-1)(2) + 2(0)(0) + 1(4)(1)]$ $= 0 + 16 + 0 - [-6 + 0 + 4] = 16 - (-2) = 18$
IV. Reflection:	Answer each question. - How did the structure of the matrix affect your choice of method? - Can both methods always be used for a 3×3 determinant? Why? - Why is it important to choose an efficient method?

*****END OF LESSON 3.6***

Lesson 3.7 MATRIX INVERTIBILITY	
Objectives:	At the end of the lesson, learners should be able to: - state the condition for matrix invertibility; and - test if the given 2×2 and 3×3 matrices are invertible using determinants.
Key Ideas and Examples:	Determinant tells us whether a matrix is invertible or not invertible. - A matrix is invertible (non-singular) if $\det(A) \neq 0$ - A matrix is not invertible (singular) if $\det(A) = 0$ This rule applies to all square matrices (2×2, 3×3, 4×4, etc.) 2. Connect to Structure Emphasize that the determinant reflects row/column independence: - If all rows/columns give independent information, the determinant is non-zero, so the matrix has an inverse. - If any row/column is: ✓ identical ✓ proportional ✓ dependent ✓ zero the determinant becomes zero, and the matrix is not invertible. Show Simple Examples Example 1. 2×2 Case $A = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$ $\det(A) = 3(4) - 5(2) = 12 - 10 = 2$ Because the determinant is 2 (not zero), A is invertible. Example 2. 3×3 Case

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

Compute determinant (learners can use Sarrus or Cofactor Expansion).

Observation:

Rows are **multiples**, so determinant = **0.B is NOT invertible.**

4. Connect to Real-Life Meaning

Explain:

- If the determinant is zero, the matrix **cannot be reversed**.
- In real-world applications (systems of equations, graphics, cryptography), invertibility determines whether a solution or transformation is possible.

Provide this as an example:

In computer graphics, matrices perform transformations such as:

- rotation,
- reflection,
- scaling,
- stretching.

Why invertibility matters

An invertible matrix allows the computer to reverse a transformation.

Example:

- Zooming in and then zooming out
- Rotating an image and rotating it back

If the determinant is zero:

- the image may flatten into a line or point,
- information is lost,
- the original image cannot be recovered.

Video Lessons:

<https://youtu.be/GuFvSwriy1Q?si=JPEjAvqRkxPtNtcs>

I. Activity 1:

Practicing Learned Skills

II. Materials:

pen and paper

III. Instructions:

Compute the determinant of the following matrices and determine whether they are invertible or not invertible. Show your complete solution. Submit your work and be ready to explain one solution to the class.

Matrices for the Activity:

1. $A = \begin{bmatrix} 4 & 7 \\ 2 & 5 \end{bmatrix}$

2. $B = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix}$

3. $C = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 1 & 4 \\ 2 & 5 & 1 \end{bmatrix}$

4. $D = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 4 & 8 & 12 \end{bmatrix}$

5. $E = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 0 & 0 \\ 1 & 4 & 2 \end{bmatrix}$

6. $F = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 4 \\ 5 & 2 & 1 \end{bmatrix}$

IV. Reflection:

Answer each question:

1. Which matrices were easiest to classify?

	2. What common patterns appear in non-invertible matrices?
I. Activity 2:	Assessing Learning Outcomes
II. Materials:	
III. Instructions:	Compute the determinant of each matrix. Write your full solution and conclusion. - Determine whether each matrix is invertible or not invertible. - Show computations clearly ($2 \times 2 \rightarrow ad - bc$; $3 \times 3 \rightarrow$ Sarrus' Rule or cofactor expansion). - State your final verdict for each matrix. Given: $A = \begin{bmatrix} 4 & 2 \\ 7 & 5 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$ $C = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix}$ PART II. Answer the following in one complete sentence: - When does the determinant of a matrix become zero? - How does the determinant help you identify whether a matrix is invertible? - Why does a determinant of zero make certain systems or processes impossible to reverse? PART III. Analyze the real life matrix scenario: Sales Performance of Three Branches A company, ABC Inc., tracks the monthly sales (in thousands of pesos) of its three beverage products—Juice, Smoothie, and Iced Tea—across three branches: Branch E, Branch F, and Branch G. The data is represented by the matrix $S = \begin{bmatrix} 20 & 25 & 22 \\ 10 & 12 & 13 \\ 15 & 18 & 17 \end{bmatrix}$ where: Row 1 → Branch E sales Row 2 → Branch F sales Row 3 → Branch B sales Column 1 → Juice, Column 2 → Smoothie, Column 3 → Iced Tea Task: - Compute $\det(S)$. - Is the matrix invertible? Explain what this means in a real-life situation (business, graphics, cryptography, systems, etc.).
IV. Reflection:	Here are three reflection questions for the activity: - What strategies did you use to compute the determinant, and how did they help you organize your solution? - What do you notice about the meaning of a zero or non-zero determinant in real-life contexts such as data or sales patterns? - How can determinant properties help you make faster or more efficient decisions when working with larger matrices?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	pen and paper, GeoGebra
III. Instructions:	1. Review your previous work on determinants and invertibility. 2. Think of a real-life situation where matrices can represent measurable data (e.g., business sales, graphics transformations, cryptography, or system modeling). 3. Construct a 3×3 matrix S using realistic numerical values drawn from your chosen context. 4. Compute $\det(S)$ using Sarrus' Rule or cofactor expansion. 5. Determine whether your matrix is invertible or not invertible based on the determinant. 6. Write a short paragraph (3–5 sentences) explaining what the determinant's value means in your chosen real-life scenario. • Example: In a business context, a nonzero determinant means each branch's performance is independent and uniquely contributes to total sales. 7. Present your matrix, computation, and interpretation neatly on paper or digitally.
V. Reflection:	Answer each question. 1. What real-life situation did you choose for your 3×3 matrix, and why is it meaningful in representing measurable data? 2. How does the value of the determinant (zero or non-zero) affect your interpretation of the relationships within your chosen context? 3. What difficulties did you encounter while computing the determinant or interpreting its meaning, and how did you address them?

*****END OF LESSON 3.7***

Lesson 3.8 CRAMER'S RULE	
Objectives:	At the end of the lesson, you should be able to apply Cramer's Rule to solve 2×2 and 3×3 systems of linear equations.
Key Ideas and Examples:	Activity-First Approach 1. Present the system of equations on the board. $2x+3y=5$ $4x-y=7$ 2. Ask learners to attempt solving it before introducing Cramer's Rule. 3. Demonstrate how to form the coefficient matrix A, then replace columns to obtain D_x and D_y. 4. Identify the coefficient matrix and label it clearly: $D = \begin{bmatrix} 2 & 3 \\ 4 & -7 \end{bmatrix}$ to find $\det(D)$ 5. Replace the x-column with the constants to find D_x. 6. Replace the y-column with the constants to find D_y 7. Explain the formula $ad - bc$ step-by-step. 8. Emphasize that Cramer's Rule only works if $\det(A) \neq 0$. 9. Compute each determinant and show how $x=D_x/D$ and $y=D_y/D$. 10. Compute each determinant and show how to substitute them into: $x = \frac{D_x}{D}, y = \frac{D_y}{D}$ 11. Repeat with a 3×3 system, if needed, using Sarrus' Rule or cofactor expansion. 12. Highlight key ideas during the demonstration: 13. Determinants tell us if the system has a unique solution. 14. Replacing columns isolates variables. Cramer's Rule is systematic and avoids long Option 2. Rule-First Approach 1. Introduce the General Form of an Augmented Matrix For a 2×2 system of linear equations: $\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$

	the augmented matrix is written as: $\left[\begin{array}{ccc c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \end{array} \right]$ For a 3×3 system: $\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$ the augmented matrix is: $\left[\begin{array}{ccc c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array} \right]$ 2. Present the general form of Cramer's Rule for a 2×2 system: $1. \quad x = \frac{D_x}{D}, y = \frac{D_y}{D}$ Where $D = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \quad D_x = \begin{bmatrix} c_1 & b_1 \\ c_2 & b_2 \end{bmatrix} \quad D_y = \begin{bmatrix} a_1 & c_1 \\ a_2 & c_2 \end{bmatrix}$ 3. Model determinant computation for D, D_x, and D_y, given these equations: $\begin{aligned} 2x + 3y &= 5 \\ 4x - y &= 7 \end{aligned}$ the augmented matrix is written as $\left[\begin{array}{cc c} 2 & 3 & 5 \\ 4 & -1 & 7 \end{array} \right]$ $D = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix} \quad D_x = \begin{bmatrix} 5 & 3 \\ 7 & -1 \end{bmatrix} \quad D_y = \begin{bmatrix} 2 & 5 \\ 4 & 7 \end{bmatrix}$ 4. Reinforce the relationship $x = D_x/D$ and $y = \frac{D_y}{D}$ through guided questioning. 5. Let learners compute and verify their own solutions.: $x = \frac{13}{7}, y = \frac{3}{7}$
Video Lesson	https://youtu.be/AUqeb9Z3y3k?si=-mHYbuYg64pxV1c-

I. Activity 1:	Practicing Learned Skills
II. Materials:	worksheet or paper, pen, calculator
III. Instructions:	1. Work individually or with a partner. 2. Solve each system of equations using Cramer's Rule only. 3. Show complete and clear determinant computations. 4. After solving, check if your answers satisfy the original equations. 5. Submit your work and be ready to explain one solution to the class. Given: 1. $3x - 2y = 11$ $5x + y = 29$ 2. $x + 2y - z = 5$ $3x - y + 2z = 10$ $2x + y + z = 8$ 3. $x + 2y + 3z = 14$ $2x - y + z = 5$ $3x + y - 2z = 4$
V. Reflection:	Answer each question. 1. How did Cramer's Rule help you organize and solve the systems of equations, and what role did the determinants play in finding the solutions? 2. What differences did you notice between solving a 2×2 system and a 3×3 system using Cramer's Rule?

	3. Why is it important to substitute your answers back into the original equations, and what does this tell you about the accuracy of your solution?
I. Activity 2:	Assessing Learning Outcomes
II. Materials	
III. Instructions:	PART I. (2×2 Systems) Solve each system using Cramer's Rule. Show your determinant computations. $2x+3y=8$ $x-y=1$ $3x+2y=7$ $x+4y=10$ PART II. COMPUTATION (3×3 Systems) Solve each system using Cramer's Rule (Sarrus' Rule or cofactor expansion). $x + y + z = 6$ $2x - y + 3z = 14$ $3x + 2y - z = 4$ PART III. CONCEPT CHECK (Write your answer in one sentence) When is Cramer's Rule applicable to a system of linear equations? What does it mean when the determinant of the coefficient matrix is zero? Why do we replace one column at a time when applying Cramer's Rule? PART IV. QUICK CHECK (Multiple Choice) 8. The determinant of the coefficient matrix of a 2×2 system is 0. What can you say about the system? A. It has a unique solution B. It has no solution or infinitely many solutions C. It always has three solutions D. Cramer's Rule still works 9. In Cramer's Rule, $x = \frac{D_x}{D}$. What does D_x represent? A. The determinant of the original matrix B. The determinant after replacing the x-column with constants C. The determinant after swapping rows D. The determinant of the identity matrix 10. Which method is BEST for computing determinants of 3×3 matrices when using Cramer's Rule? A. Sarrus' Rule B. Long division C. FOIL method D. Completing the square
V. Reflection:	Answer each question: - How did the determinant of the coefficient matrix help you determine whether Cramer's Rule could be used to solve the system? - What similarities and differences did you observe when applying Cramer's Rule to 2×2 and 3×3 systems of equations? - How did checking your solutions by substitution help you verify the accuracy and reliability of your answers?
I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	worksheet or paper, pen, calculator
III. Instructions:	Solve the following systems using Cramer's Rule. Problem 1 $\begin{cases} 2x + 3y = 13 \\ x - 2y = -4 \end{cases}$ Problem 2

	$\begin{cases} 3x + y = 11 \\ 2x - y = 4 \end{cases}$ Problem 3 $\begin{cases} x + 2y = 7 \\ 4x - y = 5 \end{cases}$ Problem 4 $\begin{aligned} x + y + z &= 6 \\ 2x - y + z &= 3 \\ x + 2y - z &= 3 \end{aligned}$
V. Reflection:	Answer each question: 1. How did Cramer's Rule help you find the values of the unknowns, and what role did the determinants play in the solution process? 2. Which problem was the most challenging to solve and why? What strategies did you use to overcome the difficulty? 3. How can you verify that the solutions obtained using Cramer's Rule are correct, and why is this step important? 4. Compare solving the 2×2 systems and the 3×3 system. What similarities and differences did you notice in the procedures and computations? *****END OF LESSON 3.8*****

Unit 4. INTRODUCTION TO LINEAR PROGRAMMING

Lesson 4.1. FUNDAMENTAL CONCEPT OF LINEAR PROGRAMMING

Objectives:	At the end of the lesson, learners are will be able to: 1. define linear programming (LP) and its importance in decision-making; 2. identify situations suitable for LP modeling; 3. identify variables, write objective functions, and convert verbal conditions to inequalities; and 4. formulate complete LP models, and create own LP problem scenario.
Key Ideas:	Linear programming is defined as a type of optimization problem that involves maximizing or minimizing a linear objective function, subject to linear equality and inequality constraints. Key Terms: Decision Variables – unknown quantities that need to be determined in order to achieve the optimal solution. Objective Function – a mathematical expression that represents the goal to be maximized or minimized. Constraints – linear inequalities or equations that represent the limitations or restrictions that the decision variables must satisfy. Play a short teacher-selected video or presentation introducing Linear Programming (optional). Ask learners to share key points they learned. The teacher may list their responses on the board. Example 1: Recognizing LP Situations A farmer wants to plant corn and beans on 10 hectares of land and maximize profit. Learners identify: • Decision variables: area planted to corn and area planted to beans • Objective function: maximize profit • Constraint: total land area is 10 hectares?

Example 2: Business Production

A bakery produces brownies and cookies. It aims to maximize profit but has limited flour and time.

- **Decision variables:** number of brownies and number of cookies produced
- **Objective function:** maximize profit
- **Constraint:** limited flour and available time

Determine whether each scenario can be modeled using LP.

- a. Planning meals with nutritional requirements (LP)
- b. Choosing clothes for a party (Not LP)
- c. Scheduling workers with limited working hours (LP)
- d. Allocating budget across school projects (LP)

Part 2. Linear Programming Model Construction

1. Identify the Decision Variables

- Decision variables are the unknown quantities that we want to determine. They represent the choices available in the problem.
- Each variable must represent a measurable quantity.
- Variables must be clearly and simply defined.
- They are usually represented by x , y (or more variables if needed), but the description of what they represent is more important than the symbol.

2. Write the Objective Function

The objective function expresses what we want to **maximize** or **minimize**. It is based on the goal of the problem.

There are two types:

- **Maximization** = e.g., profit, output, revenue, efficiency
- **Minimization** = e.g., cost, time, waste

How to write it:

1. Multiply each decision variable by its contribution to the objective.
2. Add them together to create the objective function.

1. Write the Constraints

Constraints represent the **limitations** in the problem.

These may refer to:

- Available resources (materials, time, labor, land, budget)
- Maximum production limits
- Storage, safety, or capacity restrictions
- Demand limits

Constraints must be expressed as **linear inequalities or equations**, commonly:

- “At most” $\rightarrow \leq$
- “At least” $\rightarrow \geq$
- “No more than” $\rightarrow \leq$
- “No less than” $\rightarrow \geq$
- “Exactly” $\rightarrow =$

How to write constraints:

1. Identify the resource being limited.
2. Determine how much of that resource each product uses.
3. Combine them into an inequality.

4. Add the Non-Negativity Restrictions

In real life, we cannot produce negative amounts. Therefore, all variables must be restricted:

$$x \geq 0$$

$$y \geq 0$$

Key notes:

- This ensures the solution stays realistic.

- This condition is ALWAYS present in every LP problem.

5. Combine All Parts into a Full LP Model

After identifying variables, objective function, constraints, and non-negativity, we group them logically into one clear model.

Example 1. Baking Scenario

A baker wants to determine how many cupcakes and cookies to produce in order to maximize profit. Each cupcake yields a profit of ₱15, while each cookie yields a profit of ₱10. The baker has at most 300 minutes available for production. Producing one cupcake requires 5 minutes, and producing one cookie requires 3 minutes. In addition, the bakery can sell no more than 40 cupcakes.

1. Identify the Decision Variables

Example:

A bakery makes cupcakes and cookies.

Let:

- x = number of cupcakes to produce
- y = number of cookies to produce

2. Write the Objective Function

Example:

Profit = ₱15 per cupcake, ₱10 per cookie

Objective:

Maximize $P = 15x + 10y$

3. Write the Constraints

a. Total baking time available: 300 minutes

Cupcake uses 5 minutes; cookie uses 3 minutes

Time Constraint: **$5x + 3y \leq 300$**

b. The bakery can sell no more than 40 cupcakes

Demand Constraint: **$x \leq 40$**

4. Add the Non-Negativity Restrictions

$x \geq 0, y \geq 0$

5. Combine All Parts into a Full LP Model

LP Model:

Decision Variables: x = cupcakes, y = cookies

Objective Function: Maximize $P = 15x + 10y$, where P is total profit in pesos

Constraints:

- **$5x + 3y \leq 300$**
- **$x \leq 40$**
- **$x, y \geq 0$**

Example 2. Repair Shop

A repair shop needs to minimize its weekly tool rental cost. It rents soldering irons at ₱120 each, with each unit handling 5 repair jobs per week, and multimeters at ₱180 each, with each unit handling 3 repair jobs per week. The shop must complete at least 30 repair jobs weekly, rent at least 4 multimeters for complex work, and can rent no more than 8 soldering irons due to limited supply. Formulate a linear programming model to determine how many of each tool should be rented

1. Decision Variables

Let I = number of soldering irons rented per week

M = number of multimeters rented per week

	Objective Function (Minimization). Minimize $C = 120I + 180M$, where C is the total weekly rental cost in pesos. Constraints. Job-capacity requirement: $5I + 3M \geq 30$. Minimum multimeters for diagnostics: $M \geq 4$. Maximum available soldering irons: $I \leq 8$. Non-negativity. $I \geq 0, M \geq 0$. LP Model. Decision variables: I = number of soldering irons rented per week; M = number of multimeters rented per week. Objective function: Minimize $C = 120I + 180M$. Constraints: $5I + 3M \geq 30$; $M \geq 4$; $I \leq 8$. Non-negativity: $I \geq 0, M \geq 0$.
Video Lesson:	https://youtu.be/hdqcbFWJxwE?si=LdoHB7cnn4X-9gu3

I. Activity 1:	Practicing Learned Skills																							
II. Materials:	pen and paper																							
III. Instructions:	Part I. Categorize each statement as decision variable, objective function, constraint, or non-negativity condition.																							
	1. A factory earns ₱30 profit per chair and ₱50 per table it produces.																							
	2. The company can use at most 120 hours of labor this week.																							
	3. Each chair requires 3 hours while each table requires 5 hours to build.																							
	4. Let x be the number of chairs and y be the number of tables produced.																							
	5. The warehouse can store no more than 60 finished products in total.																							
	6. The business wants to maximize its total profit from chair and table production.																							
	7. The production of chairs cannot exceed 20 units due to the material shortages.																							
	8. Both x and y must be greater than or equal to zero.																							
	Part II.																							
1. The group will receive an LP scenario with an intentionally flawed model.																								
2. Identify the errors in the LP model.																								
3. Write the corrected model and explain why each correction is needed.																								
4. If there is no mistake in any part of the model, write “No Error” but still write an explanation why.																								
Lamp Problem																								
1. 1A workshop produces desk lamps and floor lamps. Each desk lamp earns ₱80 profit while each floor lamp earns ₱120. The workshop has at most 240 labor hours. A desk lamp needs 6 hours; a floor lamp needs 9 hours. They can sell no more than 20 floor lamps.																								
<table><tr><th>LP model</th><th>Intentionally Flawed LP</th><th>Corrected LP model</th><th>Reason for the correction</th></tr><tr><td>Decision variables</td><td>Let x – number of desk lamps y = number of floor lamps z – profit</td><td></td><td></td></tr><tr><td>Objective Function</td><td>Maximize Profit $z = 80x + 120y^2$</td><td></td><td></td></tr><tr><td>Constraints</td><td>Labor hour constraint $6x + 9y \geq 240$ Floor lamp constraint $y \leq 20$</td><td></td><td></td></tr><tr><td>Non-negativity</td><td>$x \leq 0$ $y \geq 0$</td><td></td><td></td></tr></table>					LP model	Intentionally Flawed LP	Corrected LP model	Reason for the correction	Decision variables	Let x – number of desk lamps y = number of floor lamps z – profit			Objective Function	Maximize Profit $z = 80x + 120y^2$			Constraints	Labor hour constraint $6x + 9y \geq 240$ Floor lamp constraint $y \leq 20$			Non-negativity	$x \leq 0$ $y \geq 0$		
LP model	Intentionally Flawed LP	Corrected LP model	Reason for the correction																					
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Constraints	Labor hour constraint $6x + 9y \geq 240$ Floor lamp constraint $y \leq 20$																							
Non-negativity	$x \leq 0$ $y \geq 0$																							
Pastry Problem																								
2. A bakery makes cupcakes (c) and cookies (k). Profit is ₱15 per cupcake and ₱10 per cookie. There are 300 minutes available. A cupcake takes 5 minutes, a cookie 3 minutes. Due to demand, you can sell at most 40 cupcakes and at least 10 cookies.																								
<table><tr><th>LP model</th><th>Intentionally Flawed LP</th><th>Corrected LP model</th><th>Reason for the correction</th></tr><tr><td>Decision variables</td><td>Let c = number of cookies k = number of cupcakes</td><td></td><td></td></tr></table>					LP model	Intentionally Flawed LP	Corrected LP model	Reason for the correction	Decision variables	Let c = number of cookies k = number of cupcakes														
LP model	Intentionally Flawed LP	Corrected LP model	Reason for the correction																					
Decision variables	Let c = number of cookies k = number of cupcakes																							

	Objective Function	Maximize Profit $p = 10c + 15k$		
	Constraints	Time constraint $5c + 3k \leq 300$ Sale constraint $c \leq 40$ $k \leq 10$		
	Non-negativity	$x \geq 0$ $y \leq 0$		
	Satisfaction Problem A group of students want to maximize their weekend satisfaction by optimally allocating their time between studying and engaging in leisure activities. Each hour spent studying contributes 5 units to their satisfaction, while each hour spent on leisure activities contributes 10 units. The have identifies 30 hours available over the weekend that can be dedicated to either studying or leisure. To ensure academic preparedness, they must dedicate a minimum of 8 hours to studying. To avoid burnout and maintain well-being, they also want to ensure at least 6 hours spent on leisure. Furthermore, for very two hours spent studying, they believe it’s beneficial to have at least one hour of leisure to maintain focus and balance. The student aims to determine how many hours to studying and leisure to achieve highest possible total satisfaction.			
	LP model	Intentionally Flawed LP	Corrected LP model	Reasoning
	Decision variables	Let x = time for studying y = time for leisure		
	Objective Function	Maximum Satisfaction $z = 5x - 10y$		
	Constraints	1. Time constraint $x + y \geq 30$ 2. Study Hour $x \leq 8$ 3. Leisure Hour $y \leq 10$ 4. Balance Requirement $2x \leq y$		
	Non-negativity Constraints	$x \leq 0$ $y \leq 0$		
IV. Reflection:				

I. Activity 2	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	Choose the letter of the correct answer. 1. What is Linear Programming? A. A way of graphing all equations B. A method for choosing random decisions C. A tool for converting words into numbers D. A process for optimizing a goal under given constraints 2. Why is Linear Programming important in real-life decision making? A. It determines the best solution when resources are limited B. It replaces mathematical reasoning C. It guarantees perfect decisions D. It allows unlimited resources

	3. Which situation can be modeled using Linear Programming? A. Choosing your favorite sport. B. Choosing which movie to watch in a limited time. C. Maximizing profit from selling two products under a limited budget. D. Choosing the color of your notebook out of 10 notebooks in a store. 4. In a Linear Programming model, decision variables represent: A. The total cost B. The restrictions or limits C. The graph of the solution D. The quantities that must be chosen 5. Which of the following is a correct inequality from this statement, "The total working time of 60 hours must not be exceeded." A. $2x + 3y < 60$ B. $2x + 3y = 60$ C. $2x + 3y \geq 60$ D. $2x + 3y \leq 60$ Constructed Response Analyze each situation and provide what is required. 6. A student wants to allocate study time between Math and Science. Each hour of Math (x) increases preparedness by 5 points. Each hour of Science (y) increases preparedness by 3 points. Write the objective function that maximizes total preparedness points. Answer: _____ 7. A farmer grows peppers and tomatoes to earn profit. Each hectare planted to peppers yields a profit of ₱1,000, and each hectare planted to tomatoes yields a profit of ₱600. The farm has a total of 20 hectares of land available. The farmer has only 30 bags of fertilizer; peppers use 3 bags per hectare and tomatoes use 1 bag per hectare. Let x be the number of hectares planted to peppers and y be the number of hectares planted to tomatoes. Define variables: _____ Objective function: _____ Constraints: a. _____ b. _____ Non-negativity: _____ 8. Create your own employment-related situation where a manager must make a decision under limited resources. a. Describe the situation (1–2 sentences). b. Define the decision variables x and y. c. Write the objective function (maximize or minimize). d. Write at least two constraints as inequalities. e. State the non-negativity conditions.
V. Reflection:	Answer each question. 1. How did identifying decision variables, objective functions, constraints, and non-negativity conditions help you understand how a real-life situation can be translated into a mathematical model? 2. What common mistakes did you notice in the incorrect LP models, and how did correcting them improve the meaning and accuracy of the solutions? 3. How can linear programming be useful in real-life decision-making situations such as business production, budgeting, or time management? Explain using one of the problems you solved.

I. Activity 3:	Extending and Reinforcing Learning
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II. Materials:	pen and paper, GeoGebra
III. Instructions:	Perform the indicated tasks accurately. 1. Below are incomplete Linear Programming models. Identify the decision variables, objective function, and constraints. Maximize $P = 8x + 12y$ Subject to: $x + y \leq 20$ $x, y \geq 0$ Questions: - What are the decision variables? - What is the goal of the objective function? - What does the constraint mean? 2. Fill in the blanks. - In LP, the quantities to be determined are called the _____. - The expression we want to maximize or minimize is called the _____. - Restrictions like time, materials, or budget are called the _____. - Variables cannot be negative because _____. 3. Convert the verbal condition into an inequality. - You must spend no more than 50 pesos on materials. - At least 10 items must be produced. - You have 30 minutes to work, and each task takes 5 minutes. 4. Write the objective function for each situation. - Cupcakes give ₱10 profit; cookies give ₱15 profit. - A shop wants to minimize cost: Item A costs ₱40, Item B costs ₱20. - A student wants to maximize study points: Math gives 6 points, Science gives 4 points. Enhancement Activities Instruction: Perform the indicated tasks accurately. 1. Create a Linear Programming model involving three decision variables. Choose a real-life theme (business, events, school projects, agriculture) Required output: $3 \text{ decision variables}$ $1 \text{ objective function}$ $\text{At least 3 constraints}$ Non-negativity 2. Below is a proposed Linear Programming model. Identify all expressions that make it non-linear and rewrite it correctly. Maximize: $Z = 20x + 3y + xy$ Subject to: $2x + y^2 \leq 40$ $x + y \geq 10$ 3. Choose a real business, school group, or community project (real or hypothetical). Write: - Its possible objective function. - At least two real constraints. - A short explanation of why Linear Programming is helpful in this context. 4. Exchange LP models with a classmate. Review your partner's LP model by: (a) identifying the decision variables and confirming they are defined with units and a time period; (b) checking that every constraint is linear, has correct coefficients, and uses the inequality direction implied by the scenario's verbal conditions; (c)

	suggesting one specific improvement to the model (for example, tighter wording of the objective, a missing non-negativity restriction, or a coefficient rounding issue).
V. Reflection:	Answer each question. 1. How did translating real-life situations into mathematical expressions (variables, objective function, and constraints) help you understand the structure of a linear programming model? 2. What difficulties did you encounter when converting verbal statements into inequalities or objective functions, and how did you overcome them? 3. How can linear programming help in making better decisions in real-life situations such as business planning, budgeting, or resource allocation? Give one example from the activity.
*****END OF LESSON 4.1***	

Lesson 4. 2. GRAPHICAL METHOD FOR SOLVING LINEAR PROGRAMMING

Objectives:	By the end of the lesson, learners are able to: 1. identify the components of linear programming such as the decision variables, objective function, constraints, non-negativity restrictions, and feasible region; 2. illustrate the feasible region of a Linear Programming (LP) problem; 3. identify the corner points (vertices) of the feasible region; 4. determine the optimal solutions to a LP problem using the Corner Point Theorem; and 5. solve linear programming problems of two variables by graphical method, with or without technology
Key Ideas:	Feasible region is the solution set of the system of linear inequalities derived from the given constraints. It forms a polygon on the coordinate plane. The corner points of this region are the vertices of the polygon Steps of Graphical Method Using the Problem 1. Decision Variables (These represent the quantities to be determined.). 2. Objective Function (This represents the goal of the problem, usually maximizing profit.) 3. Constraints (These are the limitations given in the problem.) 4. Graph the Constraints 5. Corner Points of the Feasible Region (Find intersections) 6. Evaluate the Objective Function 7. Optimal Solution
Examples:	Feasible region is the solution set of the system of linear inequalities derived from the given constraints. It forms a polygon on the coordinate plane. The corner points of this region are the vertices of the polygon Graphical Method Using the Problem Example 1. During a school fair, a class plans to sell burgers and hotdogs to raise funds. Each burger yields a profit of ₱10.00, while each hotdog yields a profit of ₱5.00. However, the following constraints must be considered: 1. The booth can prepare at most 80 food items per day. 2. Due to limited ingredients, at most 40 burgers can be prepared. 3. The number of hotdogs must be at most 50 to meet customer demand. Solutions: Step 1. Decision Variables (These represent the quantities to be determined.) • Let x be the number of burgers to be sold. • Let y be the number of hotdogs to be sold. Step 2: Objective Function (This represents the goal of the problem, usually maximizing profit.) Each burger gives a profit of ₱10, and each hotdog gives a profit of ₱5.

Maximize:

$$z = 10x + 5y$$

where z is the total profit.

Step 3: Constraints (These are the limitations given in the problem.)

- a. The booth can prepare at most 80 food items per day, which can be expressed as:

$$x + y \leq 80$$

- b. Due to limited ingredients, at most 40 burgers can be prepared.

$$x \leq 40$$

The number of hotdogs must be at most 50 to meet customer demand: $y \leq 50$.

Non-negative constraints

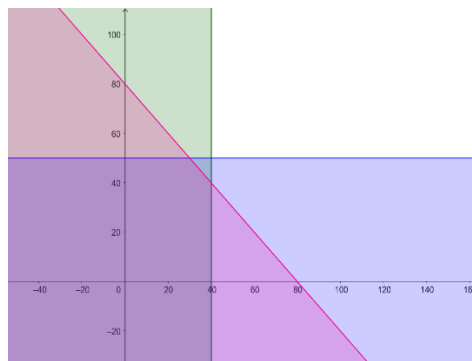
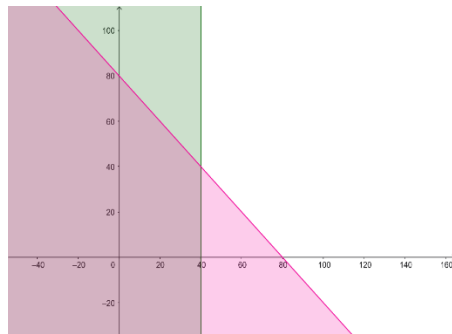
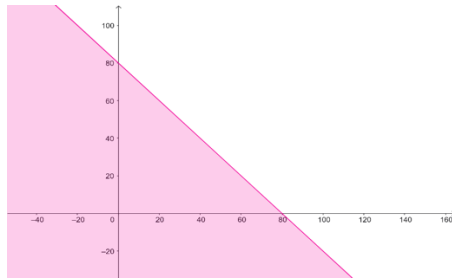
$$x \geq 0, y \geq 0$$

Step 4: Graph the Constraints

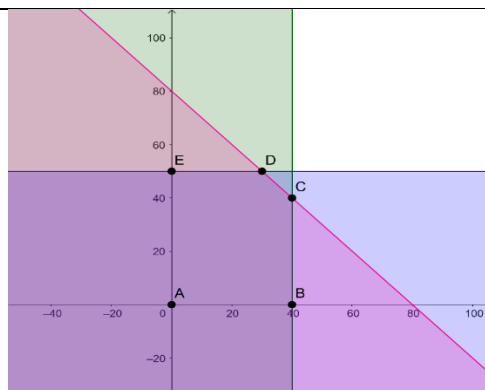
Convert each into boundary lines.

Graphing $x + y \leq 80$

$$y \leq 50$$



Step 5: Corner Points of the Feasible Region (Find intersections)



Point of intersections.

$A(0,0)$, $B(40,0)$, $C(40,40)$, $D(30,50)$, $E(0,50)$

Step 6: Evaluate the Objective Function

$$z = 10x + 5y$$

Corner	(x,y)	Profit
A	(0,0)	$10(0) + 5(0) = 0 + 0 = 0$
B	(40,0)	$10(40) + 5(0) = 400 + 0$
C	(40,40)	$10(40) + 5(40) = 400 + 200 = 600$
D	(30,50)	$10(30) + 5(50) = 300 + 250 = 550$
E	(0,50)	$10(0) + 5(50) = 0 + 250 = 250$

Step 7: Optimal Solution

The maximum profit is 600 at $(x,y) = (40, 40)$. This means that selling 40 burgers and 40 hotdogs yields the highest profit while satisfying all constraints. This solution remains valid since it also satisfies the condition for hotdogs which is $y \leq 50$, since $40 \leq 50$.

Even with the restriction of at most 50 hotdogs, the optimal solution is still 40 burgers and 40 hotdogs, with a total profit of ₱600.

Video Lesson

<https://youtu.be/hdqcbFWJxwE?si=LdoHB7cnn4X-9gu3>

I. Activity 1:

Practicing Learned Skills

II. Materials:

pen and paper

III. Instructions:

Solve the general LP problem.

Minimize:

$$z = 3x + 5y$$

Subject to

$$\begin{cases} x + y \leq 6 \\ 2x + y \geq 8 \\ x, y \geq 0 \end{cases}$$

During a school play, a theater sells VIP seats and regular seats. The theater management aims to determine the number of each type of seat to sell in order to maximize profit. Each VIP seat generates a profit of ₱120, while each regular seat generates a profit of ₱80. However, the following limitations must be considered:

1. The theater can accommodate at most 300 people.
2. Due to seating arrangements, at most 120 VIP seats can be sold.
3. To ensure affordability for students, at least 100 regular seats must be sold or offered

V. Reflection:

Answer each question.

1. Why is it necessary to test the corner points instead of any point inside the feasible region when finding the optimal solution?
2. How does your process differ when solving a minimization problem compared to maximization problem?

I. Activity 2:

Assessing Learning Outcomes

II. Materials:	pen and paper
III. Instructions:	Part I. Write TRUE if the statement is correct and FALSE if it is not correct. 1. In linear programming, the objective function represents what we aim to maximize or minimize. 2. The constraints in a linear programming problem are always represented by linear inequalities. 3. The feasible region may include points that do not satisfy all constraints. 4. If a linear programming problem includes non-negativity restrictions $x \geq 0$ and $y \geq 0$, then the feasible region lies in the first quadrant of the Cartesian plane (including the positive x- and y-axes). 5. Graphing the constraints correctly is not important because the optimal solution can be solved algebraically. Solve the linear programming problem. Show your solution. You are managing a small snack stand. You sell two items: sandwiches and fruit cups. Each sandwich earns ₱25 profit and each fruit cup earns ₱15 profit. However, you have limited resources: You can make at most 40 items total You only have ingredients for 30 sandwiches You must make at least 10 fruit cups Task: Define Variables: _____ Write the constraints: _____ Graph the constraints: _____ Identify corner points: _____ Evaluate each corner point: _____
IV. Reflection:	Answer each question. 1. Which combination gives the maximum profit? 2. What is the maximum profit in pesos? 3. Why does the best solution occur at a corner point of the feasible region?

I. Activity 3:	Extending and Reinforcing Learning
II. Materials:	pen and paper, GeoGebra
III. Instructions:	Use <i>GeoGebra</i> (or any equivalent graphing tool) to graph the constraints of each linear programming problem below and show the resulting feasible region. If a device or internet access is not available, complete the same task on graph paper. Submit a screenshot, a saved image, or a neat hand sketch. Then (a) identify the corner points of the feasible region; (b) evaluate the objective function at each corner; (c) state the maximum and the minimum of the objective function, or report that the LP has no feasible region. $1. z = 8x + 3y$ $\begin{cases} 2x - y \leq 1 \\ 3x + 2y \leq 6 \\ x, y \geq 0 \end{cases}$ $2. z = 30x + 20y$ $\begin{cases} x + y \leq 10 \\ x \geq 50 \\ y \leq 6 \\ x, y \geq 0 \end{cases}$
IV. Reflection:	1. Which corner gives the maximum, and what is its value? 2. Which corner gives the minimum, and what is its value? 3. If the feasible region is empty, what change to the constraints would create a feasible region? 4. How does using GeoGebra change the way you locate the corner points compared to graphing by hand in B.2 and B.3?

*****END OF LESSON 4.2*****

Lesson 4.3. SOLVING LINEAR PROGRAMMING PROBLEMS USING SIMPLEX METHOD

Objectives:

By the end of the lesson, you are expected to:

1. identify and set up a linear program in standard maximization form;
2. convert inequality constraints to equations using slack variables;
3. set up the initial simplex tableau by performing pivoting operations; and
4. identify the optimal solution from the optimal simplex tableau

Key Ideas:

Scope and Conditions of the Simplex Method

1. The LP must be a MAXIMIZATION problem (we maximize $Z = c_1 x_1 + c_2 x_2 + \dots + c_k x_k$).
2. Every structural constraint must be of the form $a x_1 + b x_2 + \dots \leq r$, with the right-hand side r non-negative.
3. Every decision variable must satisfy non-negativity: $x_1, x_2, \dots, x_k \geq 0$.
4. There is one slack variable per $\leq$ constraint, so the slack variables form the initial basis with values equal to the RHS of each constraint.

Steps in Solving Linear Programming Using Simplex Method

1. Set up the problem.
2. Introduce slack variables.
3. Set up the initial simplex tableau.
4. Identify the entering variable.
5. Identify the leaving variable.
6. Normalize the pivot row by dividing the s_1 row by 6.
7. Eliminate the other entries in the x_3 column.
8. Repeat the entering-variable test.
9. Identify the leaving variable.
10. Pivot and eliminate the other x_1 -column entries.
11. Check optimality and read the solution.

Examples:

Solve using the Simplex method.

Maximize:

$$Z = 10x_1 + 15x_2 + 20x_3$$

Subject to constraints:

$$2x_1 + 4x_2 + 6x_3 \leq 24$$

$$3x_1 + 9x_2 + 6x_3 \leq 30$$

$$x_1, x_2, x_3 \geq 0$$

Solution:

Step 1.

Maximize:

$$Z = 10x_1 + 15x_2 + 20x_3$$

Subject to:

$$2x_1 + 4x_2 + 6x_3 \leq 24$$

$$3x_1 + 9x_2 + 6x_3 \leq 30$$

$$x_1, x_2, x_3 \geq 0$$

Step 2. Add slack variables s_1 and s_2 to convert the $\leq$ constraints into equations.

Standard form:

$$2x_1 + 4x_2 + 6x_3 + s_1 = 24$$

$$3x_1 + 9x_2 + 6x_3 + s_2 = 30$$

$$Z - 10x_1 - 15x_2 - 20x_3 = 0$$

$$x_1, x_2, x_3, s_1, s_2 \geq 0$$

Step 3.

Basis	x_1	x_2	x_3	s_1	s_2	RHS
s_1	2	4	6	1	0	24
s_2	3	9	6	0	1	30
Z	-10	-15	-20	0	0	0

Step 4. The most negative Z-row entry is -20 under x_3 , so x_3 enters the basis.

	Step 5. Apply the minimum-ratio test to the x_3 column: $24/6 = 4$ and $30/6 = 5$. The smallest positive ratio is 4, so s_1 leaves the basis. The pivot element is 6. Step 6. The new x_3 row is $[1/3, 2/3, 1, 1/6, 0 \mid 4]$. Step 7. $R2_new = R2 - 6 \cdot R1_new$. $Z_new = Z + 20 \cdot R1_new$. Tableau after iteration 1 <table><tr><th>Basis</th><th>x_1</th><th>x_2</th><th>x_3</th><th>s_1</th><th>s_2</th><th>RHS</th></tr><tr><td>x_1</td><td>$1/2$</td><td>$2/3$</td><td>1</td><td>$1/6$</td><td>0</td><td>4</td></tr><tr><td>s_2</td><td>1</td><td>5</td><td>0</td><td>-1</td><td>1</td><td>6</td></tr><tr><td>Z</td><td>$-10/3$</td><td>$-5/3$</td><td>0</td><td>$10/3$</td><td>0</td><td>80</td></tr></table> Step 8. The most negative Z-row entry is $-10/3$ under x_1, so x_1 enters the basis. Step 9. Apply the minimum-ratio test to the x_1 column: for the x_3 row, $4 / (1/3) = 12$; for the s_2 row, $6 / 1 = 6$. The smallest positive ratio is 6, so s_2 leaves the basis. The pivot element is 1. Step 10.. The s_2 row already has pivot value 1, so it becomes the new x_1 row. $R1_new = R1 - (1/3) \cdot R2_new$. $Z_new = Z + (10/3) \cdot R2_new$. Final tableau: <table><tr><th>Basis</th><th>x_1</th><th>x_2</th><th>x_3</th><th>s_1</th><th>s_2</th><th>RHS</th></tr><tr><td>x_3</td><td>0</td><td>-1</td><td>1</td><td>$1/2$</td><td>$-1/3$</td><td>2</td></tr><tr><td>x_1</td><td>1</td><td>5</td><td>0</td><td>-1</td><td>1</td><td>6</td></tr><tr><td>Z</td><td>0</td><td>15</td><td>0</td><td>0</td><td>10</td><td>100</td></tr></table> Step 11. All Z-row entries are non-negative, so the solution is optimal. Basic variables: $x_1 = 6$ and $x_3 = 2$. Non-basic variable: $x_2 = 0$. Maximum value: $Z = 100$. Verification against the original constraints: $2(6) + 4(0) + 6(2) = 12 + 0 + 12 = 24$ (binding) $3(6) + 9(0) + 6(2) = 18 + 0 + 12 = 30$ (binding) $Z = 10(6) + 15(0) + 20(2) = 60 + 0 + 40 = 100$	Basis	x_1	x_2	x_3	s_1	s_2	RHS	x_1	$1/2$	$2/3$	1	$1/6$	0	4	s_2	1	5	0	-1	1	6	Z	$-10/3$	$-5/3$	0	$10/3$	0	80	Basis	x_1	x_2	x_3	s_1	s_2	RHS	x_3	0	-1	1	$1/2$	$-1/3$	2	x_1	1	5	0	-1	1	6	Z	0	15	0	0	10	100
	Basis	x_1	x_2	x_3	s_1	s_2	RHS																																																		
	x_1	$1/2$	$2/3$	1	$1/6$	0	4																																																		
	s_2	1	5	0	-1	1	6																																																		
	Z	$-10/3$	$-5/3$	0	$10/3$	0	80																																																		
	Basis	x_1	x_2	x_3	s_1	s_2	RHS																																																		
	x_3	0	-1	1	$1/2$	$-1/3$	2																																																		
	x_1	1	5	0	-1	1	6																																																		
	Z	0	15	0	0	10	100																																																		
	Video Lesson:	https://youtu.be/hdqcbFWJxwE?si=LdoHB7cnn4X-9gu3																																																							

I. Activity 1:	Practicing Learned Skills
II. Materials:	pen and paper
III. Instructions:	1. Maximize $z = 8x_1 + 10x_2 + 7x_3$ 2. Maximize $z = 9x_1 + 12x_2 + 15x_3$ Subject to $\begin{cases} x_1 + 3x_2 + 2x_3 \leq 10 \\ x_1 + 5x_2 + 2x_3 \leq 8 \\ x_1, x_2, x_3 \geq 0 \end{cases}$ Subject to $\begin{cases} x_1 + 2x_2 + 3x_3 \leq 18 \\ 2x_1 + x_2 + x_3 \leq 12 \\ x_1, x_2, x_3 \geq 0 \end{cases}$
V. Reflection:	Answer each question. - How did working with three decision variables (x_1, x_2, x_3) change your understanding of linear programming compared to problems with only two variables? - What strategies did you use to interpret and manage multiple constraints at the same time, and how did these constraints affect the possible solutions? - How does maximizing the objective function in these problems help you understand decision-making in real-life situations involving limited resources?

I. Activity 2.	Assessing Learning Outcomes
II. Materials:	pen and paper
III. Instructions:	True or False In a linear programming problem written in standard maximization form, all constraints must be expressed using “less than or equal to ($\leq$)” inequalities and all decision variables must be non-negative.

2. Converting an inequality constraint of the form
3. $x + y \leq 10$ into an equation for the simplex method is done by subtracting a slack variable.
4. In the initial simplex tableau for a maximization problem, the slack variables are the initial basic variables, and the right-hand side (RHS) represents their starting values.
5. Pivoting operations in the simplex method are performed only once, because the optimal solution can always be obtained in a single iteration.
6. An optimal simplex tableau for a maximization problem is reached when there are no negative coefficients in the Z-row, and the values of the decision variables can be read from the RHS of the basic variable rows.

Multiple Choice

1. Consider the linear programming problem:

Maximize: $Z = 3x + 5y$

Subject to

$$x + 2y \leq 8$$

$$3x + y \leq 9$$

$$x \geq 0, y \geq 0$$

Basis	x	y	s_1	s_2
Z	0	-1	2	0

Slack variables s_1 and s_2 are added.

Which of the following represents the correct initial simplex tableau?

- | Basis | x | y | s_1 | s_2 | RHS |
|-------|-----|-----|-------|-------|-----|
| s_1 | 1 | 2 | 1 | 0 | 8 |
| s_2 | 3 | 1 | 0 | 1 | 9 |
| Z | -3 | -5 | 0 | 0 | 0 |
- | Basis | x | y | s_1 | s_2 | RHS |
|-------|-----|-----|-------|-------|-----|
| s_1 | 1 | 2 | 1 | 0 | 8 |
| s_2 | 3 | 1 | 0 | 1 | 9 |
| Z | 3 | 5 | 0 | 0 | 0 |
- | Basis | x | y | s_1 | s_2 | RHS |
|-------|-----|-----|-------|-------|-----|
| s_1 | 1 | 2 | 1 | 0 | 8 |
| s_2 | 3 | 1 | 0 | 0 | 9 |
| Z | 3 | 5 | 1 | 1 | 0 |
- | Basis | x | y | s_1 | s_2 | RHS |
|-------|-----|-----|-------|-------|-----|
| s_1 | 1 | 2 | 0 | 0 | 8 |
| s_2 | 3 | 1 | 0 | 0 | 9 |
| Z | -3 | -5 | 1 | 1 | 0 |

2. The objective row of a simplex tableau for a maximization problem is shown below. Which variable should enter the basis in the next iteration?

Basis	x	y	s_1	s_2
Z	-2	-6	0	0

- A. x B. y C. s_1 D. None; the solution is optimal

3. For the entering variable y , the constraint rows have the following RHS and column entries. Using the minimum ratio test, which row is the pivot row?

- A. Row 1, because $12 \div 3 = 4$
- B. Row 2, because $8 \div 2 = 4$
- C. Either row, since the ratios are equal
- D. No row; the solution is unbounded

Row	RHS	y
1	12	3
2	8	12

4. After performing one simplex pivot, the new objective row becomes:
What does this indicate about the current solution?

- A. The solution is optimal
- B. Another iteration is required
- C. The solution is infeasible
- D. The problem is unbounded

	5. The final simplex tableau gives the values: $x = 2$, $y = 3$, $Z = 21$ The objective row has no negative coefficients. Which statement is most accurate? A. The solution is optimal and feasible B. The solution violates at least one constraint C. A better solution is still possible D. The optimal solution occurs when all slack variables are zero
IV. Reflection:	Answer each question. 1. How does the simplex method help you systematically find the best solution in a linear programming problem compared to guessing or trial-and-error? 2. What do the pivot operations and tableau changes represent in terms of improving a real-life decision (such as profit or resource use)? 3. How can you tell from a final simplex tableau that an optimal solution has already been reached, and why is this condition important?

I. Activity 3.	Extending and Reinforcing Learning
II. Materials:	pen and paper, GeoGebra
III. Instructions:	Analyze the scenario and perform the indicated tasks. Situation: Andrea is a senior high school student who runs a small online gadget shop. She assembles and sells two products: wireless earbuds and Bluetooth speakers. She wants to plan her weekly production to earn the highest possible profit. Decision variables: Let x = number of pairs of wireless earbuds produced per week. Let y = number of Bluetooth speakers produced per week. Production data: • Each pair of earbuds earns a profit of ₱350. Each speaker earns ₱600. • Each pair of earbuds takes 2 hours of assembly time. Each speaker takes 4 hours. Andrea has at most 100 assembly hours per week. • Each pair of earbuds uses ₱200 of capital. Each speaker uses ₱300 of capital. Andrea has at most ₱9,000 of capital per week. LP model: Maximize $P = 350x + 600y$ Subject to $2x + 4y \leq 100$ (assembly time, hours) $200x + 300y \leq 9,000$ (capital, pesos) $x, y \geq 0$ Task A. Solve using the graphical method (LE15). 1. Rewrite the constraints in simpler form: $x + 2y \leq 50$ and $2x + 3y \leq 90$. 2. Graph the constraints. 3. Identify the feasible region. 4. Determine the corner points. 5. Evaluate $P = 350x + 600y$ at each corner. 6. Identify the optimal solution. Task B. Solve using the simplex method. 1. Convert the inequalities into standard form using slack variables. 2. Set up the initial simplex tableau. 3. Perform iterations until the optimal solution is reached. 4. Read the final values of x, y, and P from the final tableau. Questions: • How many earbuds and speakers should Andrea produce? • What is her maximum profit? • Which resources are fully used at the optimum? • Which method did you find more efficient for this problem? Explain.

V. Reflection:	Answer each question. 1. Why does the optimal solution in linear programming usually occur at a corner point of the feasible region? 2. How do the constraints in this problem reflect real-world limitations in business decision-making? 3. Which method (graphical or simplex) do you think is more practical for real business problems, and why?
*****END OF LESSON 4.3*****	